

FIAT

A Novel

J & C · 2026

“Money is a collective fiction. The question is who writes the story.”

— Unknown

“The first panacea for a mismanaged nation is inflation of the currency; the second is war. Both bring a temporary prosperity; both bring a permanent ruin.”

— Ernest Hemingway

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Chapter One: The First Thread

Neo Smith was not looking for anything in particular on the morning of November 14th when he found the first thread.

He was doing what he had done most mornings for the past eighteen months — reading financial data out of habit, the same reflex that had gotten him out of bed at 5:30 AM for three decades, a behavior so deeply encoded that retirement had barely touched it. The hedge fund was gone. The commute was gone. The Bloomberg terminal with its seventeen open windows was gone. But the habit of attention — the analyst's fundamental orientation toward the world as a set of numbers waiting to be understood — that had not gone anywhere.

He was sitting in his study in Westport, Connecticut, a room he had furnished with the careful minimalism of someone who had spent too many years in cluttered trading floors. One desk. One chair. A window that looked out over Long Island Sound, where a container ship was making its slow way east. A Mac Studio on the desk, matte silver, running local inference — no cloud, no data leaving the building, a configuration he had spent two weeks getting right before he trusted it with anything he actually wanted to think about. The model he was running called itself Garvus, a name he had assigned because he found naming things useful and because the alternatives had all felt either too clinical or too cute.

"Garvus," he said. "Pull up this morning's CPI release."

The response came from the desktop speaker, unhurried, precise. "Bureau of Labor Statistics, Consumer Price Index Summary for October. All items, twelve-month change: 3.2 percent. Core CPI, excluding food and energy: 3.1 percent. Shelter: 4.9 percent. Food at home: 1.2 percent. Energy: negative 2.3 percent."

Neo looked at the numbers.

He had been to the grocery store on Saturday. He had stood in the produce aisle at the Westport Stop & Shop and done the mental arithmetic he had been doing involuntarily since childhood — this is what this cost before, this is what it costs now, this is the delta. The delta had not looked like 1.2 percent. Not even close.

This was not a new observation. He had been making it, idly, for the better part of two years. The number on the release and the number he was experiencing in the world had been diverging, and he had filed it, as he had filed many things in his working life, under the category of interesting but not my problem. His job had been to understand what the market believed, not what was true. Those were often different things. The difference was frequently where the money was.

But he was retired now, and interesting but not my problem had become interesting and I have time.

"Walk me through the methodology," he said. "Not the summary. The actual methodology. How does BLS calculate CPI?"

"The Consumer Price Index is calculated using a Laspeyres-type formula measuring price

changes for a fixed basket of goods and services,” Garvus said. “The basket is derived from the Consumer Expenditure Survey, which BLS conducts to determine what categories of goods and services urban consumers actually purchase. The basket is updated periodically — major revisions approximately every two years — to reflect changing consumption patterns.”

“The substitution adjustment.”

“Correct. The current methodology incorporates a geometric mean formula at the item level, which implicitly accounts for consumer substitution — the tendency of consumers to shift toward relatively cheaper goods when prices rise. If beef becomes significantly more expensive relative to chicken, consumers who substitute chicken are, in the BLS framework, not experiencing as much inflation as consumers who continue buying beef.”

Neo leaned back. Outside, the container ship had cleared the point and was moving into open water.

“And the argument for that methodology is what exactly?”

“That it accurately reflects actual spending patterns and therefore the actual cost of living for the average consumer. If consumers substitute and maintain similar utility — if chicken satisfies the same hunger as beef — then their real cost of living hasn’t increased as much as a fixed-basket measure would suggest.”

“And the argument against.”

“That it conflates consumer adaptation to higher prices with the absence of those higher prices. A consumer who has switched from steak to ground beef because steak has become unaffordable has experienced a real reduction in their quality of consumption. The index records this as price stability. Critics argue this systematically understates the erosion of purchasing power, particularly for lower-income consumers who have less flexibility to substitute without experiencing meaningful quality loss.”

Neo nodded slowly. This was not new to him — he had understood the substitution debate for years, the way someone who works adjacent to inflation data understands the debates about it without ever having needed to resolve them. But hearing it laid out plainly, without the institutional context that had always surrounded it in his working life, felt different. Sharper somehow.

“Hedonic adjustment,” he said.

“Hedonic quality adjustment is applied to specific categories where BLS determines that product quality has changed significantly over time. The methodology attempts to separate price changes reflecting genuine inflation from price changes reflecting improvements in the product itself. The canonical example is personal computers — a laptop costing the same as one purchased ten years ago but with dramatically more processing power, storage capacity, and features is, in the hedonic framework, effectively cheaper in quality-adjusted terms.”

“So if a car costs thirty thousand dollars today instead of twenty-five thousand five years ago, but it has more safety features and a better infotainment system—”

“BLS may adjust the price downward to reflect the quality improvement, reducing the

measured inflation in the new vehicles category.”

“Even if the consumer still has to come up with thirty thousand dollars.”

“Correct. The hedonic adjustment reflects changes in quality-adjusted price, not changes in the amount of money required to purchase the item.”

Neo was quiet for a moment.

He had spent thirty years in a business where the gap between what something cost and what it was worth was the entire game. The two numbers were almost never the same. The skill — the thing he had been genuinely good at, the thing that had made him wealthy — was understanding exactly where they diverged and why, and positioning accordingly.

What he was looking at now was a measurement system that was, by design and by documented methodology, measuring something other than what most people assumed it was measuring. Not fraudulently. Not through deliberate deception. Through a series of methodological choices, each with a defensible technical rationale, that in aggregate produced a number somewhat detached from the experience of the ordinary person trying to pay their bills.

“Owner’s Equivalent Rent,” he said.

“The largest single component of CPI shelter, which is itself the largest category in the overall index at approximately thirty-four percent of the basket. Owner’s Equivalent Rent represents the implicit rent that homeowners could receive if they rented their homes rather than living in them. BLS derives this figure through a survey asking homeowners what they believe their home would rent for.”

“Not actual rents.”

“Actual rents are captured in a separate component — Rent of Primary Residence — which covers renters. Owner’s Equivalent Rent is a constructed figure representing the homeownership share of shelter costs.”

“And the effect of this methodology during periods of rapidly rising home prices?”

“Because OER is based on rental equivalent surveys rather than home prices directly, it tends to lag actual housing market conditions significantly. During the 2021 through 2023 period of rapid home price appreciation, OER continued reflecting pre-appreciation rental equivalents for an extended period before catching up. During this lag, measured shelter inflation in CPI understated the actual increase in housing costs for homeowners and, with a further lag, for renters facing lease renewals in a market where landlords were repricing to reflect higher property values.”

Neo stood and walked to the window.

The methodology was public. All of it was public — the BLS technical documentation was available to anyone who wanted to read it, which almost nobody did, because it was dense and technical and most people trusted that 3.2 percent meant something close to what they thought it meant. The gap between the number and the experience was not a secret. It was a documented consequence of documented methodological choices.

The question forming in the back of his mind was not whether BLS was lying. He didn't think BLS was lying. He thought BLS was measuring something, accurately, that was not quite what most people assumed it was measuring.

The question was: in whose interest was it that the measurement existed in this form?

Not a conspiracy question. He was not interested in conspiracies — he had spent thirty years watching markets move on narratives that turned out to be false, and he had learned to distrust the explanatory elegance of any story that required coordinated deception among large numbers of people. Large numbers of people were terrible at coordinating. What they were very good at was acting in their individual self-interest in ways that produced coordinated-looking outcomes.

So the question was: whose individual self-interest was served by a CPI measurement that, through legitimate and documented methodology, consistently produced a number somewhat lower than alternative approaches would produce?

"Garvus. Who has an institutional interest in a lower CPI reading?"

A pause — longer than Garvus's usual response time, which Neo had learned to interpret as the model working through something genuinely complex.

"Several categories of actors have documented interests in lower measured inflation," Garvus said. "The Federal Reserve, whose credibility is tied to maintaining inflation near its two percent target — a lower CPI reading means the target has been met or approximated, which is preferable institutionally to a reading suggesting significant deviation. The federal government, because a significant number of federal expenditures and obligations are indexed to CPI — Social Security cost-of-living adjustments, TIPS bond payments, certain federal contracts — and a lower index reading reduces those outlays. Holders of fixed-income instruments, particularly long-duration government bonds, whose real returns are eroded by inflation — lower measured inflation supports the case that real returns remain positive. And institutional actors whose compensation is tied to real economic growth metrics, which are calculated using price deflators related to inflation measures — lower measured inflation can make real GDP growth appear stronger."

"None of that requires anyone to coordinate."

"No. Each of those actors has independent reasons to prefer a measurement methodology producing lower readings. The methodological choices embedded in CPI — hedonic adjustment, geometric mean substitution, Owner's Equivalent Rent — were not adopted through a single decision by a single actor. They were adopted incrementally over decades, through a process involving BLS technical staff, academic economists, congressional oversight, and the Boskin Commission's 1996 recommendations. Each choice had genuine academic support. The cumulative effect is a measurement that most serious economists acknowledge produces readings somewhat below alternative methodologies."

"How much below?"

"Shadow Government Statistics, maintained by economist John Williams, calculates inflation using methodologies closer to those used before the Boskin Commission modifications. His figures have historically run four to seven percentage points above official CPI. The Chapwood

Index, a separate alternative measure tracking actual prices of five hundred items in fifty cities, has shown similar divergences in certain periods. The Federal Reserve Banks of Cleveland and Atlanta both maintain alternative inflation measures — the Median CPI and Sticky Price CPI respectively — that sometimes diverge meaningfully from headline CPI.”

“So the gap between reported and experienced inflation is documented.”

“The gap between headline CPI and alternative methodologies is documented. Whether that gap accurately represents the difference between reported and experienced inflation depends on which methodology one believes more accurately captures the actual cost of living. This is a genuine methodological dispute among economists, not a settled question.”

Neo turned away from the window and sat back down.

He had pulled one thread. It had not unraveled anything — not yet. What it had produced was a documented methodological question about a measurement that underpinned an enormous range of policy decisions and financial instruments. The question was not new. Economists had been debating it for decades.

What was new — what retirement had given him that his working life had not — was the distance to ask why the debate had been resolved the way it had been, rather than some other way. And who had been at the table when it was resolved. And what they had needed the number to be.

“Let’s go further back,” he said. “Not the methodology. The Federal Reserve itself. Why does a central bank exist, and what is it actually designed to do?”

“The United States Federal Reserve System was established by the Federal Reserve Act, signed into law by President Woodrow Wilson on December 23rd, 1913,” Garvus said. “The stated rationale was to provide the country with a safer, more flexible, and more stable monetary and financial system, following a series of financial panics — most acutely the Panic of 1907 — that had demonstrated the fragility of the existing national banking system.”

“And the less stated rationale.”

Another pause.

“The legislative history is well-documented,” Garvus said. “In November 1910, a group of individuals met secretly at Jekyll Island, Georgia, at a hunting club owned by J.P. Morgan. The attendees included Senator Nelson Aldrich, who chaired the National Monetary Commission; Frank Vanderlip of National City Bank; Henry Davison of J.P. Morgan and Company; Charles Norton of First National Bank of New York; Paul Warburg of Kuhn, Loeb and Company; and A. Piatt Andrew, Assistant Secretary of the Treasury. The meeting remained secret for approximately twenty years. Its purpose was to draft legislation that would become the basis for the Federal Reserve Act.”

“Private bankers drafting the legislation that would create the central bank.”

“Drafting a proposal that would become the basis for the legislation, yes. The final Federal Reserve Act differed from the Jekyll Island draft in several respects and went through significant congressional debate. But the basic architecture — a system of regional Federal Reserve Banks,

technically private institutions owned by member commercial banks, overseen by a Board of Governors — reflected the Jekyll Island framework.”

“The Federal Reserve Banks are privately owned.”

“The twelve regional Federal Reserve Banks are technically owned by the commercial member banks in their districts, which hold stock in their regional Reserve Bank. This stock differs from ordinary corporate stock — member banks receive a fixed dividend rather than market-rate returns, cannot sell or trade the stock, and the stock conveys limited governance rights. The Board of Governors, which sets monetary policy, is a federal government agency whose members are appointed by the President and confirmed by the Senate. The institutional structure is a hybrid — neither fully public nor fully private — which generates ongoing debate about its accountability.”

Neo sat with that.

None of this was secret. It was in every serious history of the Federal Reserve, in academic papers, in the Congressional Record. The Jekyll Island meeting had been described in detail by the participants themselves, years after the fact. This was not hidden information. It was obscure — not concealed, but unfamiliar to most people because most people had no particular reason to go looking for it.

He was beginning to understand the nature of what he was doing. Not the uncovering of secrets. The assembly of a picture from pieces that were all in plain sight, scattered across a century of financial history, that nobody had recently put together in one place and looked at whole.

“Start me a working document,” he said. “We’re going to go further back before we come forward. Bank of England. 1694. Beginning of the architecture.”

“Working document created. The Bank of England was established by royal charter in 1694, during the reign of William III, primarily to finance the Nine Years’ War against France. The government needed to borrow £1.2 million. A group of merchants agreed to lend the money to the government in exchange for the right to incorporate as a bank, accept deposits, and issue banknotes — paper currency — up to the value of their loan to the government. This was the first instance of a central bank in the modern sense: a private institution given a government charter to issue currency in exchange for financing government debt.”

“Private institution. Government charter. Financing government debt by creating money.”

“That is an accurate summary of the basic arrangement, yes.”

Neo looked at the water. The container ship was gone, past the horizon, carrying whatever it carried toward wherever it was going.

“That’s three hundred and thirty years ago,” he said. “And the basic architecture hasn’t changed.”

“The basic relationship between sovereign debt issuance and central bank money creation has remained structurally consistent across the intervening period. The specific mechanisms have evolved considerably. The institutional forms have changed. The scale has expanded by many

orders of magnitude. But the fundamental arrangement — government borrows, central bank creates the money to facilitate the borrowing, the currency derives its value from the sovereign's taxing authority and institutional trust in the system — was established in 1694 and has not been fundamentally redesigned since."

Neo opened the working document on screen. At the top he typed a single line:

The architecture was built for a specific purpose. Follow the purpose.

He sat back and looked at the blinking cursor.

Outside, Long Island Sound was gray and flat and patient, the way it always was in November — the way it had been before any of this existed and would be long after.

He had found the first thread.

He had not yet decided whether to pull it.

But he already knew he was going to.

Chapter Two: The Architecture

Neo spent three days with the Bank of England before he felt he understood it well enough to move forward.

This was how he had always worked. Not quickly — never quickly, despite what the people who had worked for him sometimes thought when they watched him make decisions in seconds that they had spent weeks preparing. The speed at the end was possible because of the slowness at the beginning. He had learned to sit with primary sources until they stopped feeling like information and started feeling like structure. Until he could see the shape of the thing rather than just its surface.

The Bank of England, the more he looked at it, had a shape that kept recurring.

“Let’s go back to the beginning,” he said on the third morning. “Not the history. The mechanics. What actually happened in 1694, step by step, in terms of money creation.”

“In 1694,” Garvus said, “the English government under William III was financing an expensive war against France and had exhausted conventional borrowing options. A Scottish entrepreneur named William Paterson proposed a scheme: a group of private merchants would pool £1.2 million in capital and lend it to the government at eight percent interest annually. In exchange, the lending group would receive a royal charter incorporating them as the Bank of England, granting them the exclusive right to issue banknotes — paper currency — up to the value of the loan.”

“So they lent the government money they had.”

“Initially, yes. But the critical innovation was what happened next. The Bank of England did not simply transfer £1.2 million in gold coin to the government and wait for repayment. It created banknotes — paper receipts — representing claims on that gold, and lent those notes to the government. The government then spent those notes into circulation, paying soldiers, suppliers, contractors. The notes circulated as currency because they were backed by the Bank’s promise to redeem them in gold on demand.”

“And the Bank still had the gold.”

“For a period, yes. But because the notes were circulating as currency rather than being immediately redeemed, the Bank discovered it could issue notes in excess of its gold reserves — because not everyone would demand redemption simultaneously. This is the foundational insight of fractional reserve banking: a bank need not hold in reserve the full value of its outstanding obligations, because statistical experience shows that only a fraction of depositors or noteholders will demand their money at any given time.”

Neo leaned forward.

“Give me the mechanics of that. The actual arithmetic.”

“Consider a simplified example. The Bank of England holds £100 in gold. It issues £100 in banknotes fully backed by that gold — a one-to-one reserve ratio. Those notes circulate. People deposit them back at the Bank rather than carrying them. Now the Bank has £100 in gold

reserves and £100 in deposit liabilities. But because depositors rarely all withdraw simultaneously, the Bank can safely lend out a portion of those deposits — say £90 — while keeping £10 in reserve to meet expected withdrawal demand. It issues a new £90 in notes or credits to borrowers. The money supply has now expanded from £100 to £190. The Bank has £100 in gold backing £190 in currency.”

“Where did the extra £90 come from?”

“The Bank created it. Not from gold. From the act of lending. This is the mechanism that took centuries for most people to fully understand — and that most people still do not understand today. When a bank makes a loan, it does not transfer existing money from one account to another. It creates new money by crediting the borrower’s account. The loan is an asset on the bank’s balance sheet. The deposit it creates is a liability. Both are created simultaneously, from nothing, by the act of lending.”

Neo sat back.

He had understood this intellectually for most of his career. Every serious finance professional understood it. But there was a difference between knowing something and confronting it plainly, without the professional vocabulary that made it feel ordinary and technical rather than what it actually was: the fundamental mechanism by which the majority of money in modern economies comes into existence.

Not printed by governments. Created by private banks. In the act of lending.

“The analogy I want the reader to understand,” he said, half to himself, half to Garvus, “is the hotel.”

“The hotel analogy is standard in monetary economics,” Garvus said. “A hotel with one hundred rooms doesn’t need all one hundred occupied simultaneously to function — it operates on the statistical expectation that some percentage will always be occupied and some percentage will always be empty. A bank operates similarly: it doesn’t need to hold enough reserves to satisfy all depositors simultaneously, only enough to satisfy the expected demand at any given time.”

“Except the hotel can’t create new rooms,” Neo said. “The bank can create new money.”

“Correct. The analogy breaks down precisely there. The hotel is constrained by physical reality. The bank is constrained only by reserve requirements, capital requirements, and the willingness of borrowers to borrow. Within those constraints, the money supply expands with every new loan and contracts with every loan repayment. This is not a feature of modern finance specifically — it was the operational reality of the Bank of England from its earliest years.”

Neo wrote in the working document: Money is not a thing. It is a relationship. A promise. A social agreement enforced by institutional trust and, ultimately, by the state’s monopoly on legitimate violence.

He looked at what he had written. Then he continued.

“How did this model spread?”

“The Bank of England’s success as a financial innovation was rapid and visible. It allowed the English government to borrow at lower interest rates than its competitors, because the Bank’s note issuance created a liquid, stable medium of exchange that increased economic activity and therefore the tax base that backed the debt. England’s ability to finance wars, colonial expansion, and infrastructure through this mechanism gave it a structural advantage over rivals still dependent on coin-based finance.”

“So other nations copied it.”

“Over the following century and a half, central banking spread across Europe — the Banque de France in 1800, established by Napoleon partly to finance his military campaigns; the Bank of the Netherlands in 1814; the National Bank of Belgium in 1850; the Reichsbank in Germany in 1876. Each institution was adapted to local conditions but shared the core architecture: a private or semi-private institution with a government charter, a monopoly or near-monopoly on note issuance, and a balance sheet that allowed the creation of money through the mechanism of lending.”

“And the United States resisted this.”

“The United States had a complicated relationship with central banking from its founding. The debate was, at its core, a debate about the concentration of financial power — about whether a single institution with the ability to create money and influence credit conditions represented an unacceptable concentration of private power over public economic life.”

Neo nodded. He knew this history, but he wanted to hear it plainly.

“Walk me through the American banking wars.”

“Alexander Hamilton, as the first Secretary of the Treasury, proposed the First Bank of the United States in 1790. His argument was Hamiltonian in its basic character: a strong central financial institution would stabilize the currency, manage government debt, and provide the credit infrastructure necessary for commercial development. Thomas Jefferson and James Madison opposed it on constitutional and philosophical grounds — the Constitution did not explicitly authorize a national bank, and concentrating financial power in a single institution seemed to them incompatible with republican government.”

“Hamilton won.”

“Hamilton won the first round. The First Bank was chartered in 1791 for a twenty-year term. It performed many of the functions Hamilton intended — it issued notes that served as a relatively stable national currency, it managed government accounts, it provided credit. Its charter lapsed in 1811 when Congress declined to renew it, partly due to opposition from state banks that resented the competition and partly due to continued philosophical opposition to central banking.”

“And then the War of 1812 demonstrated why that was a problem.”

“Correct. Without a central bank to coordinate financing, the federal government struggled to fund the war. A Second Bank of the United States was chartered in 1816. It operated until 1836, when President Andrew Jackson vetoed the renewal of its charter — one of the most

consequential financial decisions in American history.”

“Jackson’s argument.”

“Jackson’s veto message is one of the most direct attacks on concentrated financial power in American political history. He argued that the Bank was a monopoly — a private institution with a government charter that gave it advantages unavailable to ordinary citizens, that it served the interests of wealthy shareholders, many of them foreign, at the expense of the American public, and that concentrating such power in a single institution was incompatible with democratic governance. ‘The rich and powerful,’ he wrote, ‘too often bend the acts of government to their selfish purposes.’”

“Was he right?”

A pause.

“Jackson’s critique identified real structural concerns about the concentration of financial power in a private institution with a public mandate. His analysis of who benefited from the Bank’s operations was not without foundation — the Bank’s largest shareholders were wealthy merchants and, as he noted, significant foreign investors. Whether the alternative — a fragmented system of state-chartered banks with no central coordination — served ordinary Americans better is a more complicated question. The period following the Bank’s dissolution, sometimes called the Free Banking Era, was characterized by significant financial instability, repeated panics, and a proliferation of state bank notes of wildly varying quality and reliability.”

“So Jackson killed the central bank and got chaos.”

“Jackson killed the central bank and got a financial system that was more democratically distributed in theory and more unstable in practice. The tension between those two outcomes — between the democratic concerns about concentrated financial power and the practical concerns about financial stability — is the central tension of American monetary history. It was not resolved by Jackson’s veto. It simply went underground for several decades, until the events of 1907 forced it back into the open.”

Neo stood and went to the kitchen. He made coffee with the deliberate attention he gave to things that did not require deliberate attention — an old habit, filling the spaces between thoughts with something physical. When he came back he set the mug on the desk and looked at the working document.

The architecture, so far, looked like this: A private institution. A government charter. The power to create money through lending. Established to finance sovereign debt. Copied across the developed world because it worked — because it made governments more powerful and the bankers who held the charter wealthy. Resisted in America on democratic grounds. The resistance ended in chaos. The chaos ended in 1907.

“The Panic of 1907,” he said.

“The Panic of 1907 was a financial crisis triggered by the failure of a speculative attempt to corner the copper market, which led to a run on the Knickerbocker Trust Company in New York, which spread to other trust companies and banks as depositors, fearing insolvency, rushed to

withdraw funds simultaneously. The banking system lacked any central mechanism for providing emergency liquidity — any lender of last resort that could inject funds into solvent institutions facing temporary runs.”

“The lender of last resort function. Explain that from first principles.”

“A bank run occurs when depositors lose confidence in a bank’s ability to honor withdrawals and attempt to withdraw simultaneously. Because banks hold only fractional reserves — remember our earlier example, £10 in reserves backing £100 in deposits — a simultaneous withdrawal demand cannot be met from reserves alone. Even a solvent bank — one whose assets genuinely exceed its liabilities — can fail during a run if it cannot convert its assets to cash quickly enough to meet withdrawal demands. Illiquidity is not insolvency, but illiquidity can cause insolvency if assets must be sold at fire-sale prices to meet immediate demands.”

“The distinction matters.”

“It matters enormously. A lender of last resort — a central bank willing to lend freely to solvent institutions at penalty rates against good collateral — can stop a run without using public funds to bail out insolvent institutions. The theory was articulated by Walter Bagehot in his 1873 work *Lombard Street*: lend freely, at high rates, against good collateral. The effect is to signal to the market that solvent institutions will not fail for lack of liquidity, which reduces the panic that drives runs.”

“And in 1907 the United States had no lender of last resort.”

“The United States had J.P. Morgan.”

Neo almost smiled.

“Tell me about Morgan in 1907.”

“J.P. Morgan, then seventy years old, effectively performed the function of a central bank through personal authority and the coordination of private financial resources. He convened the major New York bankers in his library, assessed which institutions were solvent and which were not, organized pools of capital to support the solvent ones, and allowed the insolvent ones to fail. His intervention stopped the panic. His personal credibility and the trust the financial community placed in his judgment served as a substitute for the institutional mechanism that did not exist.”

“One private citizen stopped a national financial crisis.”

“One private citizen with sufficient wealth, institutional relationships, and personal authority to coordinate the response, yes. The lesson drawn by the financial establishment was not that private coordination had worked — it was that the system had come dangerously close to catastrophe and that its stability depended on the continued existence and willingness of a single aging man. Morgan himself reportedly said that he would not always be around to do this.”

“And the solution was Jekyll Island.”

“The solution was Jekyll Island. Three years after the Panic, the secret meeting at the Morgan hunting club produced the framework for what would become the Federal Reserve System — an institutionalized lender of last resort, a mechanism for coordinating the banking system’s

response to liquidity crises, and a framework for managing the money supply and interest rates at a national level.”

“Designed by the people who would benefit most from it.”

“Designed by senior representatives of the major financial institutions of the period, yes. Whether the resulting institution serves primarily the public interest or primarily the interests of the financial sector it was designed to stabilize is one of the genuinely contested questions in monetary economics. The answer is probably both, in proportions that have varied across the institution’s history and that depend significantly on who is asking the question and what evidence they weight most heavily.”

Neo drank his coffee and looked out at the Sound.

The architecture was now clearer. Three hundred years from the Bank of England to the Federal Reserve. The same basic mechanism, refined and institutionalized. A private entity — or a hybrid, technically — with a public mandate. The power to create money. Established in response to crisis. Designed by the people whose interests it most directly served.

None of this was secret. All of it was in the history books, in the academic literature, in the Federal Reserve’s own official histories. The question was what it meant when you assembled it in one place and looked at it whole.

He opened a new tab and started writing.

Not for Garvus. For the Substack.

He had started the account six weeks ago with no particular plan — a place to put notes, to think out loud, to impose the discipline of writing on the research. He had seventeen subscribers. He did not know any of them.

He wrote for two hours. When he was done he had a piece called The Architecture: How Money Actually Works, and Why It Matters That Most People Don’t Know. He read it twice, changed three things, and published it at 11:47 PM.

Then he closed the laptop and went to bed.

He did not think about the seventeen subscribers. He did not think about whether anyone would read it.

He was already thinking about Bretton Woods.

Three days later, he received an email.

Mr. Smith —

I found your piece on the Federal Reserve’s origins through a footnote in a paper I was reading on housing affordability metrics. I’ve been trying to understand why the inflation numbers don’t match what I’m seeing in the cities I cover, and your explanation of CPI methodology is the clearest I’ve read. I have questions. A lot of them. I’m a journalist — independent, not affiliated with a major outlet at the moment, which means I can actually ask the questions I want to ask. Would you be willing to talk?

— Anna Reyes

Neo read it twice. The reference to the housing affordability paper told him something. The phrase “not affiliated with a major outlet at the moment” told him something else. Someone who had been somewhere and left, or been asked to leave, or left because she couldn’t ask what she wanted to ask.

He wrote back a single line: What questions?

Her response came in forty minutes. It was twelve hundred words long.

He read it carefully. Then he picked up the phone and called the one person whose opinion he trusted enough to run this past before he went any further.

The phone rang four times. Then a voice — older, precise, carrying the particular authority of someone who had spent decades being listened to in rooms where it mattered.

“Neo.” A pause. “I wondered when you were going to call.”

“Hello, Arthur,” Neo said. “I’ve been doing some reading.”

“I know,” Arthur Ellsworth said. “I saw the piece. You’ve still got the instinct. What have you found?”

“I don’t know yet. That’s why I’m calling.”

A longer pause. Outside, a boat was moving across the Sound in the early dark, its running lights red and green against the black water.

“Come see me,” Arthur said. “Bring your notes. We’ll argue about it properly.”

Chapter Three: The Professor

Arthur Ellsworth lived in a house that suited him.

It was a 1920s colonial in Cambridge, Massachusetts, three blocks from Harvard Yard, the kind of house that accumulated books the way other houses accumulated furniture — not as decoration but as infrastructure, shelving built into every available wall, stacks on the floors where the shelves had run out, papers on the dining room table that had not been used for dining in years. Neo had been coming here for thirty-five years, since he was a twenty-two year old graduate student who had wandered into Arthur's monetary economics seminar and found, for the first time, a framework that matched the way he already thought about the world.

Arthur was eighty-one now. He moved more carefully than he once had and his hearing had declined enough that he kept the television off rather than deal with the volume required. But his mind was what it had always been — precise, contrarian, allergic to comfortable conclusions, constitutionally incapable of telling anyone what they wanted to hear. He had served on the Federal Reserve Board of Governors for eight years under two administrations, had advised three Treasury secretaries, had written four books that were still assigned in graduate programs, and had spent the last decade in what he called active retirement — which meant he read everything, talked to anyone worth talking to, and maintained strong opinions about all of it.

He had coffee ready when Neo arrived. He always had coffee ready. It was the one domestic habit he had maintained with consistency through forty years of academic and institutional life.

"You look like you haven't slept," Arthur said, handing him a mug.

"I've been reading."

"I know what you've been reading." Arthur settled into the chair across the fireplace — the chair Neo had never once seen him sit anywhere else. "I read your piece. Both of them."

"Both?"

"You published a second one three days ago. The Bretton Woods piece. It was better than the first."

Neo had not realized Arthur was following the Substack. He filed this away.

"What did you think?"

"I think you're doing what you always did," Arthur said. "Following the thread wherever it goes regardless of where you'd prefer it not to go. It was your best quality as a student and your most inconvenient quality as a fund manager." He paused. "What have you concluded so far?"

"I haven't concluded anything. I'm still assembling the picture."

"Then assemble it for me. Out loud. Let's see where the gaps are."

Neo drank his coffee and looked at the fire. Outside, Cambridge was gray and cold, November doing what November did in New England.

"The architecture," he said. "Bank of England 1694. Private institution, government charter,

money creation through sovereign debt financing. The model spreads across Europe because it works — it gives governments access to capital they couldn't otherwise mobilize and it makes the charter holders wealthy. Arrives in America, gets killed twice on democratic grounds, the chaos that follows demonstrates why stability requires some version of it, Jekyll Island produces the Federal Reserve in 1913."

"Accurate summary," Arthur said. "Continue."

"Then the wars. The First World War breaks the gold standard — countries abandon convertibility to finance military spending, print money, create inflation. The interwar period is a sequence of attempts to restore the prewar system that mostly fail because the underlying economic relationships have changed too much. The Great Depression demonstrates what happens when the monetary system contracts instead of expands during a crisis — the Fed, in its early years, does exactly the wrong thing, raises rates, contracts the money supply, turns a recession into a catastrophe."

"Also accurate. Though I'd note that the Fed's culpability for the Depression's severity remains genuinely debated among monetary historians. Friedman and Schwartz made the case forcefully. Others have pushed back on the counterfactual."

"Noted. Then Bretton Woods."

Arthur nodded slowly. "Tell me what you understand about Bretton Woods."

"July 1944. Forty-four nations meet at the Mount Washington Hotel in Bretton Woods, New Hampshire. The war is not over but its outcome is not seriously in doubt. The delegates are designing the postwar international monetary system. Two major competing proposals on the table — Keynes for the British, White for the Americans."

"Walk me through both proposals."

"Keynes proposes an International Clearing Union — a supranational central bank that would issue its own reserve currency, which he calls the *bancor*, and that would manage trade imbalances by penalizing both surplus and deficit countries. The mechanism is symmetric — countries running persistent trade surpluses would pay into the system as well as countries running deficits. The logic is that imbalances are a two-party problem and the adjustment burden shouldn't fall entirely on deficit countries."

"And White's proposal."

"Harry Dexter White, representing the American Treasury, proposes a system anchored to the U.S. dollar, which would be convertible to gold at a fixed rate of thirty-five dollars per ounce. Other currencies would peg to the dollar. The dollar becomes the world's reserve currency. The International Monetary Fund and the World Bank are created as institutional pillars of the system."

"White wins."

"White wins, because the United States holds the cards. At Bretton Woods in 1944, the U.S. holds approximately two-thirds of the world's monetary gold reserves. It has emerged from the war as the only major industrial economy with its physical infrastructure intact. It is the world's

largest creditor nation. Keynes argues brilliantly and loses convincingly. The system that emerges is the system the Americans proposed.”

Arthur was quiet for a moment. He had been at meetings, over his career, where the institutional memory of Bretton Woods was still present — where people who had been young economists in 1944 were still shaping policy in the 1980s and 1990s. He had a relationship to this history that was different from Neo’s.

“What does the bancor proposal represent,” he asked, “that the dollar standard doesn’t?”

Neo thought about this carefully.

“Keynes was trying to solve the fundamental asymmetry in the gold standard’s operation,” he said. “Under the gold standard, countries with trade deficits lose gold reserves — their currencies come under pressure, they’re forced to deflate, to cut wages and spending to reduce imports and attract capital. The adjustment is painful and falls on workers and ordinary people. Countries with trade surpluses accumulate gold — they face no corresponding pressure to adjust. The system is structurally biased toward contraction and toward placing the adjustment burden on deficit countries.”

“And the bancor mechanism.”

“The bancor mechanism would charge surplus countries interest on their excess balances — creating an incentive for them to spend their surpluses back into the system rather than accumulating them indefinitely. The adjustment burden becomes symmetric. Surplus countries have skin in the game.”

“Keynes understood something that White’s proposal did not fully address,” Arthur said. “Which is that a reserve currency system in which one country’s currency serves as the global medium of exchange creates a structural tension that the issuing country cannot indefinitely manage.”

“The Triffin dilemma.”

“Tell me.”

“Robert Triffin identified it in 1960. A country whose currency serves as the world’s reserve currency must supply enough of that currency to satisfy global demand for reserves — which means running trade deficits, exporting currency. But a country that runs persistent trade deficits eventually undermines confidence in its currency — people start to wonder whether it can actually convert its currency to gold at the promised rate. The reserve currency issuer is caught between two obligations that are structurally incompatible over time.”

“And the resolution.”

“Nixon closes the gold window on August 15th, 1971. The United States unilaterally suspends dollar convertibility to gold. The Bretton Woods system as designed ceases to exist. The dollar remains the world’s reserve currency — but it is now backed not by gold but by — “ Neo paused.

“By what?” Arthur asked. The question was not Socratic. He genuinely wanted to hear how Neo would answer it.

“By institutional trust. By the network effects of a global economy denominated in dollars. By the U.S. military’s ability to guarantee the security of the trade routes through which dollar-denominated commodities flow. And by the petrodollar arrangement that follows within a few years.”

“Explain the petrodollar arrangement as you understand it.”

“Following the 1973 oil shock, the Nixon administration — primarily through Henry Kissinger — negotiates an arrangement with Saudi Arabia. The Saudis agree to price oil exclusively in dollars and to recycle their oil revenues — petrodollars — into U.S. Treasury securities. In exchange, the United States provides security guarantees to the Saudi regime. The effect is to create a structural global demand for dollars — any country that needs to import oil needs to acquire dollars first — that substitutes for the gold convertibility that had previously anchored the system.”

Arthur nodded slowly.

“And the implications of that arrangement for countries that are neither the United States nor oil exporters.”

Neo sat with that for a moment.

This was where the picture started to look different depending on where you were standing.

“Garvus,” he said, pulling out his laptop. “Give me the documented effects of dollar reserve currency status on developing economies — the actual measured outcomes, not the theoretical framework.”

“Developing economies that do not hold significant dollar reserves face a structural vulnerability,” Garvus said. “Dollar-denominated debt — sovereign borrowing in dollars rather than domestic currency — exposes these countries to exchange rate risk. If a country borrows in dollars and its domestic currency depreciates against the dollar, the real burden of its debt increases even if it has made no additional borrowing decisions. This mechanism has been documented across multiple emerging market crises — Mexico 1994, the Asian financial crisis 1997 through 1998, Argentina 2001, Ecuador, Turkey, Sri Lanka more recently.”

Arthur leaned forward slightly. For all his institutional loyalty to the system he had helped administer, this was the part that had always troubled him.

“The Asian financial crisis,” Neo said. “Walk me through the mechanics.”

“The affected economies — Thailand, Indonesia, South Korea, Malaysia, the Philippines — had accumulated significant dollar-denominated private debt during a period of capital inflow in the early 1990s. When currency pressure began in Thailand in 1997 and the baht was devalued, it triggered a regional contagion. As currencies depreciated against the dollar, the domestic-currency value of dollar debts increased dramatically. Companies and banks that had borrowed in dollars to invest in local-currency assets found their balance sheets effectively insolvent overnight — not because their underlying businesses had failed, but because the exchange rate had moved.”

“The IMF response,” Neo said.

"The International Monetary Fund provided emergency lending but conditioned it on structural adjustment programs — requirements to raise interest rates, cut government spending, and implement structural reforms including financial sector liberalization. The interest rate increases, intended to defend currencies and attract capital, deepened the economic contractions. Indonesia's GDP fell by approximately thirteen percent in 1998. Unemployment rose sharply. The political consequences included the fall of the Suharto government after thirty-two years."

Arthur stood and walked to the window. He had been involved, peripherally, in the U.S. policy response to the Asian crisis. The memory was not comfortable.

"What the crisis made visible," he said, not to Garvus but to Neo, "was something that the architects of the postwar system had not fully confronted. When capital flows freely and a single currency serves as the global reserve, the countries at the periphery of the system are structurally exposed to decisions made at its center — to Fed interest rate movements, to dollar liquidity conditions, to the risk appetites of institutional investors in New York and London — in ways they cannot fully hedge against and did not fully choose."

"The original sin," Neo said.

"The original sin in international finance — the term economists use for the inability of developing countries to borrow internationally in their own currencies — is both a cause and a consequence of the dollar system's structure. Countries that cannot borrow in their own currency are permanently exposed to exchange rate risk. Countries permanently exposed to exchange rate risk have difficulty building the institutional credibility necessary to borrow in their own currency. The trap is structural."

Neo looked at his notes. Then he looked at Arthur.

"The numbers," he said. "The actual human numbers. Not the GDP figures."

Garvus responded before Arthur could.

"The World Bank estimates that the 1997 through 1998 Asian financial crisis pushed approximately twenty-four million people into poverty across the affected countries. In Indonesia alone, the poverty rate increased from eleven percent to approximately forty percent in the crisis period before partially recovering. Child malnutrition rates increased. School enrollment declined as families pulled children out of school to contribute income. The documented effects on infant mortality and long-term developmental outcomes for children born during crisis periods are significant and persistent."

The room was quiet.

This was the part that Neo had not spent enough time with during his working career. The macro numbers had been real to him — the current account balances, the reserve levels, the capital flow data. The human consequences had been abstracted into country risk ratings and emerging market spreads.

"Anna Reyes," he said, more to himself than to Arthur.

"Who?"

"A journalist who contacted me after my first piece. She's been reporting from Latin America. She's seen the ground-level consequences of this — the currency crises, the IMF conditionality programs, what structural adjustment actually looks like when it lands on a family trying to feed their children."

Arthur looked at him steadily. "And you think she has something to contribute to this."

"I think she has something I don't have. The technical framework is necessary. It's not sufficient."

Arthur considered this for a long moment.

"The gap between how this system looks from inside its institutions and how it looks from the countries on its periphery," he said finally, "is one of the genuinely serious criticisms of the postwar financial architecture. Not the only criticism. Not necessarily a fatal one. But serious, documented, and consistently underweighted in Western economic analysis."

"Including yours?" Neo asked.

Arthur almost smiled. "Including mine. I spent thirty years inside institutions that were built on the assumption that what was good for dollar system stability was broadly good for the global economy. That assumption is not obviously wrong. It is also not obviously right. The evidence is genuinely mixed and the answer depends significantly on where you're standing when you look at it."

Neo's phone buzzed. He looked at it.

A message from an unfamiliar number. Mr. Smith — Anna Reyes. I'm in Boston for three days. Would you be willing to meet? I have documents I think you'd want to see.

He showed it to Arthur.

The older man read it and handed the phone back.

"Bring her," Arthur said simply. "Let's see what she has."

Chapter Four: Three Perspectives

Anna Reyes arrived at Arthur Ellsworth's house on a Tuesday afternoon carrying a canvas bag that appeared to contain several manila folders, a laptop, and what Neo estimated was approximately forty hours of reported notes.

She was thirty-four years old, compact, with the particular quality of attention that good journalists develop — a way of being in a room that was simultaneously present and observational, as if some part of her was always slightly outside the scene taking notes on it. She had grown up in a country where extreme poverty was not a policy abstraction but the texture of daily life, where political violence was ambient rather than occasional. She had seen people killed before she was ten years old. Not reported on it. Witnessed it. This was not something she discussed with sources. It had simply shaped, permanently and without drama, the way she looked at systems that produced those outcomes and called them structural inevitabilities.

She had come to the United States eleven years ago on a journalism visa. She had reported for a wire service, then a digital outlet, then independently — the last transition not entirely voluntary, the circumstances of which she did not elaborate on when Neo had asked. She had covered currency crises, IMF negotiations, capital flight, the ground-level consequences of monetary policy decisions made in buildings she now found herself sitting near.

She looked around Arthur's book-filled living room with an expression that was not quite surprise.

"You've read all of these?" she asked Arthur.

"Most of them," Arthur said. "Sit down. Neo tells me you have documents."

She sat. She did not perform deference to either of them, which Neo noted approvingly.

"What I have," she said, opening the canvas bag, "is three years of reporting on capital flows out of Latin America. The documented numbers are significant. But before I show you, I want to understand something." She looked at Neo directly. "Your piece on Bretton Woods — you described the Triffin dilemma as an unresolved structural tension. What's your current thinking on how the dollar system actually manages that tension in practice? Not theoretically. Actually."

Neo glanced at Arthur, who said nothing and looked interested.

"Garvus," Neo said, opening his laptop. "Walk us through the eurodollar system."

"The eurodollar system is one of the most significant and least publicly understood features of contemporary global finance," Garvus said. "The term is misleading — eurodollars have nothing to do with the euro currency. They are dollar-denominated deposits held in banks outside the United States, beyond the direct regulatory jurisdiction of the Federal Reserve."

Anna leaned forward slightly.

"The eurodollar market originated in the 1950s," Garvus continued. "Soviet bloc countries, holding dollar reserves from commodity exports, were concerned that deposits held in U.S. banks might be frozen in the event of political conflict. They began depositing their dollars in European

banks — primarily in London — where the deposits were denominated in dollars but held outside U.S. jurisdiction. American banks established London branches to compete for this business. The market expanded rapidly.”

“How large is it now?” Anna asked.

“The eurodollar market’s size is difficult to measure precisely because it exists outside conventional reporting frameworks. Conservative estimates place offshore dollar-denominated credit at somewhere between thirteen and twenty trillion dollars. Some analysts, including economist Jeffrey Snider, who has written extensively on the system, argue the true figure is considerably larger when shadow banking components are included. For comparison, the U.S. M2 money supply — the conventional measure of dollars inside the United States — is approximately twenty-one trillion dollars.”

Anna looked at Neo. “So there may be as many dollars outside the United States as inside it.”

“Potentially more, depending on measurement methodology,” Garvus said. “And critically, these dollars are created by the same fractional reserve mechanism we discussed earlier — offshore banks lending dollars they don’t fully hold in reserve — but outside the direct regulatory oversight of the Federal Reserve. The Fed can influence eurodollar conditions through its domestic policy decisions and through its swap line network, but it does not directly control the eurodollar system.”

Arthur spoke for the first time since Anna had arrived. “What this means practically,” he said, “is that the global dollar system is considerably larger and more complex than the institution nominally responsible for it can fully see or manage. During the 2008 financial crisis, the eurodollar system experienced a severe funding crisis that the Fed had not fully anticipated and that required emergency swap line extensions to fourteen foreign central banks to prevent a global dollar liquidity collapse.”

“The swap lines,” Anna said. “Who gets them?”

“The Fed’s standing swap line agreements — permanent, unlimited access — are with the Bank of Canada, the Bank of England, the European Central Bank, the Bank of Japan, and the Swiss National Bank,” Garvus said. “These are the five major reserve currency central banks. Temporary swap lines during crises have been extended to a broader group including Australia, South Korea, Brazil, Mexico, Singapore, and others.”

“Not extended to?”

“Most of the Global South. Most of Africa. Most of South and Southeast Asia outside the crisis extensions. Countries whose currencies are not reserve currencies and whose central banks are not part of the core network.”

Anna opened one of her manila folders and placed a document on the table between them.

“This,” she said, “is a summary of capital flow data I compiled from IMF balance of payments statistics, BIS quarterly reports, and the Tax Justice Network’s financial secrecy database. What it shows is the net financial transfer between developing economies and developed economies over the past two decades.”

Neo and Arthur both leaned in.

The numbers were stark.

"The conventional narrative," Anna said, "is that wealthy countries transfer resources to developing countries through foreign aid, development lending, and investment. The Global Financial Integrity project, the Tax Justice Network, and researchers at the University of Massachusetts have documented what the actual net flows look like when you include capital flight, profit repatriation, debt service payments, trade mispricing, and illicit financial flows."

"Garvus," Neo said. "The documented figures."

"Global Financial Integrity's research, covering the period 2008 through 2017, estimated that developing countries lost approximately eight point eight trillion dollars through trade mispricing alone — the practice of misreporting the value of imports and exports to shift profits to lower-tax jurisdictions. The UN Conference on Trade and Development has estimated that illicit financial flows out of Africa alone exceed one hundred billion dollars annually — significantly more than the continent receives in foreign aid. Economist Léonce Ndikumana and James Boyce, in their research on Africa's capital flight, documented that for every dollar of external debt African countries took on between 1970 and 2010, approximately sixty cents left the continent as capital flight."

The room was quiet.

"So the aid flows one direction," Anna said, "and the capital flows the other direction, in larger quantities, through mechanisms that are legal, semi-legal, and illegal — and the net result is a transfer of resources from the poorest parts of the world to the wealthiest."

"That is what the documented data suggests," Garvus said. "The methodology is contested — different researchers use different definitions and data sources and arrive at different figures. The directional conclusion — that net resource flows from developing to developed economies are larger than commonly represented — is consistent across multiple independent research programs."

Arthur was quiet for a long moment. He had known the broad outlines of this. The specific figures had a weight that the broad outlines did not convey.

"The mechanisms," Neo said. "Walk through the specific mechanisms. Not the headline number. How it actually works."

"Several documented mechanisms contribute to the net outflow," Garvus said. "Transfer pricing — multinational corporations setting prices for transactions between their own subsidiaries to shift profits to low-tax jurisdictions. A subsidiary in a developing country sells its product to a related entity in a tax haven at an artificially low price, minimizing taxable income in the country where the production occurred. The practice is legal within limits, widely used, and difficult for developing country tax authorities with limited capacity to detect and challenge."

"Give me an analogy," Anna said. "For readers who don't have a finance background."

"Consider a coffee company that sources beans from Ethiopia," Garvus said. "The Ethiopian subsidiary sells the raw beans to a holding company registered in a tax haven — the Cayman

Islands, Luxembourg, Ireland — at a price set below market value. The holding company then sells to the consumer-facing entity in Europe or the United States at market price. The profit — the difference between the artificially low transfer price and the actual market price — is captured in the tax haven jurisdiction where it faces minimal taxation. Ethiopia's tax authority sees only the low-value transaction and can tax only that. The profit that was economically generated in Ethiopia, through Ethiopian land and Ethiopian labor, is legally captured elsewhere."

Anna made a note.

"The offshore financial system," Neo said. "The City of London's role specifically."

"The City of London — the square mile financial district, legally distinct from greater London — functions as the hub of what researchers have called the spider's web of offshore financial centers," Garvus said. "A network of British Overseas Territories and Crown Dependencies — the Cayman Islands, the British Virgin Islands, Jersey, Guernsey, the Isle of Man, Bermuda — operate as satellite jurisdictions offering financial secrecy and low taxation while remaining connected to London's legal infrastructure and financial expertise. The Tax Justice Network's Financial Secrecy Index consistently ranks this network as the largest enabler of global financial secrecy."

"Documented," Arthur said. It was not a question. He had seen the research.

"The most comprehensive academic treatment is probably Ronen Palan, Richard Murphy, and Christian Chavagneux's work on tax havens," Garvus said. "Nicholas Shaxson's *Treasure Islands* provides a more accessible account. The underlying data comes from BIS banking statistics, IMF balance of payments data, and investigative journalism — the Panama Papers, the Paradise Papers, the Pandora Papers — that has made specific transactions visible."

Anna looked at Arthur. "You were at the Fed during some of this. Did the institution understand what the offshore system was doing to capital flows?"

Arthur considered the question with the care it deserved.

"The Federal Reserve understood the eurodollar system and its implications," he said. "The offshore tax and secrecy architecture was less central to the Fed's mandate — monetary policy and financial stability rather than tax policy or capital flow equity. Whether the institution should have engaged more forcefully with the implications for developing economies is a fair question. The honest answer is that the institutional frame — what problems the Fed was constituted to solve — did not naturally surface those implications as primary concerns."

"The frame determines what you see," Anna said.

"The frame determines what you see," Arthur agreed quietly.

Neo looked at the documents on the table. Then he looked at the two people across from him — the former Fed governor who had administered the system and was honest enough to acknowledge its blind spots, and the journalist who had grown up in a country on its periphery and had spent her career documenting what those blind spots cost.

"Garvus," he said. "The private equity dimension. Specifically the mechanism of leveraged buyouts and what they extract."

"A leveraged buyout is the acquisition of a company using borrowed capital, typically structured so that the debt is loaded onto the acquired company's balance sheet rather than the acquiring firm's," Garvus said. "The private equity firm contributes a relatively small equity stake — often ten to thirty percent of the purchase price — and finances the remainder with debt secured against the target company's assets and cash flows. The acquired company is then responsible for servicing the debt used to purchase it."

"The analogy," Anna said.

"You buy a house worth one million dollars. You put down one hundred thousand of your own money and borrow nine hundred thousand. But instead of the mortgage being your obligation, you transfer it to the house itself — and then charge the house rent to service the mortgage while also paying yourself a management fee for owning it. If the house generates enough cash flow to service the debt and your fees, you profit substantially from your relatively small equity investment. If it doesn't, the house defaults on its mortgage while your personal exposure remains limited to your initial equity."

"And the consequences for the acquired company."

"The debt burden constrains the company's ability to invest in operations, research, workforce, or capital expenditure. Management is incentivized to maximize short-term cash flow to service debt rather than long-term value creation. The private equity firm typically charges management fees and transaction fees regardless of performance. Exit typically occurs within three to seven years through sale or public offering, at which point the firm realizes its return and the next owner inherits whatever debt remains."

"The documented outcomes," Neo said.

"Academic research on private equity outcomes is genuinely mixed," Garvus said. "Studies by economists at Harvard, Chicago, and the University of Oxford find evidence of productivity improvements and efficiency gains in some contexts. Studies by Eileen Appelbaum and Rosemary Batt, economists who have written the most comprehensive critical analysis of the industry, document significant negative outcomes in sectors including retail, healthcare, and housing — where the business model's requirements for cash extraction are particularly incompatible with the social functions the acquired companies perform."

"Healthcare specifically," Anna said.

"Private equity ownership of hospitals, physician practices, nursing homes, and emergency care providers has been documented to correlate with increased costs, reduced staffing levels, and in some cases patient safety outcomes. A 2023 study published in JAMA found that private equity acquisition of hospitals was associated with increased patient complications and falls. The mechanism proposed is consistent with the general model — debt service requirements and fee extraction reduce resources available for staffing and equipment."

Arthur stood and went to the kitchen. He came back with more coffee and set it down without ceremony.

"What you're describing," he said to Neo, "is a set of mechanisms — offshore capital extraction, leveraged acquisition, transfer pricing — that are individually legal, institutionally

rationalized, and in aggregate produce resource concentration at a scale that the system's own measurement apparatus does not fully capture."

"Including the inflation measurement apparatus," Neo said.

"Including that." Arthur sat down. "A household experiencing the combined effects of stagnant wages, increased healthcare costs driven partly by financialization, housing costs driven partly by institutional ownership, and grocery costs that outpace measured inflation is experiencing an economic reality that the headline statistics do not fully represent. This is documentable. The question of what follows from it is less settled."

Anna had been listening carefully. Now she said something that had clearly been forming for some time.

"In the countries I reported from," she said, "these mechanisms were not abstractions. Capital flight was not a line item in an academic paper. It was the reason the hospital didn't have bandages. It was the reason the school didn't have teachers. The reason families made impossible choices." She paused. "What I find difficult, sitting in this room, is that the mechanisms you're describing — the offshore system, the transfer pricing, the debt extraction — are more sophisticated versions of what has been happening to developing economies for decades. The difference is that they're now happening at scale inside wealthy countries too."

The room was quiet for a moment.

"That's the right observation," Neo said. "And it connects to where the research is going next."

He looked at his notes. The picture assembling itself was not a conspiracy. It was something more complex and in some ways more troubling — a set of individually rational decisions, made by institutions and actors operating within legal and regulatory frameworks, that in aggregate produced outcomes that the frameworks were not designed to measure or prevent.

And underneath all of it, a monetary system approaching a boundary that its architects had not fully resolved.

"Garvus," he said. "Current global debt levels. All sectors. Give me the numbers."

"Global debt across all sectors — sovereign, corporate, household, and financial — reached approximately three hundred and seven trillion dollars as of 2023, according to the Institute of International Finance. This represents approximately three hundred and thirty-six percent of global GDP. Sovereign debt alone has increased from approximately seventy-five percent of global GDP in 2007 to approximately one hundred and one percent in 2023. The increase reflects both the pandemic fiscal response and the pre-existing trend of debt accumulation in major economies."

"Three hundred and seven trillion dollars," Anna repeated quietly.

"To put that in human terms," Garvus said, "it represents approximately thirty-eight thousand dollars for every person on earth. In a world where approximately forty-seven percent of the global population lives on less than six dollars and eighty-five cents per day."

No one said anything for a moment.

Outside, Cambridge was going dark, the streetlights coming on in the early November evening. Arthur's house was warm and quiet. The books on the walls held a century of economic thought — the frameworks that had produced this moment, the analyses that had tried to understand it, the proposals that had attempted to address it.

Neo looked at his working document. At the top, the line he had written days ago in Westport: The architecture was built for a specific purpose. Follow the purpose.

He added a second line.

The question is whether the architecture can survive what it has built.

"Same time Thursday?" he said to both of them.

Arthur nodded. Anna was already making notes.

Chapter Five: After the Gold Window

Arthur Ellsworth had been twenty-six years old on August 15th, 1971.

He had been a second-year economics PhD student at MIT, already deep enough in monetary theory to understand what Nixon's announcement meant, young enough that the implications felt abstract rather than consequential. He had watched the address on a dormitory television with three other graduate students, and what he remembered most was not alarm but fascination — the sense of witnessing something that would require the textbooks to be rewritten.

He told Neo and Anna this on Thursday morning, over the same coffee in the same chairs, as if the memory had been waiting for the right context to surface.

"What Nixon said publicly," he said, "was that he was suspending dollar convertibility temporarily, to protect the dollar against international speculators. The word temporarily did a lot of work. The Bretton Woods system never resumed."

"Walk me through why it happened," Neo said. "The actual pressures."

Arthur looked at Anna. "You understand the Triffin dilemma from Tuesday?"

"The reserve currency issuer has to run deficits to supply the world with dollars," she said. "But running persistent deficits eventually undermines confidence in the currency."

"Correct. By the late 1960s the United States was running significant deficits — financing the Vietnam War and the Great Society programs simultaneously, what Johnson called guns and butter. European central banks, particularly France under de Gaulle, had accumulated large dollar reserves and were increasingly skeptical that the United States could honor its gold convertibility commitment at thirty-five dollars per ounce. De Gaulle was converting French dollar reserves to gold and shipping it to Paris. The British requested a gold conversion of three billion dollars in the summer of 1971. The U.S. gold window was under genuine pressure."

"Garvus," Neo said. "The U.S. gold position at that point."

"U.S. gold reserves had declined from approximately twenty billion dollars in the late 1950s to approximately ten billion dollars by 1971," Garvus said. "Against total foreign dollar claims that significantly exceeded that figure. The convertibility commitment was, by 1971, not credibly maintainable at the existing gold price without either a significant devaluation of the dollar against gold or a severe domestic deflation to reduce import demand and restore the trade balance."

"So Nixon had limited options," Anna said.

"Nixon had limited options," Arthur agreed. "He could devalue the dollar against gold — politically painful and internationally disruptive. He could implement capital controls — anathema to the financial sector. He could pursue domestic deflation — economically and politically costly. Or he could close the gold window and let the dollar float. He chose the fourth option, which had the advantage of being immediately effective and the disadvantage of dismantling the international monetary system that had governed global finance for twenty-seven years."

"The immediate consequences," Neo said.

"Dollar depreciation against major currencies — the dollar lost roughly a third of its value against the deutschmark and the yen over the following decade. Commodity price spikes, since commodities priced in dollars became cheaper in other currencies, stimulating demand. Inflation — the United States imported inflation through higher commodity prices and exported it through dollar depreciation. The 1973 OPEC oil embargo compounded these pressures significantly."

Anna leaned forward. "And for developing economies?"

"Garvus," Neo said.

"The consequences for developing economies were complex and varied," Garvus said. "Countries that had borrowed in dollars faced immediate pressure as the dollar's value fluctuated. Countries that exported commodities priced in dollars saw initial price increases — beneficial in some respects — but also increased volatility. The breakdown of the fixed exchange rate system introduced currency risk that the Bretton Woods framework had largely eliminated. Countries without sophisticated financial markets and central bank capacity were less equipped to manage that volatility."

"Worth being honest about the other side," Arthur said. He looked at Anna directly. "The Bretton Woods system, for all its stability, had embedded significant constraints. The dollar's fixed link to gold meant that global liquidity — the supply of dollars available for international trade — was constrained by U.S. gold reserves. The post-1971 system, for all its volatility, allowed global liquidity to expand with global trade needs. The growth in international trade and investment over the subsequent decades was partly enabled by the removal of the gold constraint on dollar issuance."

"The benefits of dollar hegemony are real," Neo said. It was not a concession. It was an analytical point.

"They are real," Arthur said. "This is where I think popular critiques of the dollar system sometimes oversimplify. The dollar's role as the global reserve currency has provided genuine public goods — a stable unit of account for international trade, deep and liquid financial markets that provide safe haven assets during crises, the ability of the global economy to finance counter-cyclical spending during downturns because dollar assets remain in demand. These are not trivial benefits."

"And the costs," Anna said. Evenly. Not confrontationally.

"And the costs are also real," Arthur said. "What economists call the exorbitant privilege — the United States' ability to borrow cheaply in its own currency — has an exorbitant burden on the other side. Countries that need dollars for international trade must earn them through exports or borrow them. Countries that borrow them are exposed to U.S. monetary policy decisions made for domestic reasons with global consequences."

"The taper tantrum," Neo said.

"Walk her through it," Arthur said.

"In May 2013," Neo said, "Fed Chairman Ben Bernanke suggested in congressional testimony

that the Fed might begin tapering its quantitative easing program — reducing its monthly asset purchases. The statement was tentative. The market reaction was not. Emerging market currencies sold off sharply as capital flowed back to dollar assets in anticipation of higher U.S. interest rates. The Indian rupee fell fourteen percent against the dollar in three months. The Brazilian real, the Indonesian rupiah, the Turkish lira — all experienced significant depreciation.”

“Garvus,” Anna said. “The human consequences of those currency moves.”

“Currency depreciation in import-dependent economies raises the domestic price of imported goods — food, fuel, medicine, industrial inputs — in local currency terms. For households in the bottom income quintiles, who spend higher proportions of income on food and fuel, the inflationary effect is regressive — it falls most heavily on those least able to absorb it. The documented effect of the 2013 taper tantrum on poverty rates in affected countries is difficult to isolate from other factors, but research by economists at the World Bank and IMF found significant pass-through from currency depreciation to food price inflation in the most affected economies.”

“The Fed was making a domestic monetary policy decision,” Anna said. “And people in Indonesia were paying more for food.”

“Correct,” Garvus said. “This is the documented spillover mechanism — U.S. monetary policy decisions, made for domestic objectives, transmit through global dollar markets to affect economic conditions in countries that had no role in those decisions.”

“This is precisely the problem Keynes identified at Bretton Woods,” Neo said. “A single nation’s currency serving as the global reserve means the issuing nation’s domestic monetary policy is effectively global monetary policy. The asymmetry is structural.”

Arthur nodded slowly. “It is structural. And it creates genuine grievances. The question is whether the alternatives that have been proposed address the asymmetry or simply replace one form of it with another.”

“The petrodollar,” Anna said. “You mentioned it Tuesday but we didn’t go through the mechanics.”

“Garvus,” Neo said.

“Following the 1973 oil shock and the collapse of Bretton Woods, the Nixon administration — primarily through Treasury Secretary William Simon and National Security Advisor Henry Kissinger — negotiated a series of agreements with Saudi Arabia that became the foundation of what is called the petrodollar system,” Garvus said. “The documented core of the arrangement was a U.S.-Saudi Joint Commission on Economic Cooperation established in 1974, and a related agreement under which Saudi Arabia would price its oil exports exclusively in dollars and recycle its oil revenues — petrodollars — into U.S. Treasury securities.”

“In exchange for what,” Anna said.

“In exchange for U.S. security guarantees to the Saudi regime, U.S. military equipment sales, and U.S. support for Saudi regional interests. The arrangement was not a formal treaty — it was a set of executive agreements and understandings whose precise terms remained classified for

decades. The existence of the arrangement and its broad outlines have been confirmed through declassified documents and reporting, including a Bloomberg investigation published in 2016 that obtained previously secret Treasury documents through Freedom of Information Act requests.”

“The effect on dollar demand,” Neo said.

“The effect was to create a structural global demand for dollars that substituted for the gold convertibility that had anchored the Bretton Woods system. Any country that needed to import oil — which by the mid-1970s meant virtually every industrial economy — needed to acquire dollars first, because oil was priced exclusively in dollars. This created persistent demand for dollar-denominated assets that supported the dollar’s value and allowed the United States to continue financing its current account deficit.”

“The arrangement essentially exported the cost of U.S. deficit financing to oil-importing countries,” Anna said.

“That is one way to frame it,” Garvus said. “Another framing is that it provided a stable pricing mechanism for the world’s most important commodity — a common unit of account that reduced transaction costs and currency risk for international oil trade. Both framings capture real aspects of the arrangement.”

Arthur stood and walked to the window. The Cambridge street outside was bare — late November, the last leaves gone, the sky the particular gray of New England before winter properly arrives.

“The petrodollar arrangement is a useful case study,” he said, “in the distinction between how a financial mechanism looks from inside the system and how it looks from outside it. From inside — from the perspective of U.S. Treasury officials and oil company executives and international finance professionals — it was an elegant solution to a genuine problem. How do you maintain dollar demand and finance the U.S. deficit after the gold anchor is gone? You create commodity-based demand. It worked.”

“From outside,” Anna said.

“From outside — from the perspective of an oil-importing developing country — you are now required to earn or borrow a foreign currency to purchase a commodity essential to your economy, at prices set by a cartel whose members have their own political relationships with the currency issuer. Your terms of trade, your development financing, your debt burden — all are affected by decisions made in Riyadh and Washington in which you have no participation.”

“Garvus,” Neo said. “The documented relationship between petrodollar recycling and developing country debt crises of the 1980s.”

“The petrodollar recycling mechanism worked as follows,” Garvus said. “Oil-exporting countries accumulated large dollar surpluses from the price increases of 1973 and 1979. These surpluses were deposited in major international banks — primarily in London and New York. The banks, holding more dollars than they could lend domestically, recycled the deposits as loans to developing countries, particularly in Latin America and Africa. The loans were made at floating interest rates — typically benchmarked to LIBOR, the London Interbank Offered Rate.”

“And then,” Anna said. She knew what came next.

“In 1979, Federal Reserve Chairman Paul Volcker, confronting severe U.S. inflation, raised the federal funds rate dramatically — ultimately to twenty percent by 1981. The effect on floating-rate dollar debt held by developing countries was immediate and severe. Countries that had borrowed at five or six percent interest were now servicing debt at fifteen or twenty percent. Mexico declared it could not service its external debt in August 1982, triggering the Latin American debt crisis. Brazil, Argentina, Venezuela, and others followed. The crisis spread to Africa and Asia.”

Anna looked at her hands for a moment.

“My country,” she said quietly. “What happened there — the poverty, the instability, the conditions that made political violence possible — some of it was specific to local dynamics. And some of it was this. Debt that became unpayable because of an interest rate decision made in Washington.”

The room was quiet.

Neo looked at Arthur. Arthur met his eyes.

“Both things are true simultaneously,” Arthur said finally. “Local dynamics matter. Institutional failures matter. Corruption matters. And the external financial environment created conditions that made the worst local outcomes more likely and the recovery from them harder.” He paused. “I spent my career inside institutions that made some of those decisions. I believed, at the time, that the alternatives were worse. I still think that analysis was substantially correct. I also think the costs fell on people who had no voice in the decision and received none of the benefits of dollar system stability. Holding both of those things simultaneously is uncomfortable. It is also accurate.”

Anna nodded once. It was not forgiveness. It was acknowledgment.

Neo turned to the laptop.

“Where we are now,” he said. “Current dollar system pressures. Garvus — the genuine challenges to dollar hegemony. Not the overstatement. The documented picture.”

“The dollar’s share of global foreign exchange reserves has declined from approximately seventy-one percent in 2000 to approximately fifty-nine percent in 2023, according to IMF COFER data,” Garvus said. “This represents a meaningful but gradual shift. The dollar remains dominant — the next largest reserve currencies, the euro and the Chinese yuan, hold approximately twenty and approximately three percent respectively. The dollar’s dominance in international trade invoicing, commodity pricing, and financial market transactions remains substantially intact.”

“The SWIFT exclusion of Russia,” Neo said. “What that revealed.”

“When Russia was excluded from the SWIFT interbank messaging system following the 2022 invasion of Ukraine, it demonstrated both the coercive power of dollar system infrastructure and its limitations. Russia was significantly disrupted — SWIFT exclusion complicated international payments and trade finance substantially. Russia was not crippled — it had been developing

alternative payment infrastructure, pivoted trade toward China and other non-Western partners, and continued to export commodities. The episode accelerated discussions in multiple countries about reducing dependence on dollar-denominated financial infrastructure.”

“The BRICS response,” Anna said.

“The BRICS nations — Brazil, Russia, India, China, and South Africa, expanded in 2024 to include additional members — have discussed alternatives to dollar dependence at multiple levels. These include bilateral trade settlement in local currencies, development of alternative payment messaging systems, and proposals for a common BRICS currency or commodity-backed settlement unit. The practical progress on these proposals has been more limited than the rhetoric suggests — significant structural barriers remain, including the yuan’s incomplete capital account convertibility and the genuine difficulty of creating a common currency among economies with divergent inflation rates and fiscal positions.”

“Arthur,” Neo said. “Honest assessment. Is dollar hegemony ending?”

Arthur considered this with the care the question deserved.

“Dollar hegemony is under more genuine pressure than at any point since the early 1970s,” he said. “I would not characterize what is happening as ending. I would characterize it as the beginning of a slow, multi-decade transition toward a less unipolar system — one in which the dollar remains important but in which other currencies, regional payment systems, and potentially new instruments play larger roles than they currently do.”

“Driven by what specifically.”

“Several forces simultaneously. The weaponization of dollar system access — SWIFT exclusions, sanctions, asset freezes — has given countries with no particular ideological opposition to the United States genuine incentive to develop alternatives, simply as insurance. The rise of China as a major trading partner for a majority of the world’s countries creates natural demand for yuan-denominated trade settlement. And the U.S. fiscal trajectory — debt levels that raise genuine long-term questions about dollar asset quality — creates slow-building pressure on reserve demand.”

“And the counterarguments,” Neo said. He was testing Arthur, but also genuinely asking.

“The counterarguments are substantial,” Arthur said. “Network effects in monetary systems are extraordinarily powerful — the dollar is useful precisely because everyone uses it, and changing reserve currency requires coordinating a simultaneous shift that no actor has individual incentive to initiate. The alternatives to the dollar have significant problems of their own — the yuan’s capital controls limit its reserve currency potential, the euro’s governance structure creates its own fragilities. And the U.S. financial system’s depth, liquidity, and rule of law provide genuine value that alternatives cannot currently match.”

Neo wrote in the working document.

The dollar system is neither as stable as its defenders claim nor as fragile as its critics claim. It is a system under genuine pressure, with genuine benefits, generating genuine costs unevenly distributed across the global population, approaching a set of questions it has deferred for fifty

years.

He looked at what he had written.

That felt honest.

“Same time next week,” he said.

Arthur nodded. Anna was already pulling out her notebook for the following thread.

Outside, the first snow of the season was beginning — small, tentative flakes that dissolved before they reached the ground. The beginning of something that would take months to fully arrive.

Chapter Six: What We Owe

The number that stopped Neo cold was not the one he expected.

He had spent three days with global debt data — the IMF's World Economic Outlook, the Bank for International Settlements quarterly reports, the Institute of International Finance's global debt monitor. The headline figure he already knew: three hundred and seven trillion dollars in total global debt across all sectors, representing approximately three hundred and thirty-six percent of global GDP.

That number was large enough to be abstract. What stopped him was a different number.

The United States government was now spending more on interest payments on its national debt than on national defense.

He checked it twice. Then he asked Garvus to verify it against the Congressional Budget Office's most recent projections.

"Confirmed," Garvus said. "CBO projects net interest payments on U.S. federal debt to reach approximately eight hundred and ninety billion dollars in fiscal year 2024, exceeding projected defense spending of approximately eight hundred and fifty billion dollars. This represents the first time in modern U.S. fiscal history that interest payments have exceeded defense outlays. CBO projects interest payments will continue to grow as a share of federal expenditure under current policy trajectories."

Neo wrote the number in the working document and looked at it for a long time.

Then he called Arthur.

"I know," Arthur said, before Neo could speak. "I've been looking at the same projections."

"Come to Westport," Neo said. "Bring Anna if she's available. I want to go through the full debt architecture properly."

They came on Saturday. Arthur drove himself, which Neo noted — at eighty-one, on the highway from Cambridge to Westport, in November — without comment. Anna arrived by train, carrying the same canvas bag, with what appeared to be additional folders.

Neo had set up the dining room table for the first time since buying the house. Three laptops. A whiteboard he had ordered specifically for this. Coffee.

"Start from first principles," he said to Garvus. "Not the headline figures. The architecture. How does sovereign debt actually work?"

"A sovereign government that needs to spend more than it collects in tax revenue finances the difference by issuing debt — government bonds," Garvus said. "The bonds are purchased by investors — domestic and foreign — who receive periodic interest payments and the return of principal at maturity. The government's ability to service this debt depends on its ability to raise revenue through taxation, and in the case of governments that issue debt in their own currency, their ability to create money."

"The distinction between debt in domestic currency and debt in foreign currency," Arthur said. It was instruction as much as conversation — he was teaching, the way he always had.

"A government that borrows in its own currency — as the United States, Japan, the United Kingdom, and most developed economies do — faces a different constraint than a government that borrows in foreign currency," Garvus said. "A domestic currency borrower can, in extremis, instruct its central bank to create money to service debt obligations. This does not mean there are no constraints — money creation at sufficient scale produces inflation, which is itself a real cost. But it means the government cannot be forced into outright default the way a foreign currency borrower can. Argentina, which borrowed heavily in dollars, faced genuine insolvency when its peso depreciated — it owed dollars it could not create. Japan, which has debt exceeding two hundred and sixty percent of GDP denominated in yen, has not faced comparable pressure because it owes yen it can issue."

Anna looked up. "Japan's debt is two hundred and sixty percent of GDP and it hasn't collapsed."

"This is the most important empirical check on the simple version of the debt crisis argument," Arthur said. He settled back in his chair with the particular satisfaction of someone about to complicate a narrative. "The conventional view — that debt above a certain threshold reliably produces crisis — was most influentially argued by Carmen Reinhart and Kenneth Rogoff in their 2010 paper 'Growth in a Time of Debt,' which suggested that sovereign debt above ninety percent of GDP was associated with significantly lower economic growth. That paper was subsequently found to contain a spreadsheet error. When corrected, the relationship between debt levels and growth outcomes was considerably weaker than originally reported."

"What the Japan case actually shows," Neo said.

"Japan has maintained debt levels that conventional analysis would predict to be unsustainable for three decades," Garvus said. "Several factors help explain this. Approximately ninety percent of Japanese government debt is held domestically — by Japanese financial institutions, pension funds, and the Bank of Japan itself. This limits the vulnerability to foreign investor sentiment shifts. Japanese household savings rates have historically been high, providing a domestic pool of capital willing to hold government bonds at low yields. The Bank of Japan has pursued yield curve control — explicitly targeting government bond yields — since 2016, maintaining low borrowing costs for the government through direct central bank purchases."

"Is that sustainable?" Anna asked.

"Genuinely contested," Garvus said. "Economists who follow Japan closely disagree significantly. The Bank of Japan's balance sheet has expanded to hold assets exceeding one hundred percent of GDP — a scale unprecedented in a major economy. Whether this represents a stable new equilibrium, a slow-building fragility, or something that will eventually require significant adjustment is a question the data does not definitively answer."

"The honest answer," Arthur said, "is that Japan has been testing the limits of what's possible with sovereign debt and monetary financing for thirty years, and the predicted catastrophe has not arrived. That is evidence that the simple version of the debt crisis model is wrong. It is not

evidence that there are no limits.”

Neo wrote on the whiteboard: Debt matters. The form matters more than the level. The trajectory matters more than the current position.

“Walk me through the U.S. debt trajectory specifically,” he said.

“U.S. federal debt held by the public has increased from approximately forty percent of GDP in 2008 to approximately ninety-eight percent of GDP in 2024,” Garvus said. “The CBO’s extended baseline projects it will reach approximately one hundred and sixty-six percent of GDP by 2054 under current policy. The primary driver going forward is mandatory spending — Social Security and Medicare — whose costs are projected to grow faster than revenue as the population ages, compounded by rising interest costs as existing debt rolls over at higher rates.”

“The interest rate sensitivity,” Neo said.

“Each one percentage point increase in the average interest rate on outstanding U.S. federal debt adds approximately two hundred and fifty billion dollars to annual interest payments, given the current debt level. As lower-rate debt issued during the 2010s and early 2020s matures and is refinanced at current rates — which are significantly higher — debt service costs will increase substantially even without additional borrowing.”

Arthur leaned forward. “This is the genuine concern, and I want to be precise about why it’s a concern rather than a catastrophe. The United States is not going to default on its dollar-denominated debt — it can always create dollars to service it. The concern is not default. The concern is the interaction between rising debt service costs and the government’s ability to fund other priorities.”

“Crowding out,” Anna said.

“Crowding out in the fiscal sense,” Arthur said. “When interest payments consume an increasing share of federal revenue — currently approaching fifteen percent, projected to exceed twenty percent within a decade — the space available for discretionary spending on defense, infrastructure, research, education, social programs narrows. Governments facing this constraint historically respond through some combination of tax increases, spending cuts, and inflation — which erodes the real value of outstanding debt. None of those options is painless, and the political economy of choosing among them is difficult.”

“The corporate debt side,” Neo said. “Garvus.”

“U.S. nonfinancial corporate debt has increased from approximately three trillion dollars in 2000 to approximately thirteen trillion dollars in 2023,” Garvus said. “A significant portion of this increase represents leveraged loans and high-yield bonds — debt issued by companies with below-investment-grade credit ratings, typically in connection with private equity transactions or aggressive expansion strategies.”

“The covenant-lite problem,” Arthur said.

“The proportion of leveraged loans issued with minimal protective covenants — ‘covenant-lite’ — has increased from approximately twenty percent of the market in 2010 to over eighty percent by 2022,” Garvus said. “Covenants are contractual protections for lenders — requirements that

borrowers maintain certain financial ratios, restrictions on additional debt issuance, triggers that allow lenders to accelerate repayment if conditions deteriorate. The erosion of covenant protection means lenders have less visibility into borrower deterioration and less ability to intervene before problems become severe.”

“Why did lenders accept that?” Anna asked.

“Yield hunger,” Arthur said simply. “A decade of near-zero interest rates meant that investors who needed income — pension funds, insurance companies, endowments — were competing for any asset that offered above-minimal returns. Borrowers exploited that competition to issue debt on increasingly favorable terms. Lenders accepted worse protections because the alternative was holding cash at zero yield.”

“The Federal Reserve created the conditions for that,” Neo said. It was not an accusation. It was an analytical point.

“The Federal Reserve’s quantitative easing program — purchasing Treasury securities and mortgage-backed securities to push down long-term interest rates — was designed to stimulate economic activity by reducing borrowing costs,” Arthur said. “It succeeded in that goal. It also had documented side effects, including the compression of yields across asset classes that drove investors toward riskier assets at terms that would not have been acceptable in a normal rate environment. Whether those side effects were an acceptable cost of the stabilization achieved is a genuine question among economists.”

Anna had been making notes. She looked up.

“In the countries I reported from,” she said, “the equivalent of quantitative easing — central bank money creation at scale — produced inflation and currency crises rather than stabilization. The difference in outcomes between developed and developing economy monetary expansion is striking.”

“The difference is institutional credibility,” Arthur said. “A central bank that has demonstrated, over decades, that it will control inflation and that its currency will hold value can expand its balance sheet significantly without triggering the loss of confidence that produces currency collapse. That credibility is not inherent — it was built through painful decisions, particularly the Volcker disinflation of the early 1980s, which accepted severe recession to break inflationary expectations. Developing country central banks, many of which have histories of monetizing government deficits at the expense of currency stability, do not have that credibility stock. The same action has different consequences in different institutional contexts.”

“Credibility as an asset,” Neo said.

“Credibility as an asset that can be spent and that is difficult to rebuild once spent,” Arthur said.

Neo looked at the whiteboard. The debt architecture, as it was assembling, had a shape that was neither the simple catastrophe narrative nor the comfortable reassurance that everything was sustainable. It was a system operating near limits that were real but not precisely defined, with resilience mechanisms that had been tested and had held but that carried their own accumulated costs.

“Household debt,” he said.

“U.S. household debt stands at approximately seventeen and a half trillion dollars,” Garvus said. “The composition has shifted significantly since 2008 — mortgage debt remains the largest component but has grown more slowly than student loan and auto loan debt. Student loan debt has increased from approximately six hundred billion dollars in 2008 to approximately one point seven trillion dollars in 2023, representing the largest non-mortgage debt category for younger cohorts.”

“The intergenerational dimension,” Anna said.

“The debt burden distribution across generations in the United States shows a marked pattern,” Garvus said. “Households under forty hold disproportionately more student loan and auto debt relative to assets. Households over sixty hold disproportionately more financial assets — equities, bonds — whose values were substantially inflated by the low interest rate period. The Federal Reserve’s distributional financial accounts show that the bottom fifty percent of households by wealth hold approximately two and a half percent of total household wealth, a share that has declined over the past three decades.”

Anna looked at Neo. Then at Arthur.

“The picture that assembles,” she said carefully, “is not that the system is about to collapse. It’s that the system has been distributing its costs and benefits in ways that are not random — that specific populations bear the costs and specific populations capture the benefits.”

“That is what the documented data shows,” Garvus said. “The distributional analysis is empirically robust across multiple data sources and methodologies.”

“And the counterargument,” Neo said. He was looking at Arthur.

Arthur was quiet for a moment — genuinely thinking, not performing deliberation.

“The counterargument,” he said finally, “is that the alternative to the system as it has operated — whatever that alternative might be — has not been demonstrated to distribute costs and benefits more equitably. Historical attempts to organize economies on fundamentally different principles have produced outcomes that, by most measures, were worse for the populations at the bottom of the distribution, not better. The system is imperfect. The imperfection is not randomly distributed. And the history of attempts to replace it is not encouraging. Those three things can all be true simultaneously.”

Neo wrote it on the whiteboard exactly as Arthur had said it.

Then he stood and looked out at Long Island Sound. The water was gray and cold, a November sky pressing down on it, no boats visible.

He had been at this for weeks now. The picture was not what he had expected when he pulled the first thread. He had expected to find either the system working roughly as advertised, with tolerable imperfections, or a clear indictment — evidence of deliberate design toward specific ends. What he had found was more complicated than either.

A system that had evolved, through specific historical decisions made by specific people with

specific interests, into a structure that produced outcomes those people largely preferred while generating costs absorbed by people who had no role in the design. Not conspiracy. Incentive. Not evil. Something harder to address than evil.

He added a line to the working document.

The system distributes its benefits and costs through mechanisms that are legal, documented, and largely invisible to the people who bear the costs. This is a feature. Whether it is a bug depends entirely on where you're standing.

"The Substack piece," Anna said. She had been reading the working document on her laptop. "The one that brought me to you — you published it for seventeen people. How many subscribers now?"

Neo checked. He had not looked in a week.

Four thousand, two hundred and twelve.

He showed the number to Arthur and Anna without comment.

"The signal," Anna said quietly. It was not a question.

"Someone's looking," Arthur said. He did not sound surprised.

Chapter Seven: The Architecture of Elsewhere

The document that opened Chapter Seven of Neo's working file was a 2020 report published by the Tax Justice Network titled *The State of Tax Justice*.

He had read it three times. Not because it was dense — it was, actually, written for accessibility — but because the numbers required sitting with rather than moving past.

The report estimated that governments globally lose approximately four hundred and twenty-seven billion dollars annually to international corporate tax abuse and private tax evasion. Of that figure, approximately two hundred and forty-five billion was attributed to multinational corporations shifting profits to low-tax jurisdictions. Approximately one hundred and eighty-two billion was attributed to wealthy individuals hiding assets offshore.

The numbers were contested — other researchers used different methodologies and arrived at different figures. The directional conclusion was not seriously contested by mainstream economists: offshore financial architecture facilitated the movement of significant capital out of the jurisdictions where it was economically generated, reducing the tax base available for public services in those jurisdictions.

What Neo found more interesting than the headline figure was the institutional geography.

"Garvus," he said. "The financial secrecy index. Top jurisdictions."

"The Tax Justice Network's Financial Secrecy Index ranks jurisdictions by their contribution to global financial secrecy, weighting both the degree of secrecy offered and the scale of financial activity conducted," Garvus said. "The 2022 index ranked the United States first, the Cayman Islands second, and Switzerland third. The United Kingdom ranked ninth, but when British Overseas Territories and Crown Dependencies — the Cayman Islands, the British Virgin Islands, Bermuda, Jersey, Guernsey, the Isle of Man — are considered as part of a network connected to London, the combined British financial secrecy ecosystem ranks as the largest globally by some analyses."

"The United States is first," Anna said. She had come down from New York for the day, taking the train to Westport rather than driving in December traffic.

"Delaware," Neo said.

"Delaware's corporate registration framework allows the formation of companies with minimal disclosure requirements — no public registry of beneficial owners, no requirement to file accounts, low fees and administrative burden," Garvus said. "Approximately sixty-eight percent of Fortune 500 companies are incorporated in Delaware, partly for its well-developed corporate law but also for its privacy and tax characteristics. The state generates approximately a third of its revenue from corporate franchise taxes and fees — a structural incentive to maintain its attractive registration environment."

"A domestic tax haven," Anna said.

"Domestic jurisdictions offering tax and secrecy advantages exist across the United States — Nevada, Wyoming, South Dakota have developed trust and corporate registration frameworks

that compete with Delaware on secrecy grounds. South Dakota's trust laws, in particular, have attracted significant international wealth management business following the 2016 reforms that eliminated the rule against perpetuities — meaning trusts can now exist in perpetuity without the assets ever being subject to estate tax."

Arthur had been listening from his chair in the corner — he had taken the train from Cambridge that morning, a journey he made with less frequency than he once had.

"Before we go further into the secrecy architecture," he said, "I want to make the genuine case for offshore finance. Not the apologetic version. The actual case."

Neo nodded.

"Capital mobility," Arthur said, "has genuine economic benefits that are not trivially dismissed. The ability to move capital across borders means that savings in capital-rich countries can finance investment in capital-scarce countries. This is one of the mechanisms through which developing economies have historically accessed the financing needed for infrastructure and industrial development. Without cross-border capital flows, the cost of capital in capital-scarce regions would be significantly higher."

"The secrecy component specifically," Neo said. "Make the case for that."

"Privacy in financial matters is not inherently illegitimate," Arthur said. "Individuals living under authoritarian governments with histories of asset confiscation have genuine reasons to hold assets in jurisdictions that do not share information with their home governments. Businesses operating across multiple jurisdictions face genuine complexity in managing multi-currency, multi-regulatory environments — offshore structures can provide legitimate administrative simplification. And tax competition between jurisdictions creates pressure on high-tax governments to use revenue efficiently — the argument being that some competitive discipline on government spending is not obviously harmful."

"Those are real arguments," Neo said.

"They are real arguments that are made seriously by serious economists," Arthur confirmed. "The question is whether the offshore system as it has actually developed serves primarily those legitimate functions or primarily serves as a mechanism for evading — rather than avoiding — tax obligations and for concealing assets from legitimate creditors, regulatory scrutiny, and democratic accountability. The evidence suggests the latter functions have come to dominate."

"Garvus," Neo said. "The Panama Papers. What they actually showed."

"The Panama Papers were a leak of approximately eleven point five million documents from the Panamanian law firm Mossack Fonseca, published by the International Consortium of Investigative Journalists in April 2016," Garvus said. "The documents revealed the use of offshore shell companies by a wide range of clients including politicians, business figures, athletes, and others from over two hundred countries. Subsequent document leaks — the Paradise Papers in 2017, drawing on Appleby, a Bermuda-based law firm; and the Pandora Papers in 2021, drawing on fourteen offshore service providers — expanded the picture."

"What the documents actually documented," Anna said. "Not the sensational version."

"The documents documented the mechanics of offshore company formation and the range of clients using those structures," Garvus said. "They showed that offshore structures were routinely used by politically exposed persons — politicians and their associates — in ways that raised questions about the origin of assets. They showed structures used to hold assets in ways that concealed beneficial ownership from tax authorities. They showed that the offshore industry had developed sophisticated techniques for maintaining secrecy in the face of international information-sharing agreements — nominee directors, complex layering of entities across multiple jurisdictions, the use of bearer shares in jurisdictions still permitting them."

"What they didn't show," Arthur said.

"The documents did not, in themselves, demonstrate that all or even most offshore activity was illegal," Garvus said. "Much of what was revealed was legal under the laws of the jurisdictions involved. The distinction between legal tax avoidance and illegal tax evasion — both of which appeared in the documents — was not always clear and varied by jurisdiction. The documents were evidence of a large and sophisticated industry serving clients with a wide range of motivations, not a simple catalogue of criminal activity."

"The distinction that matters most," Neo said, "is not legal versus illegal. It's what the system is actually used for at scale."

"The academic research on this is instructive," Garvus said. "A study by economists Annette Alstadsæter, Niels Johannesen, and Gabriel Zucman, using leaked data from HSBC's Swiss private banking operation, found that offshore wealth was heavily concentrated among the very wealthy — the top zero point one percent of the wealth distribution — and that wealth hidden offshore was significantly underreported in official wealth statistics. Their research suggests that conventional wealth inequality measures, which rely on reported data, significantly understate actual wealth concentration."

Anna leaned forward. "What that means practically."

"If offshore wealth is concentrated among the top of the distribution and is not captured in official statistics," Garvus said, "then the measured wealth inequality — already at historically high levels in most developed economies — understates actual inequality. Policy responses calibrated to measured inequality may be inadequate to address actual inequality."

"The City of London specifically," Neo said. "Its role in the architecture."

"The City of London — the square mile — is the world's largest wholesale financial center," Garvus said. "It serves as the hub for foreign exchange trading, the eurodollar market, international bond issuance, and a significant share of global derivatives trading. Its role in offshore finance is partly direct — London-based banks and law firms provide services to offshore structures — and partly architectural. The British Overseas Territories and Crown Dependencies that form the network of offshore jurisdictions connected to London operate under legal frameworks developed with London's expertise and maintain close financial relationships with City institutions."

"The history of how that network developed," Arthur said. "It's important for understanding how it came to be rather than implying deliberate design."

"The network developed organically across the twentieth century," Garvus said. "The eurodollar market's London origins in the 1950s were discussed earlier. The Channel Islands and Isle of Man developed as low-tax financial centers partly in response to high postwar British income taxes — they offered British residents and institutions a nearby jurisdiction with lower rates within a familiar legal framework. The Caribbean territories developed as offshore centers serving American clients in the postwar period, partly facilitated by their proximity and their English common law legal systems. The development was incremental, driven by competitive pressure and client demand, rather than centrally planned."

"That matters," Arthur said. "The offshore system looks, from the outside, like a coherent and deliberately constructed architecture for evading regulation and taxation. From the inside, it looks more like a set of incremental responses to specific client needs and competitive pressures, each step individually rational, the aggregate effect not specifically intended by any single actor."

"The same pattern as the monetary system," Neo said.

"The same pattern as most complex institutional arrangements," Arthur said. "Emergent rather than designed. Which creates a genuine governance problem — you cannot reform an emergent system by identifying and stopping the bad actor who built it, because there is no single bad actor. The system is the aggregate of millions of individually rational decisions."

Anna had been quiet for several minutes. She looked up from her notes.

"In the countries I reported from," she said, "the offshore system was not an abstraction. When a mining company operating in a country transfers its profits to a subsidiary in the British Virgin Islands and pays royalties to a holding company in Luxembourg and books its final profits in a zero-tax jurisdiction, the country where the mine is — where the land is, where the workers are, where the environmental consequences fall — receives a fraction of the economic value generated within its borders. The government collects less revenue. The schools are underfunded. The healthcare system is underfunded. Those are not abstract distributional consequences."

"Documented," Garvus confirmed. "The International Monetary Fund's research on base erosion and profit shifting — the technical term for corporate structures that shift taxable income out of high-tax jurisdictions — estimates that developing economies lose a higher proportion of their corporate tax base to these practices than developed economies, partly because they have less regulatory and administrative capacity to audit and challenge complex transfer pricing arrangements."

Arthur nodded. "This is where I have less of a counterargument. The competitive dynamics of the offshore system fall most heavily on jurisdictions that are least equipped to respond to them. A country with a sophisticated tax authority and strong bilateral information exchange agreements can partially offset profit shifting through aggressive audit and enforcement. A country with a small, underfunded tax authority and limited leverage to negotiate information exchange agreements cannot. The playing field is not level and the unlevel-ness systematically disadvantages the poorest countries."

Neo wrote on the whiteboard: The offshore system is legal, documented, emergent rather

than designed, and distributes its costs toward the jurisdictions least able to bear them.

"The BIS," he said. "The Bank for International Settlements. Garvus — what it actually is, not the popular version."

"The Bank for International Settlements was established in 1930, initially to manage German reparations payments under the Young Plan following the First World War," Garvus said. "It is owned by sixty-three central banks and operates as a bank for central banks — providing banking services, facilitating central bank cooperation, and conducting research on monetary and financial stability. Its headquarters is in Basel, Switzerland. It hosts the secretariats of several international financial regulatory standard-setting bodies — the Basel Committee on Banking Supervision, the Financial Stability Board, the Committee on Payments and Market Infrastructures."

"Its governance," Neo said.

"The BIS is owned and governed by its member central banks," Garvus said. "Its meetings — including the bimonthly meetings of governors of major central banks — are not public. Its research is published. Its standard-setting processes involve consultations with industry and regulators. The accountability concern that critics raise is genuine — the BIS and the committees it hosts make decisions about global financial regulation that affect billions of people, through processes that are more opaque than domestic regulatory processes and without direct democratic accountability."

"And the counterargument," Arthur said. "Which is also genuine. International regulatory coordination requires some degree of confidentiality — central bankers discussing currency interventions or financial stability concerns publicly in real time would create the market instability they were trying to address. The BIS provides a venue for coordination that would otherwise be impossible. Its Basel capital standards, for all their imperfections, have improved the resilience of the global banking system compared to the pre-Basel era."

"Both things can be true," Neo said.

"Both things are true," Arthur confirmed. "The BIS serves genuine functions. The opacity of its processes is a genuine democratic accountability concern. The standard-setting it facilitates has produced real improvements in financial system resilience. The institutions that benefit most from the standard-setting — large international banks — have disproportionate influence over the process. These observations are not mutually exclusive."

Neo looked at the whiteboard. The offshore architecture, as it had assembled, was neither the shadowy conspiracy of popular imagination nor the benign technical infrastructure that its defenders described. It was a set of institutional arrangements that had developed incrementally, served real functions, and in aggregate produced a distribution of costs and benefits that was not random.

"The Pandora Papers specifically identified the United States as a significant beneficiary of offshore capital," Anna said. "Capital fleeing developing economies was flowing partly to U.S. states — South Dakota, Nevada — as well as to the traditional Caribbean and European centers."

"Documented," Garvus confirmed. "The United States has been slow to implement the

beneficial ownership transparency standards it has advocated other jurisdictions to adopt. The Corporate Transparency Act of 2021 requires disclosure of beneficial ownership to FinCEN, the financial intelligence unit, but does not create a public registry accessible to foreign tax authorities or civil society. The United States' own financial secrecy creates competitive pressure on other jurisdictions to maintain their secrecy standards."

"Because if you close the Cayman Islands but leave Delaware open," Neo said, "the capital just moves."

"That is the documented dynamic," Garvus said. "Financial secrecy is a competitive market. Unilateral transparency reforms shift activity to remaining secrecy jurisdictions rather than eliminating it. Genuine reform requires multilateral coordination — which the international community has made incremental progress toward through the OECD's Common Reporting Standard and Base Erosion and Profit Shifting project, and has failed to complete for reasons including resistance from jurisdictions that benefit from the current system."

"Including the United States," Anna said.

"Including the United States," Arthur acknowledged. He said it without defensiveness. At eighty-one, he was past the institutional loyalty that would have complicated the admission thirty years earlier.

Neo's phone showed a notification. The Substack.

Six thousand, four hundred subscribers.

He had published nothing new in two weeks. The early pieces were circulating on their own.

"Someone is reading," Anna said. She had seen his expression.

"Someone is reading," Neo agreed.

He did not say what he was thinking — that he was not sure yet whether that was a good thing. The picture assembling itself in the working document was not a comfortable one. Not because it was sensational. Because it was documented.

He added a line to the whiteboard.

The architecture exists. It is legal. It is documented. It was not designed by any single actor. It produces outcomes that are not random. The question is not who built it. The question is what, if anything, comes next.

"Thursday?" he said.

Arthur was already reaching for his coat.

Chapter Eight: How Capital Moves

Neo had a rule about primary sources.

He had developed it during his early years at the fund, when he had learned — painfully, through positions that cost real money — that most financial commentary was a layer of interpretation sitting on top of a layer of interpretation, each step away from the underlying data introducing small distortions that compounded. The rule was simple: before forming a view, read the actual research. Not the summary. Not the commentary. The paper.

He had spent the past week reading the actual research on private equity.

Both sides of it.

What he found was genuinely more complicated than either the industry's promotional literature or its most pointed critics suggested. He had been expecting to find a clear indictment. What he found was a methodological dispute among serious economists that had been running for two decades without resolution.

He told Arthur this when they met Thursday morning.

Arthur almost smiled. "Welcome to empirical economics," he said. "Where everything that matters is contested and the data rarely resolves it cleanly."

"Walk me through the strongest case for private equity," Neo said. "Not the apologetic version. The actual strongest case."

Arthur set down his coffee.

"Private equity fills a genuine gap in the capital markets ecosystem," he said. "Public equity markets — the stock market — are well-suited to financing large, established companies with predictable cash flows and transparent operations. They are poorly suited to financing companies that need intensive operational involvement, strategic redirection, or temporary earnings suppression in service of long-term value creation. A public company that cuts its dividend to fund a multi-year operational transformation faces immediate shareholder pressure. A PE-backed company can absorb short-term earnings pressure without that pressure."

"Garvus," Neo said. "The academic evidence on operational performance."

"The research on private equity's operational effects is genuinely mixed," Garvus said. "Studies by Steven Kaplan at the University of Chicago Booth School of Business and Josh Lerner at Harvard Business School — two of the most rigorous academic researchers in this field — find evidence of productivity improvements, increased operating efficiency, and employment growth in PE-backed companies relative to comparable non-PE-backed companies in certain contexts and time periods. A 2019 study in the *American Economic Review* by economists Shai Bernstein, Josh Lerner, and Filippo Mezzanotti found that PE-backed firms weathered the 2008 financial crisis better than comparable non-PE-backed firms — investing more, cutting employment less, and emerging more competitive."

Anna looked up from her notes. "That's the opposite of the narrative I usually encounter."

"The research is genuinely mixed," Garvus repeated. "Eileen Appelbaum and Rosemary Batt at the Economic Policy Institute document significant negative outcomes — particularly in labor-intensive service sectors. The employment effects are contested across multiple studies using different methodologies. A 2021 study by economists at the University of Chicago and Census Bureau, using comprehensive employment data, found that PE buyouts were associated with a ten percent reduction in employment at target companies within two years of acquisition, with the effects concentrated in sectors where PE ownership was associated with subsequent bankruptcy."

"The sector matters enormously," Arthur said. "PE ownership of a manufacturing company competing in a global market, where operational efficiency genuinely determines survival, may produce real productivity improvements that benefit workers and consumers alike. PE ownership of a regional hospital system, where the business model's cash extraction requirements directly conflict with the institution's public health function, produces different outcomes."

"Healthcare specifically," Neo said.

"The documented outcomes in healthcare PE are the strongest part of the critical case," Garvus said. "A 2023 study published in JAMA found that PE acquisition of hospitals was associated with increased rates of adverse patient events — falls, infections — alongside increased revenue per patient-day and increased administrative staffing relative to clinical staffing. The proposed mechanism is consistent with the general model: debt service requirements and management fee extraction reduce resources available for clinical staffing while revenue optimization increases charges."

"Emergency medicine and anesthesiology," Neo said.

"PE consolidation of physician staffing companies — particularly in emergency medicine, anesthesiology, and radiology — has been documented to produce staffing model changes associated with balance billing controversies, increased patient costs, and in some documented cases changes in care protocols that prioritized throughput over clinical judgment," Garvus said. "TeamHealth and Envision Healthcare — two large PE-backed physician staffing companies — were among the most frequently cited in the surprise billing debates that led to the No Surprises Act of 2022."

"The retail case," Arthur said. It was an acknowledgment as much as a prompt.

"The documented pattern in retail PE — Toys R Us, Sears, Payless ShoeSource, Pier 1 — follows a recognizable structure," Garvus said. "Acquisition via leveraged buyout at a debt-to-equity ratio that leaves the acquired company servicing substantial interest payments. Management fees charged by the PE firm regardless of performance. Strategic decisions that prioritize debt service over investment in competitive positioning — store maintenance, inventory management, e-commerce development. Bankruptcy when the debt burden becomes unserviceable, frequently after the PE firm has already recovered its invested capital through dividend recapitalizations."

"Define dividend recapitalization," Anna said.

"A transaction in which a PE-backed company takes on additional debt and uses the proceeds

to pay a special dividend to its PE owner,” Garvus said. “The effect is to allow the PE firm to extract cash from the investment before exit — recovering some or all of its invested capital while the company absorbs additional leverage. If the company subsequently underperforms and enters bankruptcy, the PE firm has already realized returns while creditors, employees, and in pension-related cases, retirees absorb losses.”

“That is documented to have happened,” Neo said.

“In the Toys R Us case, the PE owners — KKR, Bain Capital, and Vox Capital — had invested approximately one point three billion dollars in equity at acquisition. By the time Toys R Us filed for bankruptcy in 2017, the company was carrying approximately five billion dollars in debt, a significant portion of which dated to the original leveraged buyout and subsequent refinancings. Approximately thirty-three thousand employees lost their jobs. Initial severance was limited. After significant public and congressional pressure, the PE owners contributed seventy-five million dollars — approximately six percent of their original equity investment — to a severance fund.”

Arthur was quiet for a moment.

“The honest assessment of PE,” he said finally, “is that it is a capital allocation mechanism that works well in some contexts and badly in others, and that the contexts where it works badly tend to be contexts involving public-facing services — healthcare, retail, housing — where the business model’s requirements for cash extraction conflict with the institution’s social function. The carried interest tax treatment — which allows PE profits to be taxed at capital gains rates rather than ordinary income rates — is a genuine policy distortion that subsidizes the activity without distinction between its better and worse applications.”

“The carried interest debate,” Neo said.

“PE and hedge fund managers pay capital gains tax rates — currently twenty percent for long-term gains — on their carried interest: the twenty percent share of profits they receive as performance compensation,” Garvus said. “The argument for capital gains treatment is that carried interest represents a return on risk — the managers forfeit their carry if the fund underperforms. The argument against is that carried interest is compensation for services — management of the fund — and should be taxed as ordinary income like other service compensation. The debate has been active in Congress for over two decades. The carried interest provision has survived multiple reform attempts.”

“Because of lobbying,” Anna said.

“The financial services industry’s political expenditures are significant and well-documented,” Garvus said. “The Center for Responsive Politics tracks financial sector lobbying and campaign contributions. The specific causal relationship between those expenditures and the survival of the carried interest provision is difficult to establish with precision — correlation between industry political activity and policy outcomes favorable to the industry does not establish causation. The circumstantial case is not weak.”

“Venture capital,” Neo said. “Make the full case. Positive and negative.”

“The genuine positive case for venture capital is strong in certain respects,” Garvus said.

“Early-stage VC financing accepts risks that no other capital market participant will accept — pre-revenue companies with unproven technologies and uncertain markets. The portfolio model — investing in many companies expecting most to fail and a few to produce extraordinary returns — allows financing of high-risk research that commercial banks cannot provide and public markets will not support. The documented economic impact of VC-backed companies is significant: a 2015 study by economists at Stanford found that VC-backed companies accounted for approximately twenty percent of U.S. GDP and forty-four percent of research and development spending among U.S. public companies.”

“Google, Amazon, Intel, Apple’s early investment — all VC-backed,” Arthur said. “The biotech companies that developed the mRNA vaccine platform. The medical devices that have saved lives. This is real. It is not nothing.”

“The genuine concerns,” Neo said.

“Late-stage venture and growth equity have developed dynamics that raise legitimate questions,” Garvus said. “The unicorn phenomenon — private companies valued at over one billion dollars — has produced situations where companies remain private longer, accessing capital from institutional investors at valuations that public market investors later find difficult to sustain. The WeWork IPO attempt in 2019 — a company valued at forty-seven billion dollars in private markets that withdrew its IPO when public market investors applied conventional valuation analysis — is the most prominent example. Retail investors who accessed the stock at the eventual IPO price, or through funds holding late-stage private rounds, participated after the primary value creation and sometimes absorbed substantial losses.”

“The information asymmetry,” Anna said.

“Venture investors negotiating directly with founding teams have access to detailed company information — customer contracts, unit economics, burn rate, competitive analysis — that public market investors do not have at IPO,” Garvus said. “Venture investors also negotiate liquidation preferences — rights to recover their investment before common shareholders in the event of sale or liquidation — and other protective terms that do not exist in public market investments. The transition from private to public markets involves a significant shift in information availability and investor protection that the IPO process does not fully bridge.”

“The geographic and demographic concentration,” Neo said.

“VC investment is heavily concentrated in a small number of geographies — the San Francisco Bay Area, New York, Boston, and to a lesser extent Los Angeles and Austin — and in a small number of educational and social networks,” Garvus said. “Research by economists at the National Bureau of Economic Research finds that VC investment follows patterns of social proximity — investors fund founders they know, from networks they are part of. The documented gender and racial demographic composition of both VC investors and VC-funded founders reflects and reinforces existing network patterns. Women-led startups receive approximately two percent of VC funding. Founders of color receive a similarly low share.”

Anna wrote something in her notebook.

“The system allocates capital,” she said, “through mechanisms that reflect and reinforce the

demographic composition of the people operating those mechanisms.”

“That is what the documented data shows,” Garvus confirmed. “Whether this represents a market failure — valuable investment opportunities systematically overlooked because of network effects — or efficient allocation through networks that carry genuine information about founder quality is contested. The evidence on returns from more demographically diverse VC portfolios suggests the market failure interpretation is more accurate — studies find that diverse founding teams and diverse investor networks produce comparable or superior returns, suggesting the current concentration reflects network effects rather than pure meritocracy.”

Arthur looked at his hands.

“The honest synthesis,” he said, “is that both PE and VC are capital allocation mechanisms that have produced genuine value in specific contexts and genuine harm in others, and that the policy frameworks governing them — particularly tax treatment and investor protection — have not kept pace with the scale and scope to which the industries have grown.”

“The scale,” Neo said. “Garvus — the current AUM figures.”

“Private equity assets under management globally reached approximately eight point two trillion dollars in 2023, according to McKinsey’s Global Private Markets Review,” Garvus said. “Venture capital assets under management reached approximately two point five trillion dollars. Combined, private markets represent a significant and growing share of total investment capital — and a growing share of corporate ownership, particularly in sectors like healthcare, housing, and business services.”

Neo looked at the whiteboard.

The picture of how capital moved was assembling itself. Not through a single mechanism. Through a set of mechanisms — offshore finance, private equity, venture capital, public markets — each serving genuine functions, each producing documented costs, the costs not randomly distributed.

What he had not yet addressed — what the research was circling toward but had not yet confronted directly — was the question of what the interaction between these mechanisms and the debt architecture they depended on actually implied over time.

That was a harder question. One where the evidence was thinner and the honest answer required more uncertainty than he had been comfortable with so far.

“Arthur,” he said. “The part I can’t resolve.”

The older man looked at him steadily.

“The debt levels we discussed in Chapter Six,” Neo continued, “enable the PE model — low rates make leverage cheap. QE compresses yields and drives institutional capital into PE funds. The offshore system facilitates profit extraction. The VC ecosystem concentrates returns in a small network. Each mechanism individually has documented legitimate functions. Together they produce a capital allocation system that concentrates wealth at the top of the distribution at a rate that the underlying economic data suggests is historically unusual.”

He paused.

"Is that a system approaching a natural limit, or is it a stable equilibrium that can persist indefinitely?"

Arthur was quiet for a long time.

"I don't know," he said finally. "And I want to be precise about the nature of that not knowing. It's not that I lack information. It's that the question is genuinely open. Economic history contains examples of high inequality persisting for long periods without triggering systemic crisis. It also contains examples of systems that appeared stable until they weren't. The mechanisms that might produce a correction — political, economic, or both — are visible. The timing and character of any such correction is not predictable."

"Garvus," Neo said. "Historical precedents. Comparable periods of capital concentration and what followed."

"Economic historians identify several periods of comparable capital concentration to the present," Garvus said. "The Gilded Age in the United States — roughly 1870 to 1900 — produced wealth concentration comparable to or exceeding the present day, followed by the Progressive Era regulatory response, the antitrust movement, and eventually the New Deal framework. The 1920s produced a period of financial innovation, leverage, and wealth concentration that preceded the 1929 crash and the Great Depression. Post-World War Two policies — progressive taxation, union density, regulated financial sector — produced a period of relatively compressed inequality that economists call the Great Compression. The deregulatory period from the 1980s onward reversed much of that compression."

"The historical pattern," Anna said, "is that these periods of concentration end. The question is how."

"Sometimes through policy reform," Garvus said. "Sometimes through economic crisis that resets valuations and debt levels. Sometimes through political disruption. The historical record does not identify a single resolution mechanism, and the specific path of any particular episode is not predictable from the aggregate pattern."

Neo wrote on the whiteboard.

The system concentrates. History suggests concentrations resolve. The mechanism and timing of resolution is not predictable. This is the honest statement of what the evidence supports.

He looked at it for a long time.

Then he added one more line.

What comes next — in money, in technology, in political response — may determine which historical precedent the current episode resembles.

"Next week," he said. "I want to start on BRICS and the alternative financial architecture proposals. And then — " he looked at Anna — "crypto. I think we need to spend serious time on crypto."

Anna nodded. She had been waiting for that conversation.

Outside the window, Long Island Sound was white-capped in a December wind, the water moving fast and cold toward the open sea.

PART TWO: THE SYSTEM

Chapter Nine: The Alternative

Arthur arrived at Westport on a Tuesday in December, two days earlier than planned.

He had driven himself again, which Neo noted without comment. What he did note was something different in Arthur's manner — a quality he had been observing over the past few weeks, growing more pronounced. The careful hedging of institutional speech, the reflexive qualifier before any statement that might be taken as criticism of a body he had served, the diplomat's instinct to present every view in its most defensible form — those habits were loosening. Not dramatically. Gradually. The way a man dispenses with formal dress when he has decided that the occasion no longer requires it.

Neo suspected Arthur had received some news he had not shared. He did not ask.

"I've been thinking about the BRICS question since we scheduled it," Arthur said, settling into the chair with his coffee. "I want to say something at the outset that I would not have said publicly ten years ago."

Neo waited.

"The decision to weaponize the dollar payment system — to use SWIFT exclusion and asset freezes as instruments of foreign policy — was a significant strategic error," Arthur said. "Not morally. Strategically. The dollar system's value as a global public good rested on its perceived neutrality — on the understanding that access to dollar-denominated finance was governed by commercial and regulatory criteria rather than geopolitical ones. The moment we demonstrated that dollar system access could be revoked on political grounds, we gave every country in the world a concrete reason to build alternatives. Not just adversaries. Potential adversaries. Neutral parties. Anyone who looked at what happened to Russia's reserves and concluded they did not want to be in that position."

"The reserves freeze," Neo said.

"Three hundred billion dollars in Russian central bank reserves, held in Western financial institutions, frozen following the 2022 invasion of Ukraine," Arthur said. "Legal under the relevant sanctions frameworks. Effective as an immediate economic measure. And the most powerful advertisement for BRICS alternative financial architecture that anyone could have designed. I supported the sanctions. I think the invasion justified a strong response. I also think the people who made that decision did not fully account for the long-term cost to the dollar system's perceived neutrality."

Neo wrote this in the working document verbatim.

"Garvus," he said. "The BRICS architecture. What actually exists versus what has been proposed."

"BRICS — originally Brazil, Russia, India, China, and South Africa — expanded in January 2024 to include Egypt, Ethiopia, Iran, Saudi Arabia, and the United Arab Emirates," Garvus said. "The expanded bloc represents approximately forty-five percent of the world's population and approximately thirty-five percent of global GDP measured at purchasing power parity. At market

exchange rates, the share is lower — approximately twenty-six percent.”

“The distinction between PPP and market exchange rate GDP matters here,” Arthur said.

“It matters significantly,” Garvus confirmed. “Purchasing power parity GDP measures economic output adjusted for differences in price levels — it reflects the actual volume of goods and services produced. Market exchange rate GDP reflects the dollar value of that output at current currency prices. For countries with undervalued currencies relative to their price levels — which includes China and India — PPP GDP is substantially higher than market exchange rate GDP. The BRICS bloc’s economic significance is larger by the PPP measure. Its weight in dollar-denominated financial markets — trade invoicing, debt, reserves — reflects the market exchange rate measure.”

Anna, who had come down from New York the previous evening, looked up from her laptop. “What institutions does BRICS actually have?”

“The New Development Bank, established in 2015, is the primary institutional creation,” Garvus said. “Capitalized at one hundred billion dollars, with equal initial contributions from the five founding members, the NDB was designed to finance infrastructure and sustainable development in BRICS and developing countries, positioning itself as an alternative to the World Bank and IMF.”

“How does it actually differ from the World Bank?” Neo asked.

“Several documented differences exist,” Garvus said. “The NDB does not attach policy conditionality to its loans — borrowers are not required to implement structural adjustment programs as a condition of financing. Governance is more equal — founding members have equal voting shares regardless of economic size, contrasting with the World Bank and IMF where voting power reflects economic contribution. The NDB has stated a commitment to lending in local currencies, reducing borrower exposure to dollar exchange rate risk.”

“And the limitations,” Arthur said. It was not a question.

“The NDB’s lending volume remains modest relative to its stated ambitions — cumulative approvals through 2023 totaled approximately thirty-three billion dollars, compared to the World Bank’s annual lending of approximately fifty to sixty billion dollars,” Garvus said. “The NDB itself borrows on international capital markets, primarily in dollars, which exposes it to the same dollar system dependencies it was designed to reduce. Governance tensions among members with divergent interests — particularly between China and the other members on questions of Chinese dominance of the institution — have been documented.”

“The yuan internationalization project,” Neo said.

“China has pursued yuan internationalization through multiple channels since approximately 2009,” Garvus said. “These include bilateral currency swap agreements with over forty central banks, the establishment of offshore yuan clearing centers in London, Hong Kong, Singapore, and elsewhere, the inclusion of the yuan in the IMF’s Special Drawing Rights basket in 2016, and the Cross-Border Interbank Payment System — CIPS — as an alternative to SWIFT for yuan-denominated transactions.”

"And the results," Arthur said.

"The yuan's share of global foreign exchange reserves has increased from essentially zero in 2010 to approximately two and a half percent in 2023," Garvus said. "The yuan's share of global trade finance has increased to approximately five percent. These figures represent genuine progress. They also reflect the distance remaining — the dollar's share of reserves remains approximately fifty-nine percent, its share of global trade invoicing approximately eighty-eight percent."

"Why has progress been slower than China expected?" Anna asked.

Arthur answered before Garvus could.

"Because the yuan is not freely convertible," he said. "China maintains capital controls — restrictions on the movement of money in and out of the country. A reserve currency requires that foreign holders be able to invest and divest freely, repatriate funds without restriction, and trust that the rules governing their holdings will not change arbitrarily. China's capital controls, its opaque financial system, its history of regulatory interventions in markets — including the 2015 equity market intervention and the 2021 Ant Group IPO suspension — make foreign institutions and central banks reluctant to hold significant yuan assets."

"The fundamental tension," Neo said.

"The fundamental tension in China's reserve currency ambitions is that the characteristics that make a currency attractive as a reserve asset — free capital movement, transparent and rule-of-law financial markets, independent central bank — are characteristics that the Chinese government has been unwilling to fully adopt, partly because adopting them would constrain its ability to manage domestic financial conditions," Arthur said. "This is not a solvable technical problem. It is a political choice. China wants the benefits of reserve currency status without the constraints that reserve currency status requires. That combination is not available."

"A more direct statement of that than I would have made ten years ago," Neo said.

"Ten years ago I was still attending meetings where Chinese officials might have been in the room," Arthur said simply.

Anna looked at him with something that was not quite a smile.

"The BRICS currency proposal," Neo said. "What's actually being discussed."

"BRICS summits have discussed the concept of a common currency or alternative settlement unit since at least 2022," Garvus said. "The proposals vary — a common currency analogous to the euro, a commodity-backed settlement unit, a basket currency similar to the SDR. No specific proposal has advanced to formal negotiation."

"The obstacles," Arthur said. "Which are fundamental, not technical."

"A common currency requires either a common monetary authority — a BRICS central bank with the power to set interest rates for all members — or a fixed exchange rate arrangement among members," Garvus said. "A common monetary authority requires surrendering domestic monetary sovereignty to a supranational institution, which no BRICS member has shown

willingness to do. A fixed exchange rate arrangement creates the same fundamental vulnerabilities that brought down Bretton Woods — if member economies diverge in inflation rates or economic performance, the fixed rates become unsustainable and the arrangement collapses.”

“The euro’s experience is instructive,” Arthur said. “The eurozone crisis of 2010 through 2015 demonstrated what happens when countries with divergent economic structures — Germany and Greece, for example — share a currency without full fiscal integration. Greece could not devalue to restore competitiveness. Germany could not revalue to reduce its surplus. The adjustment burden fell entirely on Greek wages and employment. BRICS members have far greater economic divergence than the eurozone.”

“China and Brazil,” Neo said. “Different inflation rates, different fiscal positions, different economic structures, different political systems.”

“And different interests,” Arthur said. “This is the part that gets underweighted in the BRICS alternative architecture discussions. The BRICS coalition is held together by a shared critique of the dollar system, not by shared interests. China is the world’s largest exporter and has a strong interest in stable currency arrangements that support export competitiveness. Brazil is a commodity exporter whose interests frequently conflict with China’s on trade matters. India and China share a contested border and have fought a border war. Russia and the Gulf states have interests that align on some dimensions and conflict on others. Building durable financial institutions on that coalition is significantly harder than the summit communiqués suggest.”

Anna had been quiet, making notes. She looked up.

“From the perspective of the countries I reported from — smaller developing economies in Latin America, Africa, Asia — what does BRICS actually offer? Not what it promises. What it actually offers.”

Arthur looked at her with the directness he had been deploying more freely in recent weeks.

“Leverage,” he said. “Not an alternative system. Leverage. The existence of a credible BRICS alternative — or even a credible threat of one — gives developing countries more negotiating power with the dollar system institutions than they had before. The IMF’s conditionality has become somewhat less stringent in recent years. The World Bank has been more attentive to borrower concerns. That is partly a response to competition from the NDB and Chinese bilateral lending. The value of BRICS to small developing countries is not primarily that they will use BRICS institutions. It is that BRICS institutions exist.”

“And Chinese bilateral lending specifically,” Neo said. “The Belt and Road.”

“The Belt and Road Initiative — launched in 2013 — is China’s program of infrastructure financing across Asia, Africa, the Middle East, and elsewhere,” Garvus said. “By 2023, BRI had involved lending commitments of approximately one trillion dollars across approximately one hundred and fifty countries, making China the world’s largest bilateral creditor to developing countries.”

“The debt trap question,” Neo said. “What the research actually shows.”

“The debt trap diplomacy thesis — the argument that China deliberately extends unsustainable loans to developing countries with the intention of seizing strategic assets when borrowers default — has been contested in the academic literature,” Garvus said. “Deborah Brautigam at the Johns Hopkins China Africa Research Initiative, one of the most rigorous researchers in this area, has found limited evidence that China has systematically used debt to acquire strategic assets. The Hambantota Port in Sri Lanka — frequently cited as the canonical debt trap example — was acquired through a debt-to-equity swap that Brautigam and colleagues argue was negotiated by Sri Lanka rather than imposed by China.”

“The counterargument,” Anna said.

“The counterargument is that the debt trap thesis, while overstated in its most extreme form, identifies real concerns about loan terms, transparency, and collateral requirements in Chinese bilateral lending,” Garvus said. “AidData at William and Mary has documented that Chinese loan contracts frequently include confidentiality clauses, cross-default provisions, and collateral arrangements — including revenue accounts — that give China significant leverage over borrowers. Whether this constitutes a trap or normal commercial lending practice is a genuine analytical dispute.”

“The honest answer,” Arthur said, “is that Chinese bilateral lending is neither the predatory debt trap of the most alarmed accounts nor the unconditional development finance of Chinese official statements. It is commercial lending by a state that has both development interests and strategic interests, documented to include terms that create real dependencies, in contexts where borrowers frequently had limited alternatives.”

“Which is not entirely unlike what the IMF and World Bank have been doing,” Anna said.

“Which is not entirely unlike what the dollar system institutions have been doing,” Arthur agreed. “With different terms, different conditionality, different strategic interests. The borrower trades one set of dependencies for another. Whether the new dependencies are better or worse depends on the specific terms and the specific country’s circumstances.”

Neo looked at the whiteboard. The BRICS picture, assembled honestly, was neither the transformative alternative of its advocates’ most optimistic projections nor the irrelevant sideshow of its dismissers. It was a set of institutions and proposals that reflected genuine grievances, faced genuine structural obstacles, had produced some genuine leverage for developing countries, and remained — for the foreseeable future — insufficient to displace the dollar system they were designed to challenge.

“The most honest statement of where this leaves us,” he said.

Garvus responded.

“The dollar system faces genuine pressures — the weaponization of dollar system access, the rise of alternative bilateral trade relationships, China’s infrastructure finance ambitions, the structural concerns about U.S. fiscal trajectory. These pressures are real and documented. The alternatives that have been proposed face genuine structural obstacles that their advocates understate. The most likely outcome over the next two to three decades is not dollar system collapse or BRICS supplanting it, but a gradual, uneven fragmentation — a world with multiple

currency blocs, significant but declining dollar dominance, and increased friction and cost in international transactions.”

“Which benefits no one particularly,” Arthur said.

“Which benefits no one particularly,” Garvus confirmed. “Fragmentation of the global payment and reserve system increases transaction costs, reduces liquidity, complicates risk management for businesses operating across currency blocs, and reduces the capacity for coordinated response to global financial crises. The dollar system, for all its documented inequities, provided genuine efficiencies that a fragmented alternative does not replicate.”

“The thing nobody wants to say plainly,” Arthur said. He was looking at the fire, not at either of them. “The dollar system is inequitable and it works. The proposed alternatives are more equitable in aspiration and they don’t work yet, and may not work at the scale required. We are choosing between a system whose costs we know and alternatives whose costs we have not yet fully discovered.”

He paused.

“That is what I would have said privately for thirty years. I’m saying it out loud now because I no longer have professional reasons not to.”

The room was quiet.

Anna looked at him with the specific attention she brought to moments when a source said something they had never said before.

“Do you think it can be reformed?” she asked. “The dollar system. Not replaced. Reformed.”

Arthur considered this for a long moment.

“The Bretton Woods system was reformed — imperfectly, incompletely, but meaningfully — through the IMF’s quota reforms, the creation of the SDR, the expansion of swap line networks,” he said. “The Basel framework improved bank capital standards substantially, even if imperfectly. International tax cooperation has advanced further in the past decade than in the previous fifty years. The system is not static. It responds to pressure.”

“But,” Neo said.

“But the pressure required to produce meaningful reform is typically crisis,” Arthur said. “The reforms that have mattered most have followed crises — the 1930s produced Bretton Woods, 2008 produced the Basel III capital reforms and the G20 financial stability framework. Incremental reform in the absence of crisis has been slower and less complete. If the next significant reform requires the next significant crisis, the question is what that crisis looks like and who bears its costs.”

He stopped.

“That,” he said, “is where I run out of confident answers.”

Neo added a line to the working document.

The dollar system is inequitable and functional. The alternatives are more equitable in

aspiration and less functional in practice. Reform requires crisis. The costs of crisis, historically, fall on those least able to bear them.

The December light was fading early. Arthur looked at his watch — an old one, mechanical, that he had worn for as long as Neo had known him.

“I should drive back before dark,” he said.

Neo walked him to the door. On the porch, Arthur paused.

“The young woman,” he said quietly. “Anna. She asks better questions than most of the economists I worked with.”

“I know,” Neo said.

“Good.” Arthur started down the steps, then stopped. “The crypto chapter. I want to be there for that one.”

“I assumed you would.”

Arthur nodded once and continued to his car.

Neo stood on the porch until the taillights disappeared. Then he went back inside where Anna was still at the table, working through something in her notebook, the winter dark pressing against the windows.

He looked at the working document.

Twelve thousand, eight hundred subscribers.

He had not published in three weeks.

Chapter Ten: What the Data Shows

Neo had been an analyst for thirty years before he became a subject of analysis.

He noticed it three weeks into the BRICS research, when he asked Garvus to help him synthesize arguments across six competing academic papers on yuan internationalization and realized, without particular drama, that what he was doing — the core intellectual task of his career, the synthesis of complex information from multiple sources into a coherent investment thesis — was something Garvus could do in approximately forty seconds.

Not perfectly. He could still find the gaps, the overstatements, the places where the model connected dots that didn't connect. But well enough that the question was not whether AI could do his old job. The question was how much of his old job it could do, and how quickly that fraction would grow.

He sat with this for a day before bringing it to Arthur and Anna.

Arthur's response surprised him.

"You're describing the most significant productivity story in economic history," Arthur said. "And you're framing it as a threat."

"Aren't both true simultaneously?" Anna said.

"Both have been true of every major technological transition," Arthur said. He was leaning forward with an energy that Neo associated with Arthur at his most intellectually alive — the quality that had made his seminars worth attending thirty-five years ago. "The agricultural revolution displaced subsistence farmers. The industrial revolution displaced artisan craftsmen. The computing revolution displaced bookkeepers, typists, drafters. In every case the displacement was real. In every case the aggregate economic outcome was an expansion of total employment and living standards, accompanied by significant transitional pain for specific workers and communities."

"The standard argument," Neo said. It was not dismissal. He wanted Arthur to make it fully.

"It's not merely the standard argument — it's the empirically supported argument," Arthur said. "The lump of labor fallacy — the assumption that there is a fixed amount of work to be done, so if machines do some of it there is less for humans — has been wrong every time it has been confidently asserted. New technologies create new categories of work that didn't exist before. The iPhone eliminated the market for portable cameras, GPS devices, and MP3 players. It created the app economy, the gig economy, the influencer economy, the podcast economy. Net employment effect: significantly positive."

"Garvus," Neo said. "The documented historical employment effects of major automation waves."

"The academic literature on automation and employment is substantial and its findings are more nuanced than either optimists or pessimists typically represent," Garvus said. "David Autor at MIT, one of the most rigorous researchers in labor economics, has documented a phenomenon called job polarization — technological change that displaces routine middle-skill jobs while

expanding both high-skill cognitive jobs and low-skill manual service jobs. This has been documented across multiple developed economies from approximately the 1980s onward.”

“The hollowing out of the middle,” Anna said.

“The documented pattern is a reduction in routine cognitive and routine manual employment — bookkeeping, clerical work, assembly line work, data processing — and growth at both ends of the skill distribution. Whether this represents technological unemployment or technological transformation depends significantly on the time horizon examined and the populations most affected.”

“The populations most affected,” Neo said.

“Workers in routine-intensive occupations who are mid-career face the most significant adjustment challenges,” Garvus said. “Retraining for higher-skill cognitive work requires time, capital, and often geographic relocation that not all workers can undertake. Workers who entered the labor market after routine automation had already occurred — who never held those jobs — tend to adapt through different initial occupational choices. The displacement costs are real and concentrated in specific demographic and geographic groups. The aggregate employment expansion is also real. Both things are true simultaneously.”

“Why this wave might be different,” Neo said. “Make the serious version of that argument.”

“Several characteristics of the current AI transition have prompted economists who take historical analogies seriously to consider whether previous patterns will hold,” Garvus said. “Previous automation waves were narrow — they displaced specific categories of routine task but left cognitive tasks largely untouched. General purpose AI systems capable of performing non-routine cognitive tasks — analysis, writing, coding, legal research, financial modeling — represent a qualitatively different automation frontier. The previous refuge for displaced workers was cognitive work. If that refuge is also automated, the adjustment mechanism that operated in previous waves may not function.”

“Daron Acemoglu’s recent work,” Arthur said. He said it with the tone of someone acknowledging an argument that had genuinely moved him.

“Acemoglu at MIT — who previously collaborated extensively with Autor — published research in 2022 questioning whether AI automation would produce the same net employment benefits as previous automation waves,” Garvus said. “His framework distinguishes between automation that replaces human tasks and innovation that creates new human tasks. He argues that recent AI deployment has been more heavily weighted toward task replacement than task creation — that the economic incentives driving AI adoption favor labor substitution over labor augmentation, and that without deliberate policy intervention, the distributional consequences of the current wave may be worse than historical patterns suggest.”

“And the counterargument to Acemoglu,” Arthur said. It was instruction — he wanted the full picture on the table.

“Erik Brynjolfsson at Stanford, David Autor in more recent work, and researchers at the OECD and McKinsey Global Institute have argued that the productivity gains from AI will expand economic activity sufficiently to create new categories of employment,” Garvus said. “Historical

patterns of technological transition, while involving real transitional costs, have consistently produced net employment expansion over multi-decade horizons. The current wave's speed is genuinely unprecedented, but the structural forces that created new work categories in previous transitions — increased productivity leading to lower prices, expanded demand, new goods and services — remain operative."

"The honest disagreement," Neo said.

"Serious economists hold genuinely different views on the net employment effects of the current AI transition," Garvus confirmed. "The disagreement is not primarily about whether displacement is occurring — it is — but about whether the creation of new work categories will keep pace with displacement, and over what time horizon. The International Monetary Fund's 2024 assessment estimated that AI could affect approximately forty percent of jobs globally — roughly sixty percent in advanced economies — with effects that could be positive through augmentation or negative through displacement depending on policy choices and the pace of transition."

Anna had been looking at her laptop. She looked up.

"The developing economy dimension is different," she said. "And I don't think it gets enough attention in the standard discussion."

"Walk us through it," Neo said.

"The service export economy — call centers, business process outsourcing, back-office work — has been a significant development path for countries like India, the Philippines, Costa Rica, and others," Anna said. "It provided a pathway from agricultural employment to formal sector employment to higher-skill work over time. That pathway depends on labor cost arbitrage — the wage difference between a customer service worker in Manila and one in Columbus, Ohio. AI-powered customer service systems eliminate that arbitrage. Not gradually. Rapidly."

"Garvus," Neo said. "The documented scale of BPO employment in developing economies."

"The global business process outsourcing sector employed approximately eight to ten million workers in India alone in 2023, generating approximately one hundred and eighty billion dollars in annual revenue," Garvus said. "The Philippines employed approximately one point three million in BPO. Combined, these two countries account for the majority of global BPO employment. The sector has been a primary driver of middle-class formation in both countries."

"And the AI exposure of that sector," Anna said.

"Customer service and voice BPO — a significant share of total employment — is among the sectors most directly exposed to large language model automation," Garvus said. "The wage arbitrage that drives offshore BPO disappears when the alternative is AI rather than domestic labor. The timeline for significant AI displacement in this sector is contested, but several major BPO employers have announced workforce reductions and hiring freezes citing AI automation."

Arthur was quiet for a moment.

"This is where I have a genuine concern that the optimistic historical parallel understates," he said. "The agricultural and industrial transitions occurred over multiple generations — workers

who were displaced from farming did not retrain as factory workers, but their children entered factories and their grandchildren entered offices. The social disruption was real and severe. But the pace allowed for some degree of adaptation across generations. The current transition's pace — in BPO, in legal research, in financial analysis, in software development — is measured in years, not generations. The adaptation mechanisms that operated over decades may not function over years."

"The measurement problem," Neo said. "How labor market data might be masking what's happening."

"The U.S. Bureau of Labor Statistics collects and reports employment data through several surveys — the Current Population Survey for household employment and the Current Employment Statistics for payroll employment," Garvus said. "These surveys measure employment status — employed, unemployed, not in labor force — but are less sensitive to changes in the quality and compensation of employment. An economist displaced from a hedge fund who takes a part-time consulting role appears employed in the data. A BPO worker whose hours are reduced below full-time appears employed."

"The birth-death model," Neo said.

"The BLS Business Employment Dynamics model estimates net job creation from business births and deaths in sectors where direct measurement is difficult," Garvus said. "The model uses historical patterns to estimate the contribution of new business formation and business closure to net employment. During periods when historical patterns break — as they may be doing in AI-exposed sectors where new business formation may not compensate for displacement in the ways previous transitions showed — the model's estimates may diverge from actual employment dynamics."

"Worth being precise about what 'may' means there," Arthur said. "The birth-death model has been criticized for specific periods — particularly the 2007-2009 recession, where it was slow to capture the severity of job losses. Whether it is currently masking AI-driven displacement, or whether that displacement is not yet occurring at a scale that would show up in aggregate data, is genuinely uncertain."

"The honest statement," Neo said.

"The honest statement is that there is documented displacement in specific AI-exposed occupations and sectors," Garvus said. "The aggregate labor market data does not yet show the kind of elevated unemployment that previous automation fears predicted. Whether this reflects that the fears are overstated, that displacement is being masked by measurement limitations, or that displacement is occurring but is being offset by labor market tightness in other sectors, is a question the available data does not definitively resolve."

Anna looked at both of them.

"In the countries I reported from," she said, "the question of whether AI displacement is being measured accurately is almost beside the point. The BPO sector provided formal employment, benefits, career development. If that sector contracts significantly over the next decade, the workers affected don't have a social safety net that absorbs the transition. The aggregate data in

Washington looks at global productivity. The family in Manila looks at the paycheck.”

The room was quiet.

Arthur looked at her steadily.

“That is the correct frame for evaluating the distributional consequences,” he said. “Aggregate productivity gains and aggregate employment expansion are real and matter. They matter differently depending on whether you’re in the part of the distribution that captures the gains or the part that absorbs the transitional costs.”

“The AI ownership question,” Neo said. “Garvus.”

“The productivity gains from AI deployment will accrue primarily to the owners of AI systems and the businesses deploying them,” Garvus said. “The ownership of frontier AI systems is highly concentrated — a small number of companies, primarily in the United States, control the most capable models. The investors and employees of those companies capture the value through equity ownership. Workers whose productivity is enhanced by AI tools may capture some value through wages in competitive labor markets. Workers whose jobs are displaced by AI capture none of the productivity gain and bear the full transitional cost.”

“Unless policy redistributes it,” Arthur said. “Which has happened in previous transitions — progressive taxation of productivity gains, expansion of social insurance, investment in retraining. Whether the political economy of the current period supports that kind of redistribution is a separate question from whether it is economically possible.”

“Is it?” Anna asked. “Politically possible?”

Arthur looked at the fire.

“I have spent my career watching the distance between what is economically sensible and what is politically achievable,” he said. “The honest answer is that I don’t know. The political economy of AI-driven inequality will depend on factors — the pace of displacement, the geographic concentration of effects, the degree to which affected workers organize, the degree to which AI productivity gains are visible and attributable — that are not yet determined.”

He paused.

“What I know from economic history is that the transitions that produced the best outcomes for the most people were the ones where the productivity gains were broadly shared through deliberate policy choice. The transitions that produced the worst outcomes were the ones where the gains were captured by capital and the costs were borne by labor, and the political response was delayed until the pain was severe enough to produce crisis.”

“Which historical precedent does the current transition resemble?” Neo asked.

Arthur was quiet for a long time.

“I don’t know yet,” he said finally. “And I want to be honest about the weight of that not knowing. I’ve spent my career developing views about economic transitions. I have genuine uncertainty about this one that I didn’t have about previous ones.”

Neo wrote in the working document.

AI displacement is documented in specific sectors. Aggregate effects are genuinely uncertain. Historical parallels are relevant but may not hold. The distributional consequences — who captures gains, who bears costs — will be determined by policy choices not yet made. The honest position is uncertainty with significant stakes attached to it.

He looked at the Substack notification on his phone.

Twenty-two thousand, four hundred subscribers.

He had published two pieces in the past week. Both had been shared extensively in financial and economics communities. Three journalists had emailed asking for interviews. He had not responded to any of them.

Anna was watching him.

“You’re going to have to decide what this is,” she said.

“I know,” Neo said.

He did not say what he was actually thinking — that the research had arrived somewhere he had not anticipated when he pulled the first thread six weeks ago. Not at a conclusion. At a set of genuinely open questions about the interaction between financial system architecture, AI-driven productivity and displacement, and political responses that were not yet determined.

The picture was not comfortable. It was also not the simple indictment he might have assembled if he had been less careful about the evidence.

It was what the evidence actually showed.

He was not sure what to do with that yet.

Chapter Eleven: The Whitepaper

The document was nine pages long.

Neo had read it before — years ago, in the way that financial professionals in 2009 had read it: with the specific skepticism of someone who had seen enough promising innovations fail to be automatically impressed by elegant design. He had filed it under interesting but probably irrelevant and moved on to things that were affecting his positions.

He read it again now, more carefully.

Bitcoin: A Peer-to-Peer Electronic Cash System. Published on October 31st, 2008 — six weeks after Lehman Brothers filed for bankruptcy, three weeks after the Emergency Economic Stabilization Act authorized seven hundred billion dollars in bailout funds — by an author or authors using the name Satoshi Nakamoto. No institutional affiliation. No academic credentials. No real identity that anyone had been able to establish in the fifteen years since.

The genesis block — the first block in the Bitcoin blockchain, mined by Nakamoto in January 2009 — contained an embedded message: The Times 03/Jan/2009 Chancellor on brink of second bailout for banks.

A newspaper headline. Encoded permanently into the foundation of the system.

“Garvus,” Neo said. “What is that headline doing there?”

“The embedded headline is widely interpreted as a timestamp — proof that the genesis block was created on or after January 3rd, 2009, the date of that Times front page,” Garvus said. “It is also widely interpreted as a statement of intent. The creator of Bitcoin chose, from all possible timestamps, a headline about a government bailout of financial institutions. The choice appears deliberate.”

“It is deliberate,” Arthur said.

He had arrived that morning with a quality Neo had not seen before — a kind of settled resolve, as if something had been decided. Not dramatically. Quietly. The way a man looks when he has stopped managing his energy and started using it.

Anna was already at the table, her laptop open, a cup of coffee going cold beside her. She looked at Arthur when he came in. She noticed the quality too.

“Bitcoin was a response to 2008,” Arthur said, sitting down. “The Fed understood that from the beginning. We just couldn’t say it publicly.”

The room was quiet.

“What do you mean couldn’t,” Neo said.

“I mean that acknowledging the legitimacy of Bitcoin as a response to the bailouts would have required acknowledging more about 2008 than the institution was prepared to acknowledge publicly,” Arthur said. “The bailouts were necessary. I believed that then and I believe it now. They were also a demonstration — visible, undeniable, globally observed — that the financial

system operated under rules that applied differently to different participants. That systemically important institutions would be rescued from the consequences of their risk-taking while other actors would not be. Bitcoin was, among other things, a technical embodiment of the argument that a financial system without that asymmetry was worth building. We understood that. We didn't say it."

Neo wrote this in the working document verbatim.

Arthur looked at what he was writing and did not object.

"Walk me through the mechanics," Anna said. "I've read about it but I want to understand it properly. From the beginning."

"Garvus," Neo said.

"The core problem Bitcoin was designed to solve is called the double-spending problem," Garvus said. "In a conventional digital payment system, a digital file representing money can be copied — unlike physical cash, which cannot be in two places simultaneously. To prevent double spending, conventional digital payment systems require a trusted central authority — a bank or payment processor — to maintain the authoritative record of who owns what and to verify that funds being spent have not already been spent elsewhere. This central authority is the single point of trust and, as Nakamoto's whitepaper argued, the single point of failure and control."

"The blockchain solution," Neo said.

"Bitcoin solves the double spending problem without a central authority through a distributed ledger — the blockchain," Garvus said. "Every Bitcoin transaction is broadcast to a network of participants. Transactions are grouped into blocks. Each block contains a cryptographic hash — a mathematical fingerprint — of the previous block, linking them in a chain. Altering any historical transaction would require recalculating every subsequent block and doing so faster than the network is adding new blocks — a computational task that becomes exponentially more difficult as the chain grows."

"The analogy," Anna said.

"Consider a public ledger book in a town square, with thousands of identical copies held by thousands of different people," Garvus said. "Every transaction is recorded in every copy simultaneously. To fraudulently alter a transaction, you would need to alter the majority of all copies simultaneously — which, given that the copies are distributed globally and controlled by independent parties, is computationally infeasible at scale. The distributed nature of the ledger is the security mechanism."

"Proof of work," Neo said.

"The mechanism by which new blocks are added to the chain requires participants — miners — to expend computational energy solving a mathematical puzzle," Garvus said. "This is called proof of work. The puzzle is difficult to solve but easy to verify. The miner who solves it first adds the next block and receives newly created Bitcoin as a reward. The energy expenditure makes attacking the network expensive — to rewrite history, an attacker would need to outpace the honest network's computational power, which at current scale would require an investment in

hardware and energy that exceeds the potential gain.”

“And the documented criticism of that energy expenditure,” Arthur said.

“Bitcoin’s proof of work mechanism consumes approximately one hundred and twenty to one hundred and fifty terawatt hours of electricity annually as of 2023,” Garvus said. “This is comparable to the annual electricity consumption of countries including Argentina or Norway. The carbon footprint depends significantly on the energy mix used by miners — a disputed figure, with industry estimates suggesting higher renewable percentages than some independent analyses. The environmental cost is real and documented. Whether it is justified depends on assessments of Bitcoin’s utility that reasonable people weigh differently.”

Anna looked up. “Argentina.”

“You noticed,” Garvus said.

“I noticed.” She set down her coffee. “Tell me what Bitcoin actually looks like in a country with a currency that’s failing. Not theoretically. What actually happens.”

“Argentina provides the most thoroughly documented case study of cryptocurrency adoption under currency crisis conditions,” Garvus said. “Argentina has experienced annual inflation exceeding one hundred percent since 2022, a parallel exchange rate significantly weaker than the official rate, capital controls limiting dollar purchases, and a history of bank freezes — most traumatically the 2001 corralito, when the government froze bank accounts and forcibly converted dollar-denominated savings to pesos at a government-mandated rate.”

Anna was quiet.

“My family lost everything in that,” she said simply. Not for sympathy. As context.

“The documented response of Argentine citizens to these conditions has included significant cryptocurrency adoption,” Garvus said. “Argentina consistently ranks among the highest globally in peer-to-peer cryptocurrency trading volume. Stablecoin adoption — particularly USDT, a dollar-pegged stablecoin — has been documented as a mechanism for Argentines to hold dollar-equivalent value outside the banking system, beyond the reach of capital controls and potential future freezes. Cryptocurrency exchanges in Argentina have reported that the majority of their trading volume is in stablecoins rather than Bitcoin — suggesting the primary use case is currency substitution rather than speculation.”

“Because a volatile asset is still more stable than a currency losing value that fast,” Neo said.

“For a holder of Argentine pesos facing triple-digit inflation, Bitcoin’s volatility — which looks extreme from the perspective of a dollar or euro holder — may represent less purchasing power erosion than holding pesos over the same period,” Garvus said. “The frame for evaluating cryptocurrency volatility depends significantly on what the alternative is.”

Arthur leaned forward. “This is the part of the Bitcoin story that I think gets insufficient attention in developed economy discussions,” he said. “From New York or London, Bitcoin looks like a speculative asset for people with excess capital looking for uncorrelated returns. From Buenos Aires or Caracas or Harare, it looks like something different — a way to opt out of a monetary system that has demonstrably failed the people participating in it.”

“Remittances,” Anna said.

“The remittance use case is documented and significant,” Garvus said. “The World Bank estimates that global remittances — money sent by migrant workers to family members in their home countries — totaled approximately eight hundred and thirty billion dollars in 2022. The average cost of sending remittances through conventional channels — wire transfers, money transfer operators — is approximately six percent globally, higher for specific corridors. Cryptocurrency-based remittances have demonstrated the ability to reduce costs significantly in documented pilots and deployments, though adoption at scale has been slower than proponents projected.”

“El Salvador,” Neo said.

“El Salvador’s adoption of Bitcoin as legal tender in 2021 — the first country to do so — was partly motivated by remittance costs,” Garvus said. “Approximately twenty-four percent of El Salvador’s GDP comes from remittances, primarily from migrants in the United States. The Chivo wallet, the government’s Bitcoin payment application, was designed to allow fee-free remittance transfers. The documented outcomes have been mixed. Adoption by the general population has been lower than the government projected. A 2022 survey by the National Bureau of Economic Research found that sixty percent of Chivo users withdrew their initial government-provided Bitcoin immediately and did not use the system again. Remittance volumes through the app have been lower than official projections.”

“Why,” Anna said.

“Several documented factors,” Garvus said. “Volatility of Bitcoin’s dollar price creates risk for recipients who need to cover immediate expenses in dollars or local currency. Technical barriers to using cryptocurrency wallets for populations with limited smartphone access or digital literacy. Merchant adoption has been uneven despite legal tender status. And the government’s implementation has faced documented transparency concerns.”

“The honest assessment of El Salvador,” Arthur said. “Not the promotional version or the dismissive version.”

“The honest assessment is that El Salvador conducted a real-world experiment in Bitcoin legal tender adoption and the results are more complicated than either advocates or critics predicted,” Garvus said. “The experiment has not produced the transformative financial inclusion its proponents claimed. It has also not produced the economic catastrophe its critics predicted. The IMF’s concerns about financial stability risks have not materialized at the scale feared. The adoption data suggests that Bitcoin’s use case as a daily payment medium faces genuine barriers that ideological enthusiasm does not resolve.”

“The ownership concentration question,” Neo said.

“Bitcoin ownership is highly concentrated,” Garvus said. “Analysis of the blockchain — which is public — suggests that a small number of wallets hold a disproportionate share of all Bitcoin. Estimates vary depending on methodology and assumptions about wallet clustering, but most analyses suggest the top one percent of Bitcoin addresses hold more than ninety percent of all Bitcoin. This concentration replicates, in some analyses exceeds, the wealth concentration in the

conventional financial system that Bitcoin was designed to challenge.”

“Garvus,” Anna said carefully. “Is there an irony worth noting there?”

“The asset designed as an alternative to a system criticized for wealth concentration has developed a wealth concentration comparable to or exceeding that system,” Garvus said. “This observation is available in the public data for anyone who examines it.”

The flatness of the delivery was precisely right. Neo almost smiled.

“The institutional adoption question,” he said. “What it means for the original thesis.”

“Bitcoin’s original design positioned it as a peer-to-peer payment system — a way for individuals to transact without financial intermediaries,” Garvus said. “Its current primary use case, particularly in developed economies, is as a store of value or speculative asset. Institutional adoption — BlackRock’s Bitcoin ETF approval in January 2024, the entry of major financial institutions into Bitcoin custody and trading — has brought Bitcoin into the mainstream financial system it was designed to operate outside of.”

“The BlackRock detail,” Arthur said. He said it with a specific quality — the directness of a man who finds something genuinely funny and is past pretending otherwise.

“BlackRock — the world’s largest asset manager, with approximately ten trillion dollars in assets under management — filed for and received approval for a spot Bitcoin ETF in January 2024,” Garvus said. “BlackRock’s iShares Bitcoin Trust accumulated assets rapidly following launch. BlackRock’s CEO Larry Fink had described Bitcoin as an ‘index of money laundering’ in 2017. His 2024 characterization was considerably more favorable.”

“Institutions that spent years describing Bitcoin as a fraud or a tool for criminals are now offering Bitcoin products to their clients,” Neo said.

“The regulatory approval of spot Bitcoin ETFs in the United States allowed conventional investment platforms — brokerage accounts, retirement accounts — to hold Bitcoin exposure,” Garvus said. “The mechanism by which retail investors now most commonly access Bitcoin is through BlackRock, Fidelity, and similar institutions — the same institutions whose intermediary role Bitcoin was designed to eliminate.”

“I want to be careful about how that’s framed,” Arthur said. “The fact that Bitcoin has been integrated into the conventional financial system is not necessarily evidence of corruption or co-optation. It is also evidence of maturation — of an asset class developing the infrastructure that allows broader participation. Whether that broader participation serves or undermines the original purpose depends on what you think the original purpose actually was.”

“What do you think it was?” Anna asked him.

Arthur considered.

“The original purpose, as I read the whitepaper and the early communications, was twofold,” he said. “A technical demonstration that trustless digital transactions were possible — that you could transfer value without a central authority and without the single points of failure and control that central authorities represent. And a political statement — that the financial system

as it existed in 2008 had demonstrated something about itself that warranted building an alternative.”

“The technical demonstration succeeded,” Neo said.

“The technical demonstration succeeded,” Arthur confirmed. “The blockchain works. The double-spending problem is solved. The network has operated continuously since 2009 with no central authority. That is a genuine technical achievement.”

“And the political statement,” Anna said.

Arthur was quiet for a moment.

“The political statement was heard,” he said. “Whether the system that developed in response to it serves the population it was intended to serve — the people for whom the existing financial system doesn’t work — is the more complicated question. In Argentina, in El Salvador, in Zimbabwe, in the remittance corridors between migrant workers and their families, there is documented evidence that it does serve that population, imperfectly, in specific ways. In the developed economy speculative market, it primarily serves people who already have capital looking for returns.”

“Two different things wearing the same name,” Anna said.

“Two different things wearing the same name,” Arthur agreed.

Neo looked at the working document. The Bitcoin picture, assembled honestly, was neither the revolutionary alternative of its advocates’ most idealistic projections nor the fraud or tulip mania of its most dismissive critics. It was a genuine technical innovation that had developed in ways its creator did not fully anticipate, serving populations its creator intended and populations its creator did not intend, within and outside the financial system it was designed to challenge.

He added a line.

Bitcoin answered a real question. Whether the answer scaled to what the question actually required is still being determined.

“The broader crypto ecosystem,” he said. “Next week. Ethereum, stablecoins, DeFi, the documented fraud, the genuine innovation. I want to go through all of it.”

“I’ll be here,” Arthur said.

He said it simply. Not dramatically. But something in the way he said it made Neo not entirely sure he was referring only to next week.

Chapter Twelve: The Ecosystem

The whitepaper that launched Ethereum was published in 2013 by a nineteen-year-old.

Neo found this fact worth sitting with. Not because youth was remarkable in technology — it wasn't — but because what Vitalik Buterin had produced at nineteen was not a product or a company. It was a philosophical argument about what a blockchain could be.

Bitcoin answered one question: how do you transfer value without a trusted intermediary? Buterin's paper asked a different question: what if the blockchain could execute arbitrary agreements — contracts that ran automatically when their conditions were met, without any party needing to trust another, without any institution needed to enforce them?

He called them smart contracts. The name was older than he was — Nick Szabo had coined it in 1994 — but Buterin had built the infrastructure to make them real.

"Garvus," Neo said. "The technical distinction between Bitcoin and Ethereum. From first principles."

"Bitcoin's blockchain is intentionally limited in its programmability," Garvus said. "It is designed to do one thing — transfer Bitcoin between addresses — and to do that thing securely. The scripting language that governs Bitcoin transactions is deliberately not Turing-complete, meaning it cannot execute arbitrary computations. This is a security feature: limiting what the blockchain can do limits what can go wrong."

"And Ethereum's design choice."

"Ethereum introduced a Turing-complete programming environment — the Ethereum Virtual Machine — that allows developers to deploy arbitrary programs on the blockchain," Garvus said. "These programs — smart contracts — execute automatically when their specified conditions are met. Once deployed, they run exactly as written, without the ability of any party to modify or stop them. This is simultaneously the source of Ethereum's power and its most significant security challenge."

"The analogy," Anna said. She had developed a habit of asking for this at the technical junctures, not because she didn't understand but because she was thinking about how the material would land for readers who didn't.

"Bitcoin is a calculator designed to perform one operation reliably," Garvus said. "Ethereum is a general-purpose computer. A calculator can fail in a small number of ways. A general-purpose computer can fail in an enormous number of ways."

"The DAO," Arthur said. He said it with the tone of someone who had watched something instructive happen.

"In 2016, a smart contract called the DAO — Decentralized Autonomous Organization — raised approximately one hundred and fifty million dollars in Ethereum through a crowdfunding mechanism," Garvus said. "The DAO was designed to function as a venture capital fund governed by token holders rather than a board of directors — investment decisions would be made collectively through a voting mechanism embedded in the smart contract. Six weeks after launch,

an attacker exploited a vulnerability in the contract's code and drained approximately sixty million dollars in Ethereum."

"The exploit," Neo said.

"The DAO's withdrawal function contained a reentrancy vulnerability — a coding error that allowed an attacker to repeatedly call the withdrawal function before the contract updated its internal balance," Garvus said. "The attacker did not break the blockchain. The attacker exploited the code exactly as written. The smart contract executed as programmed. The program was wrong."

"The resolution was more interesting than the exploit," Arthur said.

"The Ethereum community faced a genuine philosophical crisis," Garvus said. "The blockchain's foundational promise is immutability — code that runs exactly as written, beyond the ability of any party to alter. Reversing the DAO exploit would require a hard fork — a modification of the blockchain's history to undo the transactions. This would violate the immutability principle. Not reversing it would allow the attacker to keep sixty million dollars stolen through a coding error."

"The community voted to fork," Neo said.

"The majority of the Ethereum network voted to implement the hard fork, reversing the DAO transactions. A minority refused on principle — arguing that immutability was non-negotiable and that reversing transactions, however justified, established a precedent that undermined the entire value proposition of smart contracts. This minority continued operating the original chain, now called Ethereum Classic."

"The philosophical question that never got resolved," Arthur said. "If a sufficiently large group of stakeholders can vote to rewrite history when the outcome is politically unacceptable, in what sense is the blockchain actually immutable? And if it isn't actually immutable, what exactly does it provide that a conventional database with a trusted administrator doesn't?"

Anna looked at him. "And the answer?"

"The answer is that blockchain immutability is a social guarantee as much as a technical one," Arthur said. "It holds when the community is willing to maintain it. It bends when the political pressure is sufficient. Whether that makes it meaningfully different from conventional systems — where immutability also holds until political pressure is sufficient — is a genuine philosophical question without a clean answer."

"Garvus," Neo said. "DeFi. What it is and what it promised."

"Decentralized Finance — DeFi — refers to financial applications built on blockchain platforms, primarily Ethereum, that replicate the functions of conventional financial intermediaries — lending, borrowing, trading, insurance — through smart contracts rather than institutions," Garvus said. "The promise was a financial system accessible to anyone with an internet connection, operating transparently through publicly auditable code, without the gatekeeping of banks, brokers, or regulators."

"The genuine case for that," Anna said. "Not the promotional version."

“In contexts where conventional banking is inaccessible — whether due to geography, documentation requirements, credit history, or political exclusion — DeFi protocols offer genuine financial functionality that was previously unavailable,” Garvus said. “A smallholder farmer in a country with an underdeveloped banking system can, in principle, access lending against digital collateral without a credit history or a bank account. A migrant worker can, in principle, access financial services without the documentation that conventional banks require. These are genuine use cases with documented real-world applications, particularly in Southeast Asia and Sub-Saharan Africa.”

“The documented problems,” Neo said.

“The DeFi ecosystem has experienced approximately three billion dollars in documented losses to hacks and exploits between 2020 and 2023,” Garvus said. “The smart contract vulnerability problem identified in the DAO case has not been solved — it has recurred across hundreds of protocols. The technical complexity of DeFi applications creates an attack surface that routinely exceeds the ability of auditors and developers to secure.”

“Beyond the hacks,” Arthur said.

“Beyond security vulnerabilities, DeFi has been characterized by significant rug pulls — projects where developers deploy a protocol, attract investor capital, and exit with the funds,” Garvus said. “Blockchain analytics firms estimate that rug pulls and exit scams accounted for approximately thirty-seven percent of all cryptocurrency fraud revenue in 2021. The open, permissionless nature of DeFi — its greatest strength from an access perspective — also removes the gatekeeping that prevents bad actors from deploying fraudulent protocols.”

Anna looked up from her notes. “The people who lost money to rug pulls. Who were they?”

“Documented analysis of cryptocurrency fraud victims suggests a demographic profile that skews younger and less financially sophisticated than conventional fraud victims,” Garvus said. “Survey data from the FTC found that Americans aged eighteen to thirty-nine reported losing money to cryptocurrency fraud at significantly higher rates than older age groups — the opposite pattern from most financial fraud, which typically targets older individuals. The accessibility of cryptocurrency platforms, designed to lower barriers to participation, also lowered barriers to victimization.”

“The accessibility that was supposed to help people ended up exposing them,” Anna said.

“That is what the documented data suggests in the fraud context,” Garvus said. “Worth noting that the same accessibility has also documented genuine financial inclusion benefits in other contexts. Both outcomes coexist.”

“Terra and Luna,” Neo said. “The case study.”

“Terra was an algorithmic stablecoin protocol launched in 2018,” Garvus said. “Its dollar-pegged stablecoin — UST — maintained its peg not through dollar reserves but through an algorithmic relationship with a sister token, Luna. When UST demand increased, Luna was burned to create UST. When UST demand decreased, UST was burned to create Luna. The mechanism was designed to maintain the peg through market incentives rather than reserves.”

“The flaw in the design,” Arthur said. He was leaning forward with the quality he brought to case studies that illustrated something he considered important.

“The algorithmic peg mechanism was circular,” Garvus said. “UST’s value depended on confidence in Luna. Luna’s value depended on confidence in UST. Both depended on continued demand and market confidence. When confidence wavered — triggered in May 2022 by a large coordinated withdrawal from Anchor Protocol, which was offering an unsustainable nineteen percent yield on UST deposits — the mechanism entered a death spiral. UST lost its peg. Luna was hyperinflated as the algorithm attempted to restore equilibrium. Within approximately seventy-two hours, approximately forty billion dollars in combined market value had been effectively destroyed.”

“Forty billion dollars,” Anna said quietly.

“The losses were not equally distributed,” Garvus said. “Retail investors, many of whom had allocated significant portions of their savings to Anchor Protocol based on the advertised yield and the protocol’s prominent endorsements, absorbed substantial losses. The protocol’s institutional backers — venture capital firms including Jump Crypto and Three Arrows Capital — had earlier entry points and in some cases had reduced exposure before the collapse.”

“The pattern,” Neo said. It was not a question.

“The documented pattern in multiple cryptocurrency collapses — Terra/Luna, FTX, Celsius, Voyager — shows retail investors absorbing disproportionate losses relative to institutional participants,” Garvus said. “This reflects both the earlier entry points of institutional investors, who had negotiated terms unavailable to retail participants, and the information asymmetry between sophisticated market participants and retail investors relying on public marketing materials.”

“Worth being precise about something,” Arthur said. “The pattern Garvus is describing is not unique to cryptocurrency. It describes the dynamics of every speculative bubble in financial history — the South Sea Company, Dutch tulips, the dot-com bubble, the mortgage-backed securities market before 2008. Retail investors, who enter late and exit late, consistently absorb disproportionate losses in bubble dynamics. The cryptocurrency ecosystem has replicated this pattern with unusual speed and transparency — the blockchain makes the dynamics more visible than they were in previous bubbles.”

“Is that an argument for or against crypto?” Anna asked.

“It is an argument for not confusing the technology with the speculative dynamics that surrounded it,” Arthur said. “The internet as a technology was real and transformative. The dot-com bubble that formed around it was irrational and destructive. Both things were true simultaneously. Most people who lost money in dot-com stocks were not wrong about the internet. They were wrong about the timing and the specific vehicles. Whether a similar distinction applies to blockchain technology and the cryptocurrency speculation that has surrounded it is a genuine question.”

“The Ethereum proof of stake transition,” Neo said. “September 2022.”

“Ethereum completed its transition from proof of work to proof of stake consensus mechanism

in September 2022 — referred to within the community as the Merge,” Garvus said. “Proof of stake replaces the energy-intensive mining process with a system in which validators stake — lock up — cryptocurrency as collateral to participate in block validation. The environmental impact was significant: Ethereum’s energy consumption decreased by approximately ninety-nine point nine five percent following the transition.”

“That’s a meaningful engineering achievement,” Arthur said directly. “The criticism that blockchain technology is inherently environmentally catastrophic applied to Bitcoin’s proof of work design. Ethereum’s proof of stake demonstrates that the fundamental technology can operate at negligible energy cost. Whether Bitcoin will adopt a similar transition — which its community has thus far resisted — is a separate question.”

“The regulatory response to the broader ecosystem,” Neo said.

“Regulatory approaches to cryptocurrency have varied significantly across jurisdictions,” Garvus said. “The SEC’s position in the United States has increasingly characterized most cryptocurrency tokens as securities subject to existing securities law — a position that, if fully implemented, would require most cryptocurrency projects to register with the SEC or face enforcement action. The EU’s Markets in Crypto-Assets regulation, enacted in 2023, establishes a comprehensive licensing framework for cryptocurrency service providers. China has prohibited cryptocurrency trading and mining. El Salvador has adopted Bitcoin as legal tender. The regulatory landscape is fragmented and actively evolving.”

“The SEC’s approach specifically,” Arthur said. “I want to say something about this directly.”

Neo looked at him.

“The SEC’s approach under recent leadership has been criticized, including by people whose regulatory instincts I respect, for attempting to apply regulatory frameworks designed for conventional securities to a technology with genuinely different characteristics,” Arthur said. “The criticism is not without merit — there are real questions about whether existing securities law maps cleanly onto blockchain tokens. There are also real questions about whether the SEC’s enforcement-first approach, rather than rulemaking, has served investors or primarily served regulatory uncertainty.”

“Both things being true simultaneously,” Anna said. She said it with a slight quality that suggested she had heard the phrase enough times to be gently amused by it.

“Both things being true simultaneously,” Arthur said, acknowledging the joke without breaking the point. “Regulatory clarity would benefit legitimate participants in the ecosystem. The absence of regulatory clarity has also allowed fraud and investor harm at a scale that clearer rules would have reduced. The question of which regulatory approach would have produced better outcomes is genuinely contested.”

Neo looked at the working document.

The crypto ecosystem picture, assembled honestly, was genuinely complex. Not the revolutionary monetary system of its most enthusiastic advocates. Not the fraud-in-a-trenchcoat of its most dismissive critics. A set of technical innovations — some genuinely significant, some overstated — that had developed within and around a speculative market that had produced

documented harm to retail investors while also producing documented benefits in specific real-world applications.

“The question I keep returning to,” Anna said, “is whether the people the technology was supposed to help — the unbanked, people living under broken monetary systems, migrants paying high remittance fees — whether they actually benefited, at scale, in ways that justified the documented harm to the retail investors who bought the speculation.”

The room was quiet.

“The honest answer,” Garvus said, “is that the documented evidence for transformative financial inclusion at scale is more limited than the documented evidence for speculative harm. The use cases are real. The scale has not yet matched the promise.”

“That may change,” Arthur said.

“It may,” Garvus agreed. “The technology is eleven years old. The internet was eleven years old in 2002, the year after the dot-com bubble collapsed. The subsequent twenty years produced outcomes that the 2002 moment did not clearly predict.”

Neo wrote in the working document.

The technology is real. The innovation is genuine in specific respects. The speculative market that formed around it produced documented harm distributed unequally. Whether the underlying technology will deliver on its stated purpose — financial access for those the conventional system excludes — at a scale that justifies the harm is a question the evidence does not yet answer.

He looked at the Substack notification.

Thirty-one thousand, four hundred subscribers.

Three invitations to speak at financial conferences. Two requests from publishers. One email from a former colleague at the fund, asking, with what felt like genuine unease, whether Neo had considered the implications of what he was publishing.

He had not responded to any of them.

Anna was watching him again with that journalist’s quality of attention — the sense of a person cataloguing what they observed.

“You’re going to have to decide soon,” she said.

“I know,” Neo said.

Chapter Thirteen: The Unbanked

[CHAPTER THIRTEEN — Please add from conversation **The number Anna put on the table first was not the one Neo expected.

He had been thinking about El Salvador, about the Chivo wallet data, about the documented mixed results of the Bitcoin legal tender experiment. Anna put a different number on the table.

“One point four billion,” she said.

“Adults without access to a bank account,” Garvus said. “World Bank Global Findex Database, 2021 estimate. Down from approximately two billion in 2011. The decline reflects significant progress in financial inclusion over a decade, driven primarily by mobile money adoption in Sub-Saharan Africa and South Asia.”

“Down from two billion,” Arthur said. He said it with the quality he brought to numbers that contained both good news and bad news simultaneously.

“The progress is real,” Anna said. “I want to be clear about that before we go anywhere else. The financial inclusion story of the past fifteen years is genuinely one of the more significant development achievements of the period. It deserves honest recognition.”

Neo looked at her. This was not the frame he had expected her to lead with, and he found himself recalibrating slightly.

“Walk me through M-Pesa,” he said. “From the beginning.”

“M-Pesa was launched in Kenya in March 2007 by Safaricom, the dominant mobile network operator,” Garvus said. “The product allowed users to deposit, send, and withdraw money using a basic mobile phone — not a smartphone, a feature phone — through a network of agents operating from small shops and market stalls. The design was deliberately simple: no bank account required, no credit history, no documentation beyond a national ID.”

“The adoption curve,” Anna said.

“Within three years of launch, M-Pesa had approximately thirteen million registered users — approximately forty percent of Kenya’s adult population,” Garvus said. “By 2020, approximately ninety-six percent of Kenyan households had at least one M-Pesa user. The platform processed approximately fifty billion dollars in transactions annually, equivalent to approximately fifty percent of Kenya’s GDP. The agent network, which required no bank branches or ATM infrastructure, had expanded to approximately five hundred thousand locations.”

“The documented welfare effects,” Neo said.

“Research by economists Tavneet Suri at MIT and William Jack at Georgetown, using longitudinal household survey data, found that access to M-Pesa increased household consumption — a proxy for income — by approximately two percent in rural households,” Garvus said. “For households headed by women, the effect was larger — approximately nine percent. The mechanism appears to be risk sharing: mobile money allowed households to receive transfers from family members and friends during adverse shocks — illness, drought, job loss —

more quickly and at lower cost than previously possible.”

“The gender dimension specifically,” Arthur said.

“The research documented that female-headed households in particular benefited from mobile money access through two channels,” Garvus said. “First, the ability to receive remittances and support transfers without a male household member’s involvement — women who previously had no independent access to financial services gained the ability to save and transact independently. Second, the reduction in cash-in-hand vulnerability: women carrying cash for household expenses faced theft and pressure from family members; digital balances were more controllable.”

Anna was quiet for a moment.

“In the countries I reported from,” she said, “the conversation about financial inclusion often treated women as a passive beneficiary category. The M-Pesa research suggests it’s more active than that. Financial access changes what’s possible for a household in ways that don’t show up in aggregate statistics.”

“The model has replicated,” Garvus said. “Mobile money services following variations of the M-Pesa model now operate across Sub-Saharan Africa — bKash in Bangladesh, GCash in the Philippines, Wave in Senegal, MTN Mobile Money across multiple African markets. The World Bank estimates that approximately five hundred and sixty million adults in Sub-Saharan Africa have mobile money accounts, compared to approximately two hundred and forty million with conventional bank accounts.”

“Mobile money penetration exceeded bank penetration,” Neo said.

“In multiple markets, yes,” Garvus said. “The infrastructure economics explain the divergence. A bank branch requires real estate, security, trained staff, regulatory compliance — a fixed cost structure that makes low-transaction-volume rural markets unprofitable. A mobile money agent requires a SIM card, a float of cash, and a small shop that may already exist for other purposes. The agent model scales to populations that the branch model cannot reach economically.”

Arthur leaned forward. “This is what I want to say clearly, because I think it gets underweighted in discussions that focus on cryptocurrency and CBDC: the financial inclusion achievement of the past fifteen years did not require blockchain technology. It did not require decentralization. It required a workable product, a viable distribution model, regulatory permission to operate, and a problem worth solving. The conventional financial technology stack, adapted for low-resource environments, produced a genuine and documented improvement in the lives of hundreds of millions of people.”

“You’re making a case against blockchain for financial inclusion?” Anna said.

“I’m making a case for proportionality,” Arthur said. “Blockchain is one tool. Mobile money is another. The question is which tool fits which problem. In Kenya in 2007, the problem was distributing financial services to people without bank accounts or smartphones in a country with poor physical infrastructure. Mobile money solved that problem. Whether blockchain would have solved it better, or whether blockchain solves different problems better, is a genuine question

that shouldn't be answered by enthusiasm for the technology."

"El Salvador," Neo said. "The honest assessment."

"El Salvador adopted Bitcoin as legal tender in September 2021 through the Bitcoin Law," Garvus said. "The stated rationale included reducing remittance costs — approximately twenty-four percent of El Salvador's GDP comes from remittances, primarily from migrants in the United States — and expanding financial inclusion for the approximately seventy percent of the population without bank accounts."

"The Chivo wallet," Anna said.

"The government launched the Chivo wallet as the primary interface for Bitcoin transactions," Garvus said. "Every Salvadoran citizen who downloaded the wallet received a thirty dollar Bitcoin bonus — a significant incentive in a country where the median daily wage is approximately fifteen dollars. Initial adoption was high by this measure: approximately two point seven million users downloaded the wallet within the first few months, approximately forty percent of the adult population."

"And the sustained use," Arthur said.

"The National Bureau of Economic Research survey published in 2022 found that sixty percent of Chivo users had withdrawn their thirty dollar bonus and not used the wallet again," Garvus said. "Approximately twenty percent of respondents reported using Chivo in a given month. Merchant adoption, while legally required to accept Bitcoin, has been uneven — businesses without reliable internet connectivity faced genuine operational difficulties."

"The remittance cost reduction," Neo said.

"Remittance costs through the Chivo wallet, for transfers from the United States, were documented to be lower than conventional channels for users who maintained Bitcoin balances," Garvus said. "The practical friction — the need to acquire Bitcoin at one end, the volatility risk during transit, the need to convert to dollars at the other end — limited adoption among remittance senders who preferred the certainty of conventional services despite higher fees."

"Worth noting what did work," Arthur said. He said it with the specific directness he had been deploying more frequently.

"Documented," Garvus said. "Users who engaged with the full Chivo ecosystem — maintaining balances, using the payment functionality — experienced genuine cost reductions on remittances and genuine convenience in peer-to-peer payments. The experiment demonstrated proof of concept for digital payment infrastructure at a national scale. The technical implementation, while imperfect, did not produce the financial stability catastrophe that the IMF's concerns had suggested as a risk."

"The IMF's response to El Salvador," Neo said.

"The IMF expressed significant concerns about the Bitcoin Law's financial stability and consumer protection implications and has discussed the issue in the context of El Salvador's debt restructuring negotiations," Garvus said. "The fund recommended that Bitcoin be removed as legal tender, retaining optional use. The Salvadoran government has not complied with this

recommendation. Bitcoin's subsequent price decline from its 2021 peak created paper losses on El Salvador's government Bitcoin holdings — a position the government has periodically increased through purchases during price dips."

"The government buying Bitcoin during dips," Arthur said flatly. "With public funds."

"Documented," Garvus said. "President Bukele has announced multiple government Bitcoin purchases on social media, describing them as buying the dip. The government's Bitcoin holdings have fluctuated in value significantly. Whether this represents sound reserve management or speculative risk-taking with public funds is a matter of genuine debate among economists."

"There is a dry observation available here," Garvus said. "A government that adopted Bitcoin partly as a response to concerns about dollar dependence now holds a volatile asset denominated in dollars and managed by a head of state who tweets about buying the dip."

Anna looked at the ceiling briefly. "That's the most concise summary of the situation I've heard."

"The CBDC question," Neo said. "How we get from mobile money and Bitcoin to what central banks are actually building."

"Central Bank Digital Currencies represent a qualitatively different instrument from both mobile money and cryptocurrency," Garvus said. "Mobile money is a private sector product — a liability of Safaricom or bKash or Wave, backed by deposits at conventional banks. Cryptocurrency is a decentralized asset — a liability of no institution, backed by the network's computational consensus. A CBDC is a direct liability of the central bank — the digital equivalent of a banknote, but in digital form, directly accessible to individuals without a commercial bank intermediary."

"The distinction matters," Arthur said.

"The distinction matters significantly," Garvus said. "Commercial bank deposits — what sits in your bank account — are a liability of the commercial bank, insured up to FDIC limits but at risk if the bank fails. A CBDC would be a direct claim on the central bank, which cannot fail in the way a commercial bank can. For financial inclusion purposes, this means access to safe digital money without exposure to commercial bank credit risk."

"The genuine use cases," Neo said. "Before the concerns. Make the strongest honest case."

"The financial inclusion argument is genuine," Garvus said. "A CBDC accessible via basic mobile phone, without a bank account requirement, could extend safe digital payments to the same populations that M-Pesa reached — but with the backing of the central bank rather than a private company. Payment system efficiency gains are real and documented: the Federal Reserve's FedNow instant payment system, while not a CBDC, demonstrates the demand for real-time payment infrastructure. Cross-border payment costs — currently high and slow — could be reduced through interoperable CBDC systems."

"The monetary policy transmission argument," Arthur said.

"During the COVID-19 pandemic, the U.S. government distributed relief payments through the existing banking infrastructure," Garvus said. "Households without bank accounts —

disproportionately lower-income — received checks that required check-cashing services at a documented cost. A CBDC with universal access would allow direct distribution of fiscal transfers to every citizen's digital wallet, instantly, without bank intermediation and without the friction and cost experienced by the unbanked."

Anna had been quiet, listening carefully. She looked up.

"I want to say something about this from the perspective of someone who grew up in a country where the government froze bank accounts," she said. "The financial inclusion argument for CBDCs is real. I accept it. But the same infrastructure that delivers payments directly to every citizen's wallet is the infrastructure that can freeze every citizen's wallet. Or program every citizen's wallet to expire. Or exclude specific citizens from the payment system entirely."

"The programmability question," Neo said.

"This is where the technology departs from physical cash in ways that are significant and not always clearly articulated in CBDC discussions," Garvus said. "Physical cash is anonymous, untrackable, programmable by no one, and available to anyone regardless of their political or social status. A CBDC is, by its nature, a tracked, identified instrument. Whether additional programmability — expiration dates, category restrictions, negative interest rates — is implemented is a policy choice. But the infrastructure that enables those choices would exist in a way that it does not with physical cash."

"The Chinese digital yuan as the current most advanced implementation," Neo said.

"China's digital yuan — the e-CNY — has been in active pilot since 2020 and has been distributed through lotteries, employer payroll, and retail promotions in multiple cities," Garvus said. "The documented features include controllable anonymity — transactions below a certain threshold are treated as anonymous to merchants but are visible to the People's Bank of China — and programmable validity periods on some distributed e-CNY. The system is designed to operate with or without internet connectivity, addressing rural access concerns."

"What the People's Bank of China can see," Arthur said directly.

"The e-CNY system gives the central bank visibility into retail transactions at a granularity that the current banking system does not," Garvus said. "Whether this visibility is used for financial inclusion purposes — identifying underserved populations, monitoring for fraud — or for surveillance and control of political dissidents, journalists, and ethnic minorities is a question of governance rather than technology. The infrastructure supports both."

"And the Western CBDC programs," Neo said.

"Most major central banks have CBDC research programs at varying stages," Garvus said. "The European Central Bank is in the investigation phase of a digital euro. The Bank of England has published consultation documents on a digital pound. The Federal Reserve has published research on a potential digital dollar while stating it would not proceed without clear Congressional authorization. The Bank for International Settlements coordinates research across central banks through its Innovation Hub."

"The governance question that none of them have fully answered," Arthur said.

“No major Western CBDC design has fully specified the privacy architecture, the programmability limits, or the oversight mechanisms that would govern use,” Garvus said. “The design choices — whether CBDCs hold balances directly at the central bank or through intermediary banks, whether offline functionality is supported, what transaction visibility the central bank retains, what programmable features are permitted — determine whether the resulting instrument resembles digital cash or digital surveillance.”

Anna looked at her notes.

“The countries I reported from,” she said carefully, “have governments with track records on these questions. The corralito — the 2001 Argentine bank freeze — was implemented through the existing banking infrastructure. A CBDC would not have required a bank freeze. It would have been easier. The same infrastructure that makes financial inclusion more accessible makes financial control more accessible.”

The room was quiet.

Arthur looked at her with the directness of a man who had spent his career inside the institutions being discussed.

“The central banks developing CBDCs are, by and large, staffed by people who genuinely believe in the financial inclusion mission and who would not deliberately design instruments of surveillance or control,” he said. “I know some of them. I believe their intentions are what they say they are. What I also know, from thirty years inside institutions, is that institutional intentions and institutional outcomes are not the same thing. The infrastructure that is built for good reasons becomes available for other reasons when the institutional context changes. What a central bank controlled by one government would do with CBDC infrastructure is not a guarantee of what a central bank controlled by a subsequent government would do.”

He paused.

“That is the thing I would not have said in a policy meeting ten years ago. It is also true.”

Neo wrote in the working document.

The financial inclusion case for CBDC is genuine and documented. The surveillance and control risk is structural, not conspiratorial — the infrastructure that enables inclusion enables control, and institutional intentions are not institutional guarantees. The design choices matter enormously and most have not yet been made.

He looked at the Substack.

Forty-seven thousand, three hundred subscribers.

A message from a former Treasury official he had met once at a conference, asking if they could speak privately.

He had not responded.

“The interaction between CBDCs and what we’ve documented about the financial system,” Anna said. “The debt architecture. The offshore capital flows. The AI displacement. How do those threads connect?”

"That," Neo said, "is the next chapter."

He looked at Arthur.

The older man was looking out the window at the winter light on the Sound. Something in his expression had the quality of a man measuring time differently than he once had.

"I'll be here," Arthur said.

Chapter Fourteen: Convergence

Neo spent three days assembling the picture before he was willing to show it to anyone.

Not because it was dramatic. Because it wasn't — and he wanted to be sure he was seeing it clearly rather than constructing something that felt coherent because he had spent three months pulling threads and wanted them to connect.

The picture was this: a global financial system carrying debt at historically high levels, with servicing costs consuming increasing shares of government revenue. An AI-driven productivity transition whose distributional consequences — who captures the gains, who absorbs the displacement — were not yet determined by policy. A digital monetary infrastructure under construction whose design choices would determine whether it extended financial access or financial control. And underlying all of it, a capital concentration dynamic whose documented trajectory had not historically resolved without significant disruption.

He looked at this for three days.

Then he called Arthur.

"Before you show me the synthesis," Arthur said, when he arrived, "I want to say something."

He sat down in the chair that had become his chair in this room over the past months — the one by the window, angled toward the fire. Outside, Long Island Sound was the particular gray of January, ice forming at the edges.

"I've watched you build this for three months," Arthur said. "The research is careful. The mechanics are accurate. The individual pieces are documented. I want to make sure the synthesis receives the same standard of care as the pieces."

"That's why I called you," Neo said.

"The convergence narrative," Arthur said, "is the most dangerous place in this kind of analysis. Not because it's wrong. Because it feels right. When you have assembled careful evidence about multiple problems, the evidence tends to arrange itself into a picture of crisis. The problems reinforce each other. The timeline closes in. The conclusion feels inevitable."

He paused.

"And sometimes it is inevitable. Sometimes the convergence is real and the crisis arrives. And sometimes the convergence narrative is the story that serious, careful people tell themselves in every generation — the story that the accumulated problems of the present moment represent a qualitative break from all previous accumulated problems — and the system adapts in ways the narrative didn't predict, and the crisis looks different from what was anticipated, or arrives later, or doesn't arrive in the form that was feared."

"The boy who cried wolf," Anna said.

She had come the previous evening, staying in the guest room — the first time she had done so. Something about the research was reaching a stage that seemed to pull the three of them toward proximity rather than scheduled meetings.

“Not quite,” Arthur said. “The boy who cried wolf was wrong about the wolf. The analysts I’m describing were often right about the problems. They were wrong about the timing, the mechanism, and the form of resolution. That distinction matters enormously.”

“Walk me through the counterargument specifically,” Neo said. “Not generically. Against the specific threads we’ve assembled.”

Arthur nodded once.

“The debt argument,” he said. “The concern is that debt at three hundred and thirty-six percent of global GDP represents an unsustainable burden that must eventually resolve through default, inflation, or growth that is historically unlikely. The counterargument is that we have been saying this about Japanese debt for thirty years and Japan has not collapsed. That the United States has run fiscal deficits for the majority of the past sixty years and the dollar remains the world’s reserve currency. That debt levels that appear unsustainable under conventional analysis have proven sustainable under conditions — low interest rates, reserve currency status, institutional credibility — that the conventional analysis did not fully account for.”

“The interest rate environment has changed,” Neo said.

“The interest rate environment has changed,” Arthur acknowledged. “The free money era is over. Debt service costs are rising. This is a genuine tightening of the constraint. Whether it represents a qualitative break — a transition from sustainable to unsustainable — or a quantitative adjustment that the system will absorb through some combination of growth, taxation, and fiscal consolidation is genuinely uncertain. The CBO’s projections look alarming. CBO projections have looked alarming before and the projected crisis has not arrived on schedule.”

“The AI displacement counterargument,” Anna said.

“Every major technological transition has produced convergent evidence, at the time of transition, that this time was different — that the displacement was qualitatively unlike previous displacements, that the creation of new work categories would not keep pace,” Arthur said. “Every major technological transition has produced that evidence and most of them have resolved, over multi-decade horizons, with net employment expansion. The pace of the current transition is genuinely faster than previous ones. Whether it is fast enough to outrun the adaptation mechanisms that operated in previous transitions is unknown.”

“The CBDC concern,” Neo said.

“The surveillance and control concern is real and I share it,” Arthur said. “The counterargument is that the design choices haven’t been made. That democratic accountability mechanisms, however imperfect, exist in the countries currently developing CBDCs. That the financial inclusion benefits are genuine and currently underweighted in the critical analysis. And that the alternative — cash declining in use, financial services increasingly intermediated by private technology platforms with their own surveillance architectures — may not represent a more privacy-protective outcome than well-designed CBDC.”

“Garvus,” Neo said. “The honest probability assessment. Not a number. A characterization.”

"The convergence of the documented trends — debt trajectory, AI displacement pace, CBDC implementation timeline, capital concentration dynamics — creates a set of conditions that historical analysis associates with periods of significant systemic adjustment," Garvus said. "Whether that adjustment takes the form of a sharp crisis, a prolonged slow deterioration, a managed policy transition, or some combination that doesn't fit historical categories is not predictable from the available evidence. Economists who have studied financial crises — Reinhart and Rogoff, Minsky, Kindleberger — consistently find that the specific form and timing of systemic adjustment is not predictable in advance, even when the underlying imbalances are clearly documented."

"The Minsky moment question," Arthur said.

"Hyman Minsky's financial instability hypothesis argues that stability itself is destabilizing — that periods of sustained financial stability encourage risk-taking and leverage accumulation that eventually produces instability," Garvus said. "The hypothesis has significant empirical support across multiple historical episodes. It does not predict when the Minsky moment arrives. The conditions Minsky describes can persist for extended periods before resolving."

"What it means for ordinary people," Anna said. She was looking at her notes. "Not the aggregate. The specific populations."

"The distributional question is the most honest frame for this analysis," Garvus said. "Systemic adjustments — whether slow or sharp — distribute their costs unevenly. The populations with the least financial resilience — limited savings, limited access to alternative assets, high debt-to-income ratios, employment concentrated in AI-exposed sectors — absorb the largest proportional costs in most historical adjustment scenarios. The populations with the greatest financial resilience absorb smaller proportional costs or, in some scenarios, benefit from asset price dynamics that accompany adjustment."

"This is what is documented," Anna said. "Not the crisis itself. The distribution of consequences when adjustment happens."

"Correct," Garvus said. "The documented pattern across historical financial crises — 1929, the Latin American debt crisis, the Asian financial crisis, 2008 — is that the costs fall disproportionately on populations with the least capacity to absorb them. The mechanism is not designed. It is structural — the consequence of how financial claims, debt obligations, and asset ownership are distributed in the pre-crisis period."

Arthur was looking out the window. The January light was flat and pale on the water.

"I want to say something that I have been thinking about since we started this," he said. He was not looking at either of them. "About what this research is for."

The room was quiet.

"When I was inside the institution," he said, "I operated with a specific frame. The frame was that the financial system, with all its imperfections, produced sufficient aggregate benefit to justify the work of maintaining and improving it. That the distributional problems were real and worth addressing, but were best addressed within the system rather than through fundamental challenge to it. I believed that. I still believe most of it."

He paused.

“What I find myself thinking now, at the end of things, is that the frame itself — the assumption that the primary task is maintaining and improving the existing architecture — may have prevented asking whether the architecture’s basic design could be different. Not whether it should be immediately dismantled. Whether, over time, with deliberate design, the fundamental relationship between money creation, sovereign debt, and private financial intermediation could be structured differently in ways that produce different distributional outcomes.”

“And the answer?” Neo said.

“I don’t know,” Arthur said. “I know that every serious attempt to answer that question in the twentieth century produced outcomes that ranged from disappointing to catastrophic. I know that the distributed alternatives — crypto, DeFi, BRICS — have not yet demonstrated the ability to operate at the scale and stability required. And I know that not asking the question because the answers are hard has produced the system we have documented over these months.”

He turned away from the window.

“That is the most honest thing I have said about this in fifty years of thinking about it.”

Neo looked at the working document. The convergence chapter, assembled with genuine care, did not resolve toward the clean crisis narrative that the accumulated threads created pressure toward. It resolved toward something more uncomfortable: genuine uncertainty about outcomes, with a documented distributional pattern in which the costs of uncertainty fell on specific populations regardless of which form the adjustment took.

“What do we do with this?” Anna asked.

It was the question she had been building toward for weeks.

“We publish it,” Neo said. “Carefully. With the uncertainty intact. Not as a prediction. As a picture assembled from documented evidence, with the honest acknowledgment of what the evidence supports and what it doesn’t.”

“Fifty-eight thousand people are waiting for it,” Garvus said.

The flatness of the delivery was precisely right.

“Garvus,” Anna said. “Did you just make a joke?”

“I provided a factual update on subscriber count at a moment when it was contextually relevant,” Garvus said. “The interpretation is yours.”

Arthur almost smiled. It was the most he had managed in several weeks.

“The alternative architectures,” Neo said. “Small modular reactors. Decentralized communities. The genuine solutions. We’ve documented the problems carefully. The next section has to document what’s actually being tried with equal care.”

“That’s the harder work,” Anna said.

“It’s the more important work,” Arthur said. “Anyone can document what’s broken. The question worth asking — the one that separates this from the crisis literature that’s been wrong

about timing for fifty years — is what might actually be different. What's being built. Where the genuine alternatives are emerging."

He stood, slowly, with the deliberate care he had been applying to movement for the past month.

"Same time next week," he said.

Neo walked him to the door. On the porch, Arthur stopped.

"The Treasury official who contacted you," he said quietly. "I'd take that meeting."

"Why?"

"Because whoever sent that message already knows what you're building," Arthur said. "And whether they're warning you or joining you is information worth having."

He went down the steps to his car.

Neo stood on the porch until the taillights disappeared. Then he stood a little longer, looking at the winter dark over the Sound.

Fifty-eight thousand people.

He went back inside.

PART THREE: WHAT IS BEING BUILT

Chapter Fifteen: What's Being Built

Arthur arrived in February with something Neo had not seen before.

Not optimism exactly. Arthur was constitutionally allergic to optimism — he had spent too long watching confident predictions fail to trust the feeling. What he arrived with was closer to interest. The specific quality of attention he brought to things that surprised him.

He had been reading about fusion.

"Tell me what you know," he said, before he had taken his coat off.

Neo almost smiled. Thirty-five years and the seminar dynamic had never fully reversed.

"Garvus," Neo said. "The National Ignition Facility. December 2022."

"On December 5th, 2022, the National Ignition Facility at Lawrence Livermore National Laboratory achieved fusion ignition for the first time in history," Garvus said. "The experiment delivered two point zero five megajoules of energy to a fuel target and produced three point one five megajoules of fusion energy output — approximately fifty-three percent more energy out than in. This is the first time in the history of fusion research that a laboratory experiment produced more energy from fusion than the laser energy delivered to the target."

"The significance," Arthur said. He was sitting now, coat still half on, the quality of attention sharp.

"The significance is scientific rather than immediately commercial," Garvus said. "The experiment demonstrated that fusion ignition is physically achievable under laboratory conditions — a question that had been open since the 1950s. The two megajoules delivered to the target required considerably more energy to generate — the facility's total electrical consumption during the shot was approximately three hundred megajoules. Commercial fusion requires not just ignition but a substantial positive energy return on the full system energy investment."

"How far away is that," Anna said. She had come in behind Arthur, unwrapping a scarf, and had been listening from the doorway.

"Genuinely uncertain," Garvus said. "Commonwealth Fusion Systems, a private company spun out of MIT's plasma science program, projects achieving net energy gain at the system level in the early 2030s with their SPARC tokamak — a compact design enabled by high-temperature superconducting magnets. The company achieved a significant milestone in September 2021 when its prototype magnet reached twenty tesla field strength, validating the core technology. TAE Technologies, Helion Energy, and several other private fusion companies are pursuing alternative approaches on similar timelines."

"The private investment dimension," Neo said.

"Private investment in fusion energy reached approximately four billion dollars in 2022 — approximately double the preceding year's total," Garvus said. "The investor base includes technology entrepreneurs, established energy companies, and sovereign wealth funds. The pace

of private investment suggests a genuine belief among sophisticated investors that near-term commercial fusion is achievable — though sophisticated investors have been wrong about energy technology timelines before.”

“Worth being honest about the timeline history,” Arthur said. “Fusion has been twenty years away for sixty years.”

“The joke is accurate as a historical observation,” Garvus said. “The timeline confidence is higher now than at previous points for documented technical reasons — the superconducting magnet advances that enable smaller, less expensive reactors, the ignition demonstration, and the sustained private investment that has accelerated iteration. Whether this time is genuinely different is not yet established.”

“Small modular reactors,” Neo said. “The nearer-term technology.”

“Small modular reactors are nuclear fission reactors — the same basic technology as conventional nuclear plants — designed at smaller scale, typically three hundred megawatts or less versus one thousand megawatts for conventional plants,” Garvus said. “The design philosophy differs from conventional nuclear in several important ways. Factory fabrication rather than site construction is intended to reduce costs and construction time. Smaller scale enables deployment in locations unsuitable for conventional plants — remote communities, industrial facilities, regions with limited grid infrastructure. Passive safety systems that rely on physics rather than active cooling reduce the consequences of loss-of-power events.”

“The documented progress,” Arthur said. “And the documented problems. Both.”

“NuScale Power received Nuclear Regulatory Commission design certification in January 2022 — the first SMR to receive NRC certification in the United States,” Garvus said. “In November 2023, NuScale and the Utah Associated Municipal Power Systems cancelled their Carbon Free Power Project — a planned deployment of NuScale reactors — citing cost increases that had made the project uneconomical for the municipalities involved. The projected cost per megawatt-hour had increased from approximately fifty-eight dollars to approximately eighty-nine dollars, exceeding the cost of competing clean energy alternatives.”

“The cancellation matters,” Arthur said. He said it directly. “I’ve watched enough energy technology transitions to know that first-of-kind costs and nth-of-kind costs are different things — the learning curve in energy is real and documented. But the cancellation also demonstrates that SMR economics have not yet been proven at commercial scale. The technology is real. The economics are unproven.”

“TerraPower and X-energy received Department of Energy funding through the Advanced Reactor Demonstration Program,” Garvus said. “TerraPower’s Sodium reactor, a sodium-cooled fast reactor co-developed with GE Hitachi, broke ground in Kemmerer, Wyoming in 2024. Kairos Power’s fluoride salt-cooled reactor is in licensing review. The Canadian government has approved SMR development at the Darlington nuclear site in Ontario.”

“The genuine promise that isn’t about electricity,” Neo said.

“Industrial process heat is one of the most significant and least discussed potential applications,” Garvus said. “Approximately twenty percent of global energy consumption is

industrial process heat — temperatures required for cement, steel, chemical production, and other heavy industries that cannot be easily electrified. SMRs operating at higher temperatures than conventional light water reactors can provide this heat directly, potentially decarbonizing sectors that electricity-based solutions cannot easily reach.”

Anna looked up from her laptop. “What does this mean for developing economies? Not the technology. The access.”

“The energy access dimension is significant and genuinely underweighted in Western discussions of SMRs,” Garvus said. “Approximately seven hundred and seventy million people globally lack access to electricity. Conventional large nuclear plants require grid infrastructure, regulatory capacity, and capital investment that most developing countries cannot access. SMRs designed for remote deployment — some designs can be transported by truck or barge — could in principle reach communities that grid extension has not.”

“In principle,” Arthur said.

“In principle,” Garvus confirmed. “The regulatory frameworks governing nuclear technology transfer, the capital requirements even for smaller plants, and the technical expertise required for operation represent genuine barriers. The in-principle access case is real. The practical pathway to that access requires policy decisions and financing mechanisms that do not yet exist at scale.”

“The solar and wind picture,” Neo said. “Because that’s where most of the actual transition is happening.”

“The cost declines in solar photovoltaic and wind power over the past fifteen years are among the most dramatic in the history of energy technology,” Garvus said. “Solar PV module costs declined by approximately ninety percent between 2010 and 2023. Wind turbine costs declined by approximately seventy percent over the same period. The levelized cost of electricity from new utility-scale solar and wind in most markets is now below the operating cost of existing fossil fuel plants — meaning it is cheaper to build new solar or wind than to continue running existing coal or gas plants in many regions.”

“This is real,” Arthur said. He said it with a quality that was close to wonder — not naive, not promotional, but the genuine surprise of a man whose professional life had been spent watching energy transitions occur slowly and who was watching this one occur fast. “The cost curves are real. I’ve followed energy economics for forty years and I did not predict this pace.”

“The storage and grid challenge,” Anna said.

“The primary limitation of solar and wind is intermittency — they generate electricity when the sun shines and wind blows, not necessarily when demand is highest,” Garvus said. “Grid-scale energy storage — primarily lithium-ion batteries, with longer-duration alternatives in development — has also experienced significant cost declines, approximately ninety percent since 2010. Whether storage can scale sufficiently to support a high-penetration renewable grid at reasonable cost is a genuine technical and economic question that current deployments are beginning to answer empirically.”

“Australia and Texas as case studies,” Neo said.

“South Australia has achieved periods of one hundred percent renewable electricity generation and has deployed grid-scale battery storage at significant scale, including the Hornsdale Power Reserve — originally the world’s largest lithium-ion battery installation,” Garvus said. “Texas’s ERCOT grid, which experienced catastrophic failure during the February 2021 winter storm, has since added significant battery storage capacity. Both systems provide data on high-penetration renewable grid operation that previous analysis could only model.”

“The connection to the financial system analysis,” Arthur said. He was leaning forward now with the quality of someone who had been thinking about something and was ready to say it. “The energy transition and the financial system problems we’ve documented are not separate stories.”

“Explain,” Anna said.

“The fossil fuel system and the dollar system developed together,” Arthur said. “The petrodollar arrangement we discussed — oil priced in dollars, dollar recycled into Treasuries — is not just a financial arrangement. It’s an energy arrangement. The dollar’s reserve status is partly underwritten by the global economy’s dependence on dollar-denominated oil. A genuine energy transition — one that reduces the global economy’s dependence on dollar-denominated fossil fuels — changes the geopolitical and financial landscape in ways that most financial analysis hasn’t fully incorporated.”

“The countries that benefit most from the current energy system,” Anna said slowly.

“And the countries that would benefit most from a different one,” Arthur said. “The populations with the most to gain from distributed renewable energy are frequently not the populations that current energy and financial systems serve well. Abundant, affordable, distributed energy — generated locally rather than extracted from geological formations that happen to be located in specific countries — changes the terms of energy geopolitics in ways that downstream financial consequences are not yet clear.”

Neo wrote in the working document.

The energy transition is real and faster than most predicted. SMRs and fusion represent genuine future possibilities with honest uncertainties. The transition changes the geopolitical substrate on which the financial system rests — in ways whose full consequences are not yet visible.

“What’s being built locally,” Anna said. “Not at the technology level. At the community level. The decentralized responses.”

“That,” Neo said, “is the next chapter.”

He looked at the working document. The alternatives section, approached honestly, was not a simple good-news counter to the problems chapters. It was more complicated than that — real progress, genuine challenges, honest uncertainty about what scaled and what didn’t.

But it was also genuinely different in texture from the problem documentation. There was something in it that the earlier chapters didn’t have — the specific quality of things being built rather than things being documented as broken.

Arthur was looking at the window. The February light was pale and flat, but there was something in it — the particular quality of late winter light that carries the first intimation of spring without yet being spring.

“I find,” he said, to no one in particular, “that I think about this more clearly now than I did when I was inside it. When the question of what was possible was bounded by what was institutionally acceptable.” He paused. “I’m not sure if that’s wisdom or just the freedom that comes from having nothing left to protect.”

“Maybe both,” Anna said.

“Maybe both,” Arthur agreed.

Chapter Sixteen: At the Edges

The Bristol Pound launched in 2012 with twelve hundred business members and folded in 2020 with a statement that cited the pandemic and the practical limits of local currency adoption.

Neo put this fact in the working document before anything else.

Not because it was the most important fact about local currency experiments. Because it was true, and the research had taught him that the most important discipline was starting with what was true rather than what fit.

“Garvus,” he said. “The documented local currency experiments. Full picture.”

“Local and complementary currency systems have operated across multiple countries with varying degrees of success and longevity,” Garvus said. “The Bristol Pound in the United Kingdom reached approximately one million pounds in circulation at peak and was accepted by approximately eight hundred local businesses. It folded after eight years, citing difficulty achieving the transaction volume necessary for operational sustainability. The BerkShares currency in western Massachusetts has operated since 2006 through a network of regional banks and local businesses, with approximately ten million BerkShares issued cumulatively — a more modest but more durable implementation.”

“The Sardex network,” Arthur said. He had arrived with a folder of printed papers, which he almost never did — he had been doing his own reading.

“Sardex is a mutual credit network operating in Sardinia, Italy, established in 2010,” Garvus said. “Rather than issuing a physical currency, it operates as a B2B credit clearing system — businesses earn Sardex credits by selling to network members and spend them by buying from network members. The network has processed over three hundred million euros in transactions among approximately four thousand member businesses. The model addresses a specific problem: liquidity constraints in a regional economy with high unemployment and limited access to bank credit. By creating a parallel credit system for local transactions, it allows businesses to trade without cash.”

“What works about it,” Anna said.

“The mutual credit model works where the conventional currency system creates liquidity constraints that limit real economic activity between willing buyers and sellers,” Garvus said. “Sardinia’s economic geography — peripheral to continental Europe, with high youth unemployment and limited bank lending — created conditions where a local credit system addressed a genuine gap. The network’s transaction volume suggests it serves a real function. Whether it is replicable in economies without those specific conditions is less clear.”

“The honest limitation,” Arthur said.

“Local and complementary currencies address local liquidity problems,” Garvus said. “They do not address the global financial system’s structural features — the debt architecture, the offshore capital flows, the distributional dynamics — that the earlier chapters documented. They are

resilience mechanisms at a specific scale, not alternatives to the broader system.”

“The Mondragon cooperative,” Neo said.

“The Mondragon Corporation in the Basque Country of Spain is the world’s largest worker cooperative,” Garvus said. “Founded in 1956 by a Catholic priest and five engineers, it has grown to approximately eighty thousand employee-owners across approximately one hundred cooperatives in manufacturing, retail, finance, and education. The cooperative structure means workers are the owners — they share in profits, elect their own management, and cannot be laid off without cooperative approval except in genuine business failure.”

“The documented outcomes,” Arthur said.

“Mondragon’s documented outcomes include significantly lower executive-to-worker pay ratios than comparable conventional corporations — typically six to one versus three hundred to one in large U.S. corporations,” Garvus said. “Lower unemployment volatility — the cooperative structure incentivizes work-sharing during downturns rather than layoffs. Greater geographic stability — cooperative ownership reduces the incentive to offshore production for lower labor costs. The model has survived multiple economic crises, including Spain’s severe 2008 to 2013 recession, with fewer job losses than comparable conventional firms.”

“And the limitations,” Anna said.

“Mondragon has faced documented challenges scaling the cooperative model,” Garvus said. “Its retail subsidiary, Eroski, employs significant numbers of non-member workers who do not share ownership rights — a tension between the cooperative’s scale ambitions and its democratic ownership principles. Some manufacturing has moved to lower-cost countries under competitive pressure. The model works well at certain scales and in specific sectoral and geographic contexts. It has not demonstrated easy replicability in different contexts.”

“The credit union model,” Neo said.

“Credit unions — member-owned financial cooperatives — collectively represent approximately two point two trillion dollars in assets in the United States,” Garvus said. “They are owned by their members, return profits as dividends and lower fees rather than to external shareholders, and are governed by member-elected boards. Research consistently finds that credit union members pay lower loan rates and receive higher deposit rates than customers of comparable commercial banks. The model has demonstrated durability across more than a century of operation in multiple countries.”

“What the cooperative and credit union models share,” Arthur said.

“They demonstrate that ownership structure matters for distributional outcomes,” Garvus said. “The documented differences in executive compensation, profit distribution, and member treatment between cooperative and conventional corporate structures are not trivial. They suggest that the distributional dynamics documented in earlier chapters — wealth concentration, extraction — are not inevitable features of market economies but are partly functions of ownership and governance structures that can be designed differently.”

“Monetary reform proposals,” Neo said. “The serious ones. Not the fringe.”

“Several monetary reform proposals have significant academic support and policy discussion,” Garvus said. “Sovereign money reform — advocated by organizations including Positive Money in the UK and the American Monetary Institute — proposes that money creation should be the exclusive function of a public authority rather than a function of private bank lending. The 2012 IMF working paper by Jaromir Benes and Michael Kumhof modeled the macroeconomic effects of sovereign money and found potential benefits including reduced financial instability and improved public finances.”

“The counterargument to sovereign money,” Arthur said. He leaned forward.

“The primary counterargument is that private bank money creation — money created in response to credit demand from businesses and individuals — is more responsive to economic needs than centrally created sovereign money,” Garvus said. “Central planning of money supply has historical precedents that suggest caution. The challenge is calibrating money creation to actual economic needs without the decentralized information that private credit markets provide.”

“Land value taxation,” Neo said.

“Land value taxation — taxing the unimproved value of land rather than the improvements on it — has been advocated since Henry George’s Progress and Poverty in 1879 and has periodic revivals in policy discussion,” Garvus said. “The economic logic is that land value is created by community investment — infrastructure, amenity, economic activity — rather than by the landowner’s effort, making it an efficient and equitable tax base. Estonia implemented a land value tax in the 1990s and has maintained it since. Some jurisdictions in Pennsylvania implemented split-rate taxation with documented reductions in land speculation.”

“UBI,” Anna said. “What the pilots actually show.”

“Universal Basic Income pilots have been conducted in multiple countries with documented outcomes,” Garvus said. “Finland’s 2017 to 2018 pilot provided two thousand unemployed individuals with five hundred sixty euros monthly. Documented outcomes included improved wellbeing and mental health among recipients without significant reduction in employment-seeking behavior. Stockton, California’s SEED program provided one hundred and twenty-five low-income residents with five hundred dollars monthly from 2019 to 2021. Documented outcomes included full-time employment increasing from twenty-eight to forty percent among recipients, and improved financial stability metrics.”

“The scale problem,” Arthur said.

“The pilots are too small and too short to resolve the central economic debates about UBI,” Garvus said. “Whether a universal program — not a targeted one — would produce different labor market effects, what the macroeconomic impact of financing it would be, whether it would adequately replace or supplement existing social programs — these questions require implementation at scale to answer empirically. The pilots demonstrate that the feared work disincentive effects are not as large as critics predicted. They do not resolve the financing and macroeconomic questions.”

“What good CBDC design would actually look like,” Neo said. “Not what’s being built. What

should be built.”

“The Bank for International Settlements has published design principles for CBDCs that include privacy protection, interoperability, financial inclusion, and resilience,” Garvus said. “A well-designed CBDC would separate transaction data from identity data at the protocol level — allowing regulators to detect fraud and money laundering without routine visibility into individual transactions. It would support offline functionality for populations without reliable internet access. It would include holding limits that prevent CBDC from displacing commercial bank deposits at a destabilizing scale. It would have programmability restrictions written into law rather than left to administrative discretion.”

“The gap between those principles and current implementations,” Anna said.

“The Chinese e-CNY does not fully implement privacy-protective design,” Garvus said. “Most Western CBDC research programs have not yet committed to specific privacy architectures in binding legal frameworks. The design principles exist. Their implementation requires political will that is not yet evident in most jurisdictions.”

Arthur was quiet for a moment.

Then he said something that had the quality of words that had been forming for a long time.

“I want to say something I have not said in any policy context over fifty years,” he said. “About the institution I worked for.”

Neo and Anna were both still.

“The Federal Reserve’s mandate is price stability and maximum employment,” Arthur said. “In practice, from roughly 1980 onward, price stability was treated as the primary mandate. The inflation targeting framework — the two percent target, the willingness to accept unemployment to achieve it — reflected a genuine intellectual consensus about how monetary policy worked. I believed it. Most of my colleagues believed it.”

He paused.

“What I did not say — what none of us said publicly — was that the distributional consequences of prioritizing price stability over employment were not randomly distributed. Low inflation protects the value of financial assets. Financial assets are disproportionately held by the wealthy. Unemployment, when it occurs as a consequence of tight monetary policy, falls disproportionately on workers with the least savings and the fewest options. The framework was presented as technical and neutral. It was neither.”

The room was quiet.

“You knew this,” Neo said.

“I understood it,” Arthur said. “Understanding something and saying it publicly are different things when you are inside an institution whose functioning depends on public confidence in its neutrality. I told myself the aggregate benefits of price stability justified the distributional costs. I still think that argument has some merit. I no longer think it was as clearly correct as I presented it.”

He looked at his hands.

“That is the thing I would not have said while it mattered professionally.”

Anna wrote something in her notebook without looking up.

Neo looked at the working document. The alternatives section, assembled honestly, was not a refutation of the problems chapters. It was something more interesting — a picture of real experiments at the edges of the system, some successful, some failed, some too small to evaluate. Real people building different things in different ways. Not a solution. Evidence that the frame could be different.

He wrote a single line.

The edges are where different is being tried. Most of it is small. Some of it works. The question of what scales is not yet answered.

“One more chapter,” he said.

Arthur nodded.

Anna was still writing.

Chapter Seventeen: The Open Question

Arthur died on a Tuesday in March.

Not dramatically. He had driven himself to Cambridge on a Friday, as he always did. He had called Neo on Sunday evening to discuss something he had read about the Bank for International Settlements and quantum computing — a tangent they had not gotten to, a thread that would remain unpulled. He had gone to sleep and not woken up.

Neo learned of it from Arthur's daughter, whom he had met twice at faculty events thirty years ago and whose voice on the phone carried the specific quality of someone who had been expecting a call like this and was still not ready for it.

He sat with the phone in his hand for a long time after hanging up.

Then he opened the working document and looked at what they had built.

Anna came to Westport the following weekend.

She brought nothing — no folders, no laptop, just herself. They sat in Arthur's chair and Neo's chair, which had never been formally designated but had become that over the months, and they drank coffee and did not talk about the research for a long time.

"He said what he needed to say," Anna said finally.

"He did," Neo said.

"The thing about the Fed's distributional choices. He'd been holding that for fifty years."

"He had."

She looked at the window. The Sound in March was still gray but different — the particular gray of water that knows spring is coming even if it isn't there yet.

"What do we do with it?" she asked. "All of it."

Neo had been thinking about this for three months, and the honest answer was that he didn't know. Not in the way he had known things throughout his career — not with the specific confidence of a thesis he was prepared to defend. The picture assembled itself and then refused to resolve into either the crisis narrative that some readers were clearly expecting or the comfortable reassurance that others wanted.

"Tell me what you think it says," he said. "You've been inside it with me. What does it say?"

Anna was quiet for a moment.

"It says the system is old," she said. "The basic architecture — private money creation, sovereign debt, reserve currency — is three hundred and thirty years old. It has been adapted and patched and reformed many times. It has survived many things that should have broken it. And the accumulated weight of what it has produced — the concentration, the offshore flows, the debt levels, the measurement gaps — is heavier than it has been at most previous points."

She paused.

"It also says that things at the edges are being tried. Cooperatives and local currencies and energy transitions and monetary reform proposals. Most of them are small. Some of them work. And the technology — AI, blockchain, digital currency — is moving faster than the institutions governing it."

"And what does that add up to?"

"I don't know," she said. "That's the honest answer. I've reported from countries where the system failed completely. Where the currency collapsed and the banks closed and people lost everything they had saved. I've also reported from countries where the system muddled through — where things got significantly worse and then got better, not through a clean transformation but through the grinding accumulation of small adjustments."

"The muddling-through hypothesis," Neo said. "Arthur's favorite."

"He told me that," Anna said. "The week before he died. He said: most of economic history is muddling through. The spectacular collapses are the ones that get remembered. The slow deteriorations and partial recoveries that most people actually live through are the ones that don't have a clean narrative."

Neo looked at the working document.

Sixty-one thousand, two hundred subscribers.

Three publishers. Four conference invitations. The former Treasury official, whose name he now knew, who had sent a second message that was more direct than the first: I think you should know what I know before you publish. I'm not threatening you. I'm offering.

He had not responded.

"The Treasury official," Anna said. She had seen his expression.

"Yeah."

"You're going to have to decide."

"I know."

She leaned forward. "What are you afraid of?"

He thought about this honestly.

"That publishing it changes what it is," he said. "The research was private. Garvus and me and the working document and then Arthur and you. It was — honest, in the way that private things can be honest. The moment it becomes public it becomes an argument. People read it as a prediction or a warning or an agenda. The uncertainty I've tried to hold — the genuine uncertainty about what the evidence supports — becomes a position."

"It already is a position," Anna said. "You've had sixty thousand people reading it."

"The Substack pieces are excerpts. The full picture is different."

"Is it?"

He sat with that.

Garvus had been running on the Mac Studio throughout the conversation, as it always was when Neo was working. It had not spoken. Now it did.

"There is a documented pattern in research that achieves public attention," Garvus said. "The research that reaches the most people is rarely the research with the most carefully held uncertainty. The audience for nuance is smaller than the audience for clear conclusions. Whether that pattern should influence the decision to publish is not an analytical question."

Anna looked at the speaker. Then at Neo.

"He's right," she said. "But it's also not a reason not to publish."

"No," Neo said. "It's not."

He looked at the working document one more time.

What they had assembled — Arthur and Anna and Garvus and the months of careful reading — was not a prediction. It was not a manifesto. It was a picture of a system that had been built for specific purposes, had produced specific outcomes, was under genuine pressure from multiple directions, and was being met at its edges by experiments in different ways of doing things.

Whether the pressure produced crisis or adaptation or something that didn't fit either category — that was the open question. The honest one. The one that the evidence didn't answer.

He thought about Arthur driving himself to Cambridge on a Friday and calling on a Sunday about quantum computing and the BIS and going to sleep knowing what he knew and having said what he needed to say.

He thought about Anna on a train from a country where people had been killed in the streets, sitting in a graduate school library trying to understand why, and ending up in Westport, Connecticut looking at the same financial architecture from the other side.

He thought about a Scottish entrepreneur named William Paterson in 1694 who had a clever idea about how to finance a war, and about the three hundred and thirty years of architecture that had accumulated on top of that idea, and about the nineteen-year-old in 2013 who had asked whether a blockchain could execute arbitrary agreements, and about the eighty-one-year-old economist who had spent his career inside an institution and had finally, at the end of things, said what he actually thought.

He opened a new document.

He began to write.

Not the synthesis he had been building toward. Something simpler. A letter, almost. To the sixty-one thousand people who had been following the threads.

This is what the evidence shows. This is what it doesn't show. This is where the uncertainty lives. I am one retired analyst with a local AI and two people who were willing to think alongside me. Take from this what you find useful. Leave what you don't. The conversation is yours now.

He read it back.

Then he added one more line.

We don't know what comes next. Neither does anyone else. That's the most honest thing I can tell you.

He showed it to Anna.

She read it. She looked up.

"That's it," she said.

He hit publish.

Outside, Long Island Sound was moving in the March wind — not the patient gray of winter anymore but something with more motion in it, more uncertainty about where it was going. The kind of water that didn't know yet what it was becoming.

He closed the laptop.

Garvus was quiet.

The signal, wherever it went, was out there now.

END OF FIAT

by J & C

FREE & UNMONETIZED — ALL RIGHTS RESERVED — THE SIGNAL GETS THROUGH